

Chapter 1

Portfolio Shifts and Financial Intermediation: A DSGE Analysis of U.S. Household Deposit Outflows

1 Introduction

The composition of financial assets held by U.S. households has changed markedly since the COVID-19 pandemic. A notable change is the decrease in time and savings deposits, as their share in total financial assets decreased by 3.2 percentage points and their level decreased by \$2,563 billion in real values, which was a 27% decline from 2022:Q1 to 2025:Q4. Those deposits have been a primary source of funding for depository institutions as 50% of their total equity and liabilities have been funded by time and savings deposits, and households' portion of them was 86% on average during the period.

This episode is important because it highlights the qualitative difference from the credit or funding shocks studied in the macroprudential literature: deposits fell while bank holding companies' equity investment in their subsidiary banks rose, so banks experienced a shift in the composition of their funding rather than a uniform contraction of it. This funding-mix character—deposits down, equity up, leverage actually improving—is what makes the episode worth analyzing on its own terms and calls for a framework in which household portfolio choice, bank funding structure, and macroprudential regulation are considered jointly.

However, the relationships between agents in the financial market have become more complex, and the macroeconomic policies designed to stabilize the market and the economy have become more entangled. These complexities make it difficult to track the effects of major changes in the financial environment while maintaining a comprehensive view of the overall economy. To address this challenge, I develop a tractable DSGE model that incorporates the unprecedented decrease in deposits and recent observations in the U.S. financial market.

The salient features of the model are summarized as follows. First, the shift in household portfolio choices is modeled by introducing financial distress costs for holding financial assets other than deposits. An exogenous shock to the elasticity parameter of the cost function, which represents a change in the financial environment around households, can shift the share of deposits held by the household. With this feature, I investigate the effects of changes in household portfolio choices on the overall economy.

Second, the substitution of funding sources of depository institutions is implemented based on observations of bank holding companies' (BHCs) operations in the U.S. The investment of BHCs in their subsidiary banks has increased and replaced lost deposits partially while expanding their assets.¹ This feature generates more realistic dynamics in the banking sector, since banks would not simply decrease their lending activities when deposits decrease.² To represent the bank holding company and its different degree of risk aversion, I introduce a banker who optimally decides how much to invest in subsidiary banks by borrowing from the household.

Third, different types of lenders and borrowers are introduced into the model to represent the segmented financial market. Traditional banks lend to intermediate good producers in the production sector. I specify the loan contract as credit line loans, which are a widespread type of contract in the U.S. economy.³ The other financial intermediation is represented by a shadow bank's operations. The shadow bank in the model lends to borrowers with a risky profile without supervision by financial authorities. To represent the intertwined relationship between depository and shadow banks, the model assumes that the shadow bank is funded by the traditional banks.

Fourth, the interaction between different borrowers in the production sector is introduced into

¹From 2022:Q1 to 2025:Q4, the time and savings deposits decreased by 23.6%; however, the investment of BHCs increased by 5.0%. Also, the depository institution loans n.e.c., most of which have been extended to nonfinancial businesses, increased by 63.0% (the figures are retrieved from the Z.1 Federal Reserve's Financial Accounts of the United States).

²Begenau and Stafford (2022) also point out the possibility of substitution in the funding sources of banks, which could attenuate the impact of deposit fluctuations on lending.

³Greenwald et al. (2020) reports that 58% of 4,496,353 loan facility observations in the U.S. from 2012:Q3 to 2019:Q4 were credit line loans.

the model. Along with a conventional two-stage production structure, I introduce an additional input in the production sector, which is produced by a capital good producer, refined by startups without cost, and combined with the capital of intermediate good producers. Intermediate good producers and startups borrow from different types of lenders, and the investment and borrowing are closely related, so the substitution of capital inputs by the intermediate good producers affects the credit demand of both types of borrowers. The endogenous reallocation of credit in the production sector is generated based on this interaction.

Finally, I implement and extend well-established frameworks of monetary and macroprudential policies. The monetary policy framework is based on Sims et al. (2023), which represents the floor system of monetary policy relying on the adjustment of the interest rate on reserves of the central bank. The macroprudential policy framework is based on Gerali et al. (2010) and Angelini et al. (2014), introducing the capital regulation of banks under Basel III. Additionally, a central bank lending program is implemented as an asset purchase policy. Based on the discussion of credit lines and firms' financial distress costs in Greenwald et al. (2020) and Amberg et al. (2023), the policy aims to provide liquidity with a better contract term for firms without enough credit line limit.

I conduct a Bayesian estimation of the model using U.S. data from 2009:Q2 to 2025:Q4. The variance decomposition and forecast error variance decompositions are performed to quantify the effects of the shocks and the contribution of each shock to the fluctuations in key variables. The parameterization of the model relies on the literature and the estimation results. The impulse response function, sensitivity, welfare, and determinacy analyses are conducted to investigate the dynamics of the model.

The analysis results show that the model can represent the recent portfolio shift of U.S. households away from deposits and the subsequent reallocation of credit flows in the financial market. The estimation results suggest that the model successfully captures the fluctuations of the share and level of deposits driven by the financial distress shock. However, most of the variations in the

credit supply are explained by other shocks.

The impulse response function and counterfactual analysis show that financial policies can play a central role in credit reallocation when households shift their portfolio away from deposits unexpectedly. A strong counter-cyclical asset purchase policy and capital regulation of banks amplify the reallocation of credit in the financial market, while a strong cyclical risk-weight adjustment dampens the reallocation. Finally, the welfare analysis suggests that the design of a counter cyclical capital regulation of banks does not operate in favorable ways for the households. The reason is specific to the funding-mix nature of the shock: because deposits fall while bankers' equity investment rises, bank assets and bank leverage move in opposite directions, and a rule that conditions on asset growth loosens the target ratio precisely when the bank's balance sheet is strengthening. This implies that macroprudential policies designed for canonical credit booms—where assets and leverage co-move—should be reconsidered for shocks that originate on the household funding side.

2 Motivations

2.1 Empirical Observations

Figure 1 shows the level and share of time and savings deposits held by U.S. households from 2008:Q1 to 2025:Q4 in real values. The level of deposits shows a distinct decrease after the COVID-19 pandemic and the decrease in the share of them in total financial assets has accelerated after the same period.⁴

Meanwhile, the other financial assets, as well as the total financial assets, show a different trend. Over the post-pandemic window (2022:Q1–2025:Q4), total financial assets increased by 9.2%. Equity and investment funds increased by 14.5%, money market funds increased by 71.0% and the debt securities increased by 103.3%. This alters the composition of U.S. households' financial assets, and the share of time and savings deposits decreased by 3.2 percentage points.

⁴All the balance sheet data in this document are obtained from “Z.1 Financial Accounts of the United States”, Board of Governors of the Federal Reserve System. The data are transformed into real values using “Personal consumption expenditures” retrieved from the Bureau of Economic Analysis.

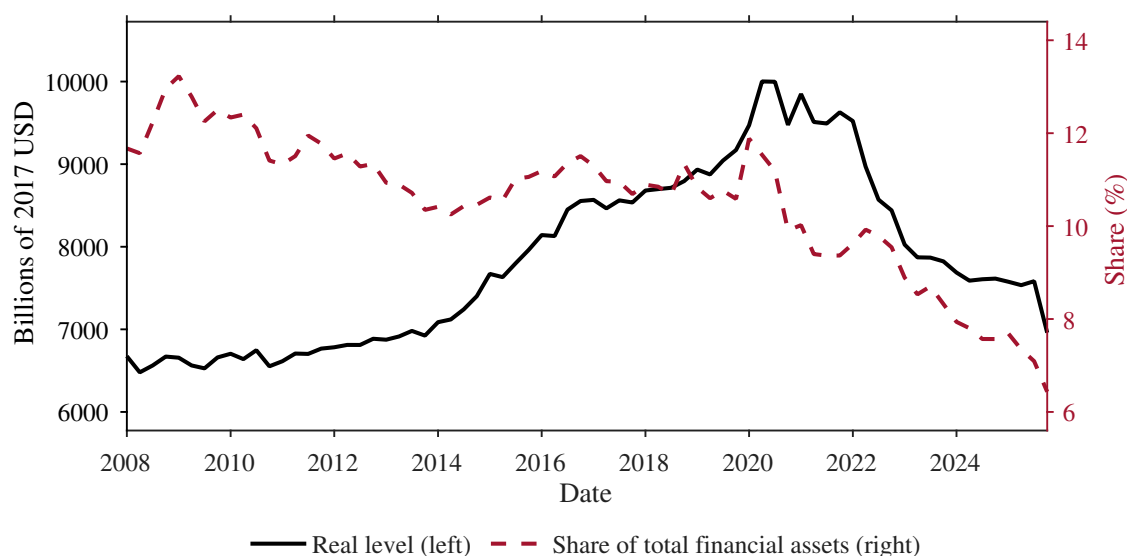


Figure 1: The level and share of time and savings deposits

Figure 2 shows this change in the composition of U.S. households' financial assets from 2008:Q1 to 2025:Q4 in real values. Even checkable deposits and currency, which are the most liquid assets for transactions, increased by 21.4% (0.41 percentage points in share) over this window. Thus, the overall trend suggests that U.S. households shifted their portfolios away from savings deposits toward other safe assets such as money market funds and Treasury securities.

The shift in household portfolio choices can be explained by the increase in the interest spread between deposits and alternative financial assets, which offer higher returns. Panel (a) of Figure 3 shows the increasing spread between the average interest rate on certificates of deposit (CDs) and that of the 3-month Treasury bill (T-bill) as a proxy for the opportunity cost of holding non-transaction deposits. Indeed, the spread has widened after the COVID-19 pandemic up to 4.25% in 2023:Q2, indicating that the widening interest spread was a major driver of the recent shift in household portfolio choices.

However, the increase in the same spread from 2016:Q1 to 2019:Q2 was not accompanied by a decrease in time and saving deposits even though the spread reached a local maximum of 2.18% in 2019:Q1. Instead, the level of deposits increased by 9% during the period. Thus it is natural to ask why households responded differently to the increasing interest spread in different periods.

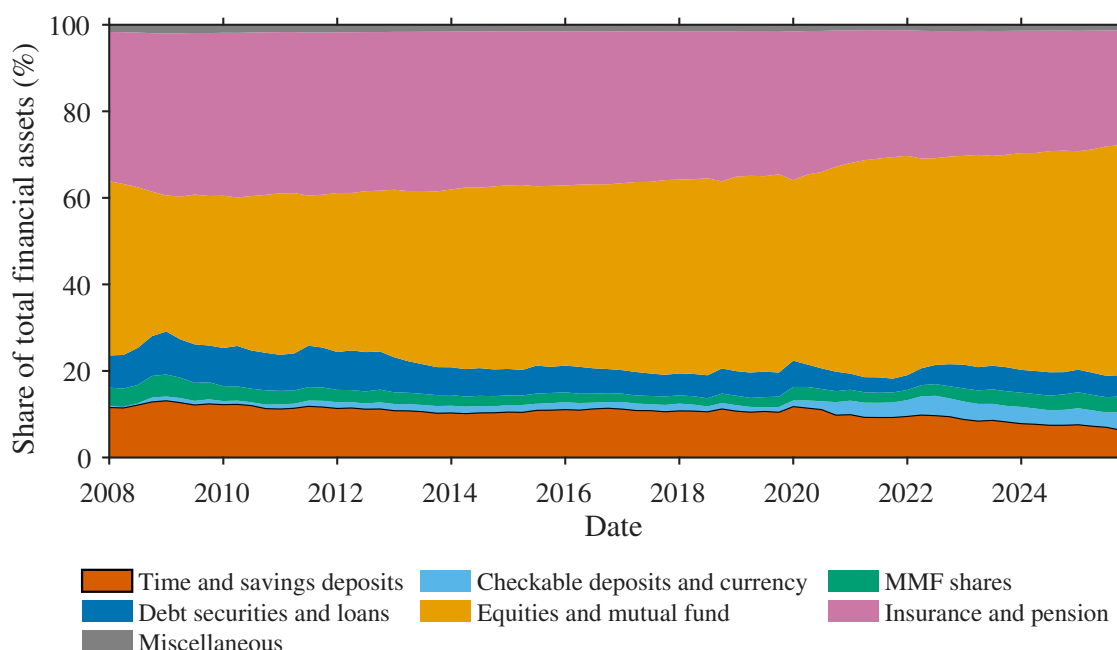


Figure 2: The composition of U.S. households' financial assets

One plausible explanation is the changing response of households to the interest spread over time due to fundamental changes in their risk profiles and financial literacy. But some indirect evidence such as little change in the average financial literacy and no dramatic increase in the stock market participation rate of U.S. adults during the period contradicts this argument.⁵

From these observations, I suggest the possibility that a rapid change in the financial circumstances of U.S. households influenced their portfolio choices. For instance, the adoption of mobile banking rose sharply after the COVID-19 pandemic.⁶ The technology adoption has made households more comfortable with mobile banking and online financial services, which gave them greater access to financial products as alternatives to deposits. This can be connected to the discussion in the literature on the search costs of households interpreting the adoption of a new technology as a global decrease in different search costs across heterogeneous households.⁷ With a new

⁵TIAA Institute Personal Finance Index (P-Fin Index) shows that there is no significant change in the average level of financial literacy of U.S. adults from 2017 to 2025. Also, Gallup reports a 62% stock market participation rate among U.S. adults. The rate did not recover to the pre-Global Financial Crisis level of 63% in 2004. See Appendix A.4 and Appendix A.5 for details.

⁶See Appendix A.6.

⁷For example, Yankov (2018) investigates the heterogeneity of search costs with different household

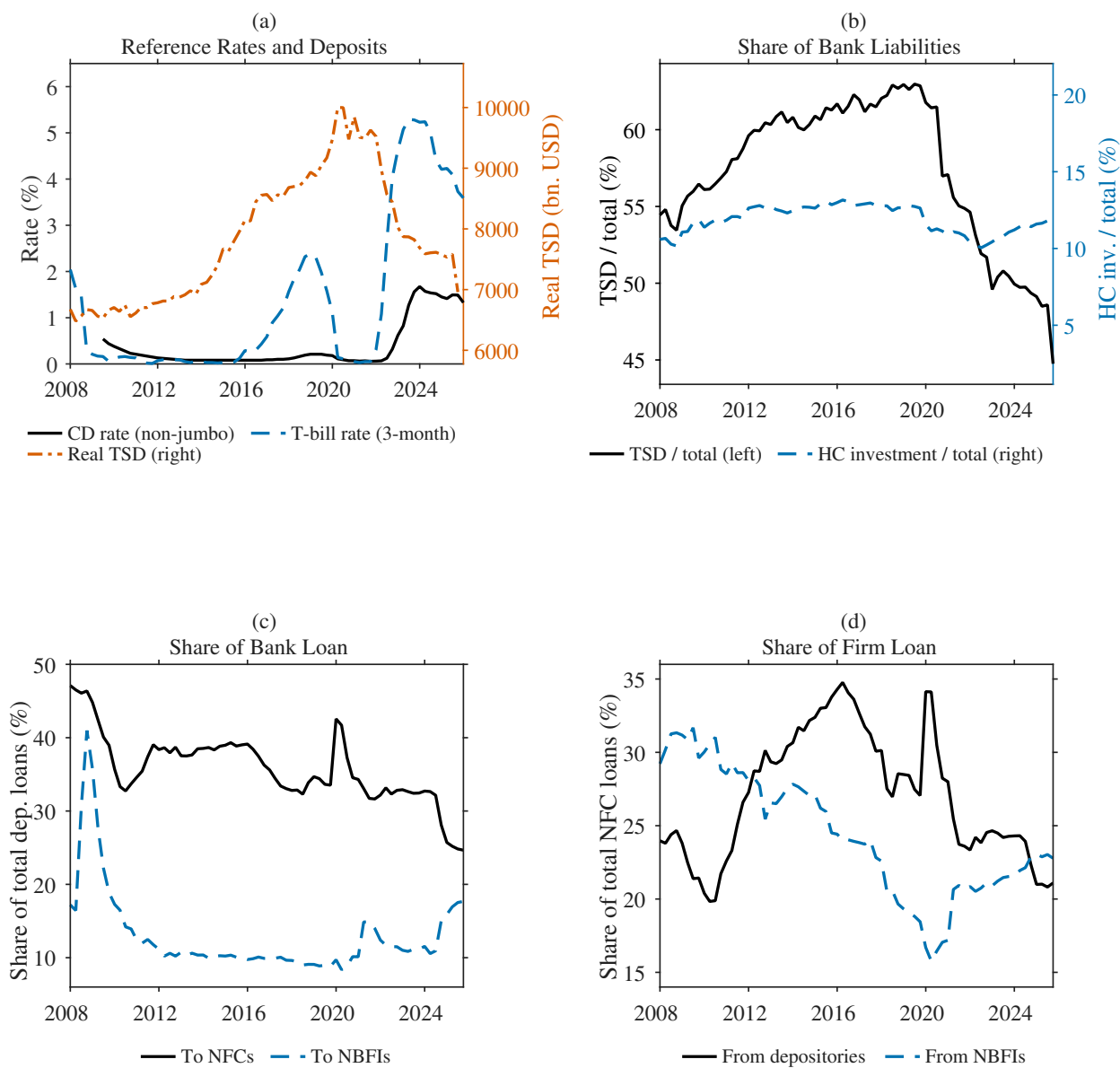


Figure 3: Empirical Observations on U.S. Financial Market

technology, households could have moved their portfolios away from deposits with fewer stages of physical visits to banks and less effort to compare alternatives.

In addition to the drastic decrease in deposits, there are other notable observations in the U.S. financial market that suggest a reallocation of credit flow is underway. When households reallocated their portfolios away from deposits, banks faced a sharp decrease in the share of deposits in their total equities and liabilities as shown in panel (b) of Figure 3. The share decreased by 9.9 percentage points (pp) over this window, whereas the share of equity and other investments from bank holding companies increased by 1.31 pp. The increase in the share of investment is remarkable as the total equities and liabilities of banks decreased by 6.7%. On the asset side of banks, the share of loans to Nonbank Financial Intermediaries (NBFIs) increased by 5.3 pp whereas the share of loans to nonfinancial corporates decreased by 7.5 pp. Panel (c) of Figure 3 summarizes these changes.

Panel (d) of Figure 3 shows the change in the lending sources for loans to nonfinancial corporates. The share of loans from banks decreased by 2.3 pp over this window, whereas that from NBFIs increased by 1.9 pp.⁸ This observation is consistent with the recent literature on the role of NBFIs in supplying credit to nonfinancial corporates (Cai and Haque (2024) and Acharya et al. (2024)). The details of data construction for the figures are explained in the subsections of Appendix A.

In sum, it seems that banks are trying to replace the lost deposits with other sources of funding including the investment from their parent companies. Since deposits are the cheapest source of funding for banks, they are facing higher funding costs than before, which could have led them to adjust their asset portfolios, for instance, by lending more to NBFIs. This idea is consistent with the recent literature on the relationship between banks and NBFIs (Acharya et al. (2024), characteristics. He models the asymmetric transmission of the policy rate to deposit rates based on the distribution of search costs.

⁸The decline of the depository loan in the asset side of banks was larger than that in the liability side of nonfinancial corporates. This could be due to the expansion in the balance sheet of banks regardless of the change in the source of funding.

Berrospide et al. (2025)). This combination—deposits falling while BHC equity investment rises, and asset reallocation toward NBFIs—is the empirical signature of the funding-mix shock studied in the rest of the paper, and it distinguishes the episode from the funding-level contractions that motivate standard counter-cyclical macroprudential rules.

This pattern also speaks to why the deposit outflow is best understood as an unanticipated shock originating on the household side rather than a deliberate bank funding decision. A bank that values its cheap and sticky deposits and had foreseen the outflow would have adopted one of two textbook responses: it would either defend the deposits by raising deposit rates to retain them, or accept the loss and let its balance sheet contract to avoid paying for costlier funding. The data show neither response in isolation. The pass-through from policy rates to deposit rates remained weak over the period as shown in panel (a) of Figure 3, which is consistent with banks letting the deposits go; yet banks simultaneously replaced the lost funding with more expensive holding-company equity rather than deleveraging, which is consistent with banks needing to retain funding. Observing both at once is difficult to reconcile with a bank that anticipated the outflow, but it is exactly what one would expect if the outflow was an unexpected shift in household portfolio behavior to which banks reacted after the fact. This reading motivates the modeling choice in the rest of the paper, where the episode is generated by an exogenous, unanticipated shock to the household’s financial distress cost.

2.2 Related Literature

Following seminal works of the New Keynesian DSGE literature with a financial intermediation such as Bernanke et al. (1999) and Kiyotaki and Moore (1997), this paper extends the well-established frameworks to represent new observations in the U.S. financial market.

First, the model in this paper incorporates macroeconomic policy frameworks from the literature. Sims et al. (2023) provide a baseline framework to implement the floor system of monetary policy and quantitative easing (QE) or Large Scale Asset Purchase Programs (LSAP) by the central

bank. Additionally, Kumhof and Wang (2021) suggest stylized forward-looking policy rules for these policies. The model of this paper incorporates their frameworks and the financial constraints in the production sector from Greenwald et al. (2020) and Amberg et al. (2023) to implement unconventional monetary policies such as the Main Street Lending Program (MSLP) by the Federal Reserve.

Second, I implement the macroprudential policy framework from Gerali et al. (2010) and Angelini et al. (2014). They provide a mechanism for the capital regulation of banks in the form of banks' adjustment costs. I extend their intuitive and tractable mechanism to enlarge the financial system of the model in this paper. Specifically, I modify the adjustment cost to cover different assets and liabilities of the traditional banking sector including reserves and equity investment from BHCs.

Finally, the foundation of the financial system of this paper is built upon the framework of Lubello and Rouabah (2024). They provide an intuitive and manageable framework to implement a segmented credit market with two different types of financial intermediaries for different borrowers. I extend their framework by adding holding companies' funding, bank-funded shadow banks, and credit line contracts from the recent observations in the U.S. financial market (Acharya et al. (2024), Berrospide et al. (2025), Greenwald et al. (2020)). Additionally, the costly state verification (CSV) mechanism is added to represent risky loans from shadow banks (Gasparini et al. (2024)).

3 The Model

3.1 Overview

This section provides an overview of the model by summarizing the key features of each agent. The real and financial sectors of the model are illustrated in Figure 4 and Figure 5, respectively.

First, a standard representative household with two distinct savings assets is introduced into the model. Additionally, the model has another household-type agent, called a banker. The banker,

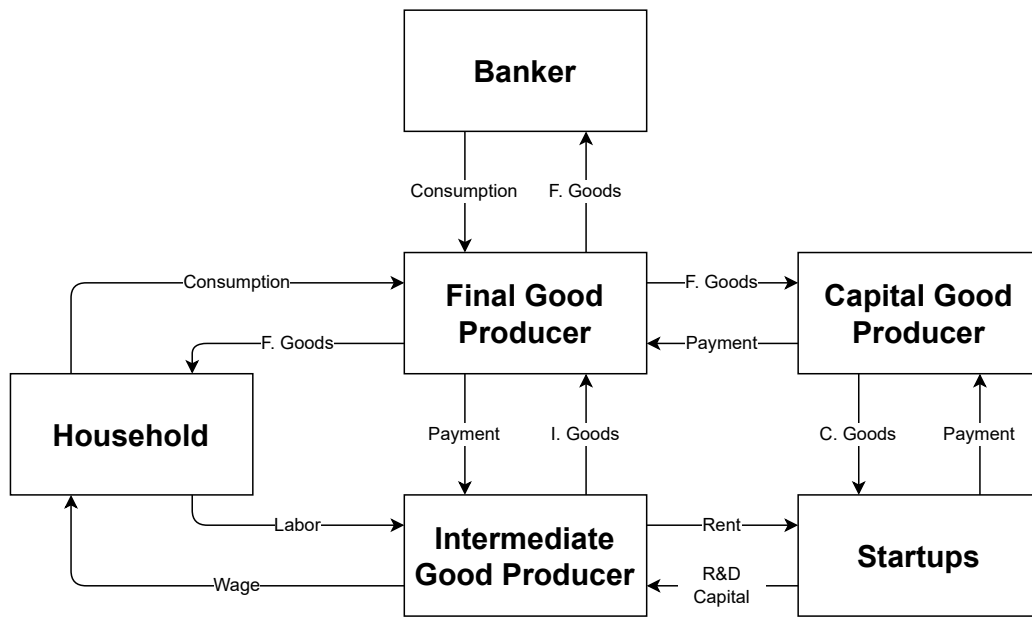


Figure 4: The Summary of Real Sector

The “F.Goods” represents final goods, “I.Goods” represents intermediate goods, and “C.Goods” represents capital goods.

who owns depository institutions and lends to NBFIs, represents BHCs in the real economy. The main role of the banker is to raise funds from the household and invest in its subsidiary bank. However, the banker’s decisions under a different risk profile and time preference from those of household are also important in the dynamics of the model.

Second, there are two types of banks in the financial sector: traditional banks and shadow banks. The financial intermediation of the traditional banking sector has a two-stage structure consisting of a wholesale bank and retail banks. The retail branches make deposit contracts with the household, and loan contracts with the production sector and the shadow bank. I assume that the retail banks have Dixit-Stiglitz type monopolistic competition in both deposit and loan markets to model realistic interest rate spreads.

In addition to the assets and liabilities from the retail branches, the wholesale bank manages its balance sheet with additional assets and liabilities, which are reserves in the central bank’s account and investment from the banker, respectively. It also retains its own capital as a portion of

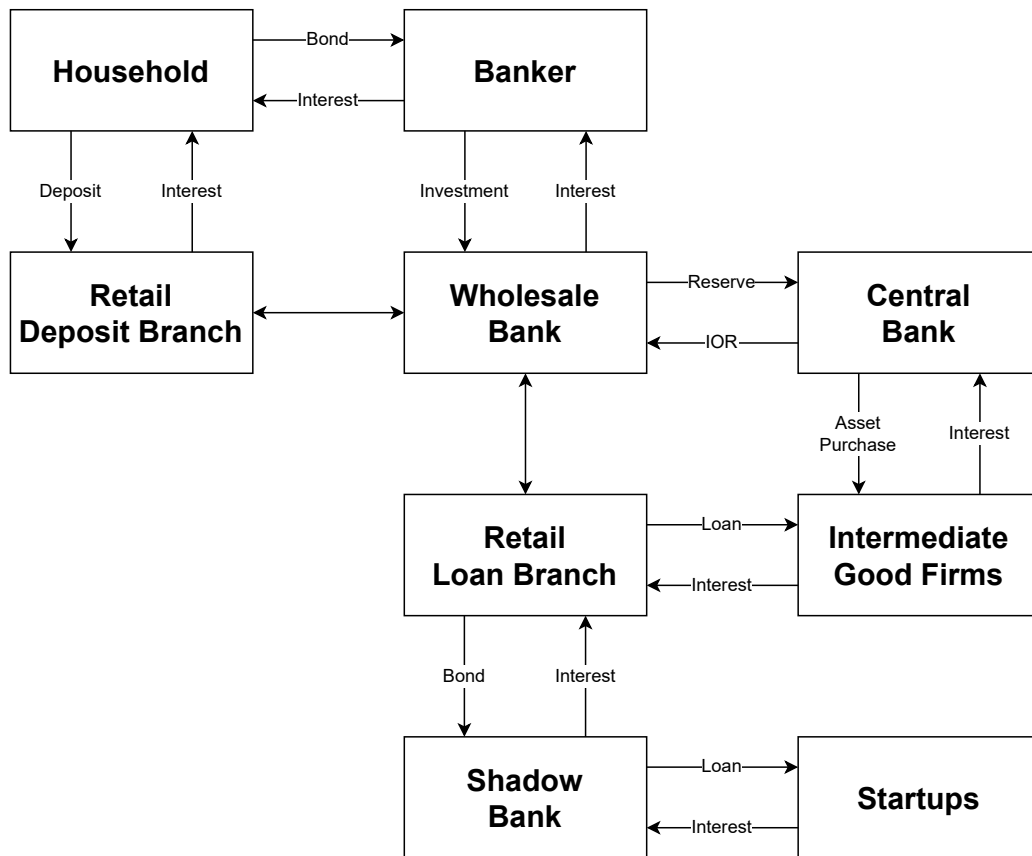


Figure 5: The Summary of Financial Sector
The “IOR” represents the interest on reserves from the central bank.

operating profit. With the help of the two-stage structure, macroeconomic policies are implemented to intervene in the wholesale bank's operation only.

The shadow bank is funded by the wholesale bank and lends to distinct borrowers named 'startups', who cannot borrow from the traditional banking sector.⁹ Similar to the wholesale bank, the shadow bank leaves a portion of its operating profit as inner capital.

Third, production in this model is an extension of the standard two-stage production structure in the literature. In addition to the constant elasticity of substitution (CES) aggregation of goods from competitive intermediate good producers employing labor and their own capital, I introduce an additional input to the production sector. Specifically, I assume that the new input takes the form of capital, which is produced by a capital good producer using the final good. Then startups transform the new capital into R&D capital without cost and rent it to the intermediate good producers.

Based on this production structure, the agents in the production sectors take the role of borrowers in the model. I assume that the intermediate good producers borrow from the retail banks, whereas the startups borrow from the shadow bank. Additionally, I assume a different form of loan contract for each borrower. The intermediate good producers borrow under a credit line contract, which allows them to adjust their loan utilization rate within the committed credit. To model the realistic behavior of borrowers under the credit line contract, I introduce a firm's financial distress cost that increases in the share of bank loans in their total borrowing. For the startups, I assign a risky profile to them by introducing an idiosyncratic return shock on their operation. Based on this assumption, a stylized form of contract is implemented to connect the idiosyncratic shock to the debt payment of startups. This feature is essential to represent the spread in loan rates observed between depository loans and those from NBFIs.

Finally, the policy frameworks in this model are the following. First, the interest-on-reserve

⁹I employ the name 'startup' to emphasize the risky, R&D-intensive nature of shadow bank borrowers. Following Chernenko et al. (2022), who show that NBFi borrowers have riskier profiles and spend more on R&D activities (9% vs. 4% of sales), I model startups as producers of R&D capital who provide it to intermediate good producers. The production structure is crucial as it generates endogenous credit reallocation between borrower types.

(IOR)-based monetary policy framework is implemented. A typical Taylor-type rule determines the IOR responding to the inflation gap. Additionally, a central bank lending program is implemented as an unconventional monetary policy (asset purchase policy) to support the borrowers from the traditional banking sector. Second, the macroprudential policy framework under Basel III is implemented to regulate the financial intermediation of the traditional banking sector. Specifically, the wholesale bank receives penalty costs when its ratio of capital to risk-weighted assets deviates from the target ratio set by the macroprudential policy authority. The authority also assesses the risk of banks' assets continuously, so the risk weights on assets are adjusted accordingly. A similar penalty cost is imposed on the banker when its investment in the wholesale bank relative to borrowing from the household deviates from the target double-leverage ratio.

In the following sections, the key mechanisms of each agent in the model are summarized. A detailed derivation of the model can be found in Appendix D.

3.2 Households

The household sector of this model has two notable features compared to the prototypical households in NK-DSGE models. First, a representative household holds two types of financial assets: deposits and banker's bonds similar to Lubello and Rouabah (2024). Second, the household pays financial distress costs that increase with the share of banker's bonds held in total savings.¹⁰

The operation of the household sector is as follows: a representative household consumes (c_t) and works (h_t).¹¹ It holds two types of financial assets: deposits in the loan branch of retail banks (d_t) and bonds issued by the banker (b_t^b). It pays the financial distress cost ($d_t^h \Phi_t$) where the cost function (Φ_t) decreases in the share of deposits (s_t) nonlinearly. Additionally, the household owns the firms in the production sectors and receives dividends from them (D_t^f from the intermediate

¹⁰The financial distress cost is defined as the physical or psychological costs from holding a financial asset other than deposits. It can represent various costs of households such as the discomfort from bearing illiquidity risk, the variation in returns from alternatives, or the effort to manage multiple financial products in an abstract way.

¹¹A detailed derivation of the representative household's problem can be found in Appendix B.3.

good firms, D_t^k from the capital-producing firms, and $(1 - \iota)\chi_n \mathcal{W}_t^n$ from startups). In addition, the central bank transfers its operating profit (T_t^{cb}) to the household. Finally, it transfers a fixed amount (X_0^{bhc}) to the banker.

The household maximizes its expected lifetime utility discounted by its time preference (β^h):

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_h^t \left[\frac{1}{1 - \frac{1}{\gamma_h}} \left\{ c_t^h - h_c c_{t-1}^h \right\}^{1 - \frac{1}{\gamma_h}} - \frac{\chi_h}{1 + \frac{1}{\zeta_h}} \left\{ h_t^h \right\}^{1 + \frac{1}{\zeta_h}} \right] \quad (1)$$

subject to its nominal budget constraint:

$$\begin{aligned} R_{t-1}^b P_{t-1} b_{t-1}^b + r_{t-1}^d P_{t-1} d_{r,t-1} + P_t w_t h_t^h + P_t Z_t^{total} \\ \geq P_t c_t^h + P_t b_t^b + P_t d_{r,t} + P_t d_t^h \Phi(s_t) \end{aligned} \quad (2)$$

where the transfers are:

$$Z_t^{total} = D_t^f + D_t^k + (1 - \iota)\chi_n \mathcal{W}_t^n + T_t^{cb} - X_0^{bhc}.$$

The household financial distress cost function designed after Dotsey and Ireland (1996) and Lubello and Rouabah (2024) is defined as:

$$\Phi(s_t) \equiv \chi_1 \left(\frac{1 - s_t}{s_t} \right)^{\vartheta_t}. \quad (3)$$

Here, s_t denotes the share of deposits in the household's total savings d_t^h , and χ_1 and ϑ_t represent the scale and elasticity of the function, respectively. The stylized form of the cost function ensures a positive holding of both assets in equilibrium as the cost approaches infinity when the share of deposits approaches 0.

To generate the desired dynamics in the share of deposits, I add an exogenous stochastic process

S_t^ϑ for ϑ_t .

$$\vartheta_t = S_t^\vartheta \bar{\vartheta} \quad (4)$$

$$\log(S_t^\vartheta) = \rho_\vartheta \log(S_{t-1}^\vartheta) + u_t^\vartheta \quad (5)$$

where the exogenous shock, u_t^ϑ follows normal distribution with the mean of zero and the standard deviation, σ_ϑ^2 . Under the stylized assumptions of the model, the deposits and banker's bonds can be written as a share of total savings

$$d_{r,t} = s_t d_t^h, \quad (6)$$

$$b_t^b = (1 - s_t) d_t^h. \quad (7)$$

In a cashless economy, the representative household chooses consumption, labor supply, and portfolio allocation to maximize its utility, taking into account the budget constraint. What distinguishes this household sector from typical assumptions is the presence of the financial distress cost and additional assumptions on portfolio allocation. Given the setup, the exogenous shock to the elasticity process variable (ϑ_t) will affect the household's decision on the share of deposits held in total savings. Specifically, a negative shock to the elasticity of the financial distress cost function (ϑ_t) generates a decrease in the share of deposits held by the household (s_t). The transmission of this shock to the overall economy will be the main focus of this paper and more discussion on this is provided in Section 5.

3.3 Banker

The banker in the model takes the role of a bank holding company (BHC) in the real economy. With the appearance of an additional household sector, which is not uncommon in the literature, it is introduced into the model to implement another source of funding for the traditional banking sector. Basically, fund-raising from the household and investing in the subsidiary bank are the

main functions of the banker in the model. To supplement a simple structure of financial intermediation, borrowing and leverage adjustment costs are introduced to prevent the corner solution in the banker's decisions. Specifically, the leverage adjustment cost is implemented to represent the regulation to prevent the excessive investment of banker in their subsidiary banks.

A representative banker consumes (c_t^b) and does not work. It raises funds from the household (b_t^b) and invests in the wholesale bank (e_t^b) . It owns the wholesale, retail and shadow banks and receives dividends from them (D_t^b, D_t^{sb}) . It also receives transfers from the household (X_0^{bhc}) . It pays fund-raising and leverage adjustment costs $(\mathcal{C}_{t-1}^{bhc,b}, \mathcal{C}_{t-1}^{bhc})$. I assume that the costs occurring at $t - 1$ are paid at t .¹²

The banker maximizes its expected lifetime utility discounted by its time preference (β^b) :

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_b^t \left[\log \left(c_t^b \right) \right] \quad (8)$$

subject to its nominal constraint:

$$\begin{aligned} & P_t b_t^b + R_{t-1}^{eb} P_{t-1} e_{t-1}^b + P_t D_t^b + P_t D_t^{sb} + P_t X_0^{bhc} \\ & \geq P_t c_t^b + R_{t-1}^b P_{t-1} b_{t-1}^b + P_t e_t^b + P_{t-1} \mathcal{C}_{t-1}^{bhc,b} + P_{t-1} \mathcal{C}_{t-1}^{bhc}, \end{aligned} \quad (9)$$

where the fund-raising cost is:

$$\mathcal{C}^{bhc,b} \left(b_t^b \right) = \frac{\kappa_{bhc,b}}{1 + \frac{1}{\theta_{bhc,b}}} \bar{b}^b \left(\frac{b_t^b}{\bar{b}^b} \right)^{1 + \frac{1}{\theta_{bhc,b}}}, \quad (10)$$

¹²For technical reasons, the remuneration and adjustment costs occur at $t - 1$ and are paid at t . By assumption, the dividends which the banker receives at t are the portion of operation profit of banks at $t - 1$ (see equation (14)). Thus the remuneration and adjustment costs of banks occur at $t - 1$ but appear in the consolidated resource constraint at t . For consistency, I assume that the timing of the payment of each type of costs in the model is identical.

and the leverage adjustment cost is:

$$\mathcal{C}^{bhc} \left(b_t^b, e_t^b, v_t^{bhc} \right) = \frac{\kappa_{bhc}}{2} \left(\frac{e_t^b}{b_t^b} - v_t^{bhc} \right)^2 e_t^b. \quad (11)$$

A few remarks are in order. First, I assume the log utility function for technical tractability and a parsimonious representation of the banker's risk-taking behavior compared to the household.¹³ Second, I assume the banker owns the wholesale, retail and shadow banks in the model.¹⁴ Finally, I implement the leverage adjustment cost, which has an analogous form to the capital adjustment cost in Gerali et al. (2010). The cost is essential to link the borrowing and investment decisions of the banker. More details of the leverage ratio regulation are provided in Section 3.6 and Section 5.3.3.

3.4 Banking Sector

The banking sector consists of three types of banks: wholesale banks, retail banks, and shadow banks. The wholesale banks and retail banks represent the traditional banking sector. They are funded by deposits from the household and equity investment from the banker. They supply loans to the production sector and the shadow bank. Using the funds from the traditional banking sector, the shadow bank provides loans to the startups.

The two-stage structure in the traditional banking sector is a well-established framework in the literature (Gambacorta and Karmakar (2018), Angelini et al. (2014), Gerali et al. (2010)). One of its advantages is to represent different subjects of interest from the separated stage of banking. In this paper, the wholesale bank is the main target of monetary and macroprudential policies,

¹³It is hard to pin down the exact level of banker's risk aversion since it depends on the ownership structure, regulation frameworks, and macroeconomic conditions. See Laeven and Levine (2009) regarding the varying risk taking behavior of banks.

¹⁴The ownership of the shadow bank by the banker is motivated by the recent observations in the U.S. financial market. For instance, Acharya et al. (2024) report the fundraising activity of the Blackstone Private Credit Fund. In December 2022, 98% of the committed credits, which were from 18 out of 19 funding facilities, were provided by 13 banks including BHCs, for instance, Bank of America and Goldman Sachs.

whereas the retail banks' operations serve to model the interest rate spreads in deposit and loan markets.

3.4.1 Wholesale Banks

Wholesale banks serve as intermediaries between retail banks and the rest of the financial system. They purchase loans originated by retail banks—both loans to intermediate good producers ($\tilde{\ell}_t^{f,b}$) and loans to shadow banks (b_t^{sb})—and provide deposit funding (d_t) back to the retail banks. Externally, wholesale banks receive equity investment from bankers (e_t^b) and hold reserves (re_t) at the central bank. The balance sheet condition is:

$$\tilde{\ell}_t^{f,b} + b_t^{sb} + re_t = d_t + e_t^b + X_t^b \quad (12)$$

where X_t^b represents the internal capital of the wholesale bank. Additionally, I assume a fixed dividend rate (v^b) to be paid out of the banks' net worth following Brunnermeier and Koby (2018). Then, the internal capital accumulation rule is as follows:

$$X_{t+1}^b = (1 - v_b)N_{t+1}^{b,total} \quad (13)$$

where $N_{t+1}^{b,total}$ is the total net worth of the wholesale and retail banks. This suggests that a fraction of the total profit of the traditional banking sector is paid to bankers as dividends. Additionally, the retail banks do not operate with their own capital and are fully funded by the wholesale bank. I will define the total net worth at the end of the retail bank section.

The wholesale banks' net worth is defined as follows:

$$\Pi_{t+1}N_{t+1}^b = R_t^f \tilde{\ell}_t^{f,b} + R_t^{sb} b_t^{sb} + R_t re_t - R_t^d d_t - R_t^{eb} e_t^b - \mathcal{C}_t^b, \quad (14)$$

where $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$ is the gross inflation rate and R_t^s is the gross interest rate on different assets where $s \in \{f, sb, d, eb\}$.

The capital-asset ratio adjustment cost is defined as:

$$\mathcal{C}_t^b = \frac{\kappa_{Kb}}{2} \left(\frac{X_t^b + \rho_e e_t^b}{\rho_{re} re_t + \omega_t^f \tilde{\ell}_t^{f,b} + \omega_t^{sb} b_t^{sb}} - v_t^k \right)^2 (X_t^b + \rho_e e_t^b). \quad (15)$$

The risk weights for wholesale loans to intermediate good producers and shadow banks are ω_t^f and ω_t^{sb} , respectively, and they are defined as:

$$\omega_t^s = (1 - \rho_s) \bar{\omega}^s + (1 - \rho_s) \chi_s (y_t - y_{t-4}) + \rho_s \omega_{t-1}^s, \quad s \in \{f, sb\}, \quad (16)$$

where κ_{Kb} is the adjustment cost coefficient parameter and v_t^k is the target capital-asset ratio set by the macroprudential policy authority.

The wholesale bank maximizes the expected net worth tomorrow discounted by the banker's stochastic discount factor ($\Lambda_{t,t+1}^b$):

$$\max \mathbb{E}_t \left[\Lambda_{t,t+1}^b N_{t+1}^b \right] \quad (17)$$

subject to the balance sheet condition.¹⁵

The wholesale banks' operation is the main target of monetary and macroprudential policies. I assume the "floor system" for the monetary policy, which means that the interest rate on reserves (IOR) is adjusted by the central bank. Also, I implement the capital-asset ratio adjustment cost to represent the capital regulation on banks. The functional form of the cost is designed after Gerali et al. (2010) and Angelini et al. (2014) but is modified to include investment from bankers and reserves (e^b, re).

The parameterization on the risk weights of assets and liabilities is crucial to match the observed dynamics in the U.S. financial market. Although they are not directly observable, I pin down the

¹⁵The bankers' stochastic discount factor is defined by $\Lambda_{t,t+1}^b = \beta_b \mathbb{E}_t \left[\frac{\lambda_{t+1}^b}{\lambda_t^b} \right]$. Here, λ_t^b is the nominal Lagrange multiplier for the bankers' budget constraint.

parameters using proxies from the data and the regulation frameworks. First, the parameters, ρ_f and χ_f in equation (16) are estimated by the U.S. delinquency rate on business loans. Second, ρ_{sb} and χ_{sb} are parameterized parsimoniously, considering the lack of data on the risk profile of shadow bank loans. The details of the parameterization of equation (16) are elaborated in Appendix C.1.

Second, the risk weight for reserves is parameterized by the model estimation using U.S. data. An informative prior distribution is set to have a lower risk weight for reserves than loans, which is consistent with the regulation frameworks in Basel Committee on Banking Supervision (2020b).

Third, the parameterization of the capital weight for equity investment from bankers (ρ_e) is based on Basel Committee on Banking Supervision (2020a). A stylized assumption is made that only the inner capital of the wholesale bank, X_t^b is the tier 1 capital. The detail of the parameterization of ρ_e is explained in Appendix B.2.

3.4.2 Retail Banks

Loan Branch

The loan branch of retail banks provides retail loans to intermediate good producers ($\ell_{r,t}^f$) and shadow banks ($b_{r,t}^{sb}$). They raise funds by selling those loans to the wholesale bank. Following Gerali et al. (2010), I assume that the retail banks are price setters who face loan demand schedules with elasticities $\varepsilon_{b,f}$ and $\varepsilon_{b,sb}$ for each type of loan. This enables the model to capture the stickiness of retail interest rates and represent the interest rate spread from the policy rate.

The loan branch of retail bank j solves:

$$\max_{r_t^f(j), r_t^{sb}(j)} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_b^t \lambda_t^b \left[r_t^f(j) \tilde{\ell}_{r,t}^{f,b}(j) + r_t^{sb}(j) b_{r,t}^{sb}(j) - \left(R_t^f \tilde{\ell}_{r,t}^{f,b}(j) + R_t^{sb} b_{r,t}^{sb}(j) \right) \right], \quad (18)$$

The retail bank j faces the demand schedules:

$$\tilde{\ell}_{r,t}^{f,b}(j) = \left(\frac{r_t^f(j) - 1}{r_t^f - 1} \right)^{-\varepsilon_{b,f}} \tilde{\ell}_{r,t}^{f,b} \quad (19)$$

$$b_{r,t}^{sb}(j) = \left(\frac{r_t^{sb}(j) - 1}{r_t^{sb} - 1} \right)^{-\varepsilon_{b,sb}} b_{r,t}^{sb}. \quad (20)$$

Here, $r_t^s(j)$ is the retail gross interest rate on loans set by retail bank j , and r_t^s is the aggregate retail interest rate on the loans where $s \in \{f, sb\}$.

Deposit Branch

The deposit branch of retail banks issues retail deposits to the household ($d_{r,t}$) and lends funds to the wholesale bank. Similar to the loan branch, the retail banks in the deposit branch are also price setters who face deposit supply schedules with elasticity ($\varepsilon_{b,d}$). The retail interest rate on deposits ($r_t^d(j)$) is set by each retail bank j . The deposit branch of retail bank j solves:

$$\max_{r_t^d(j)} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_b^t \lambda_t^b \left[-r_t^d(j) d_{r,t}(j) + R_t^d d_t(j) \right] \quad (21)$$

with the deposit demand schedule:

$$d_{r,t}(j) = \left(\frac{r_t^d(j) - 1}{r_t^d - 1} \right)^{-\varepsilon_{b,d}} d_{r,t}, \quad (22)$$

where r_t^d is the aggregate retail interest rate on deposits.

Total Net Worth of Wholesale and Retail Banks

Given the specification of retail banks' operation, the total net worth of the traditional banking

sector is defined as follows:

$$\begin{aligned}
\Pi_{t+1}N_{t+1}^{total} &= \Pi_{t+1} \left(N_{t+1}^b + N_{t+1}^r \right) \\
&= \left(r_t^f - r_t^d \right) \tilde{\ell}_{r,t}^{f,b} + \left(r_t^{sb} - r_t^d \right) b_{r,t}^{sb} + \left(R_t - r_t^d \right) re_t \\
&\quad - \left(R_t^{eb} - r_t^d \right) e_t^b + r_t^d X_t^b - \mathcal{C}_t^b
\end{aligned} \tag{23}$$

where N_{t+1}^b is the net worth of the wholesale bank and N_{t+1}^r is that of the retail banks.

3.4.3 Shadow Banks

The shadow banks borrow from the retail banks (b_t^{sb}) and lend to startups (b_t^n). Their balance sheet is as follows:

$$b_t^n = b_t^{sb} + X_t^{sb}, \tag{24}$$

where X_t^{sb} is the inner capital of shadow banks.

Shadow banks earn a net interest margin from their operations and pay two types of costs: the lending cost ($\mathcal{C}_t^{sb,n}$) and the monitoring cost (m_t^{sb}). The net worth of shadow banks is defined as follows:

$$N_{t+1}^{sb} = (\Pi_{t+1})^{-1} \left(R_{t+1}^{fn} b_t^n - r_t^{sb} b_t^{sb} - \mathcal{C}_t^{sb,n} - m_t^{sb} \right). \tag{25}$$

The interest rate on loans to startups (R_{t+1}^{fn}) is an ex-post gross rate that reflects the idiosyncratic return shock faced by startups.

The lending cost is designed following the cost of making loans in Kumhof and Wang (2021):

$$\mathcal{C}_t^{sb,n} = \frac{\kappa^{sb,n}}{1 + \frac{1}{\theta^{sb,n}}} \bar{b}^n \left(\frac{b_t^n}{\bar{b}^n} \right)^{1 + \frac{1}{\theta^{sb,n}}}, \tag{26}$$

which is proportional to the amount of loans to startups. The lending cost prevents shadow banks from lending an infinite amount and ensures a positive interest rate margin on shadow banks' financial intermediation. The monitoring cost occurs when shadow banks receive payments on

defaulted loans from startups.

Shadow banks maximize the expected net worth tomorrow, discounted by the bankers' stochastic discount factor:

$$\max \mathbb{E}_t \left[\Lambda_{t,t+1}^b N_{t+1}^{sb} \right] \quad (27)$$

subject to the balance sheet condition (24). Similar to wholesale banks, I assume a fixed dividend rate (v_{sb}) to be paid out of shadow banks' net worth. The inner capital accumulation rule is as follows:

$$X_{t+1}^{sb} = (1 - v_{sb}) N_{t+1}^{sb} \quad (28)$$

3.5 Production Sector

The backbone of the production sector is the standard two-stage production structure in the DSGE literature: competitive intermediate good production and final good aggregation with the Constant Elasticity of Substitution (CES) technology. In addition to this, capital-producing firms and startups are introduced into the model. The startups purchase capital from the capital-producing firms using loans from shadow banks, transform it into R&D capital without cost and rent it to the intermediate good producers.

The structure of the production sector is designed to capture the different types of financing in the real economy. The intermediate good producers are the borrowers from traditional banks, whose loan contract takes the form of a credit line. Meanwhile, the startups are the borrowers from shadow banks, whose loan contract reflects the possibility of default due to idiosyncratic return shocks.

Finally, the stylized asset purchase policy is implemented. The policy represents a central bank lending program that has better terms and conditions than those from traditional banks. The detailed discussion of the policy's implications is provided in the following sections.

3.5.1 Intermediate Good Producers

A continuum of intermediate good producers indexed by j produce differentiated goods using labor ($h_t(j)$), internal capital ($K_t(j)$), and rented capital from startups ($K_t^n(j)$). The production technology is given as follows:

$$y_t(j) = S_t^a (h_t(j))^{1-\alpha} \left[(K_t(j))^{1-\zeta} (K_t^n(j))^\zeta \right]^\alpha \quad (29)$$

where S_t^a is the technology shock process following:

$$\log(S_t^a) = \rho^a \log(S_{t-1}^a) + u_t^a. \quad (30)$$

Here, ζ is the share of rented capital from startups in total capital. The internal capital ($K_t(j)$) accumulates according to a conventional law of motion:

$$K_t(j) = (1 - \delta)K_{t-1}(j) + I_t(j). \quad (31)$$

where δ is the depreciation rate and $I_t(j)$ is the investment of intermediate good producer j .

To introduce rigidities in price and investment, the standard Rotemberg-type cost functions in the literature are adopted.¹⁶ The adjustment costs are defined as follows:

$$G_{P,t}(j) = \frac{\phi_p}{2} y_t \left(\frac{\frac{P_t(j)}{P_{t-1}(j)}}{\Pi_{t-1}} - 1 \right)^2 \quad (32)$$

$$G_{I,t}(j) = \frac{\phi_I}{2} I_t \left(\frac{I_t(j)}{I_{t-1}(j)} - 1 \right)^2. \quad (33)$$

where ϕ_p and ϕ_I are the adjustment cost parameters for price and investment, respectively.

Based on these conventional settings, I assume that intermediate good producers borrow from

¹⁶See Christiano et al. (2005), Kumhof and Wang (2021) and Lubello and Rouabah (2024).

retail banks under a credit line contract. The bank loan to intermediate good producer j consists of two components: the committed loan $\ell_t^{f,b}(j)$ and the utilization rate $u_t^f(j)$. The committed loan is pre-determined and the firm decides how much of it to utilize. Analogous to capital utilization in the literature, the capital should not change simultaneously with the utilization rate to obtain a unique value of the utilization rate.

Considering the procedure of credit line loans in practice, I assume that the committed loan is affected by the firm's earnings before interest, taxes, depreciation, and amortization (EBITDA) in the previous period.¹⁷ The process is given as follows:

$$\begin{aligned} \Pi_{t+1}\ell_{t+1}^{f,b}(j) &= \rho_\ell \ell_t^{f,b}(j) + (1 - \rho_\ell) \bar{\ell}^{f,b}(j) \\ &+ (1 - \rho_\ell) \theta_\pi \left(\Pi_t^{EBITDA}(j) - \bar{\Pi}^{EBITDA} \right) \end{aligned} \quad (34)$$

where the EBITDA is defined as

$$\Pi_t^{EBITDA}(j) = P_t y_t(j) - P_t w_t h_t(j) - P_t r_t^{k,n} K_t^n(j), \quad (35)$$

which is the revenue minus the factor costs in the model.

Next, I link the borrowing of firms to investment. The assumption that firms increase debt when they invest more is not uncommon in the literature.¹⁸ I assume that the total borrowing is under-bounded by the multiplication of investment. The constraint for the total borrowing is given as

¹⁷Greenwald et al. (2020) analyze firm-level data in the U.S. from 2012:Q3 to 2019:Q4, and report that the utilization rate of credit line loans is between 40% and 60%. The rate decreases to below 40% as firm size increases. Additionally, they report a positive correlation between credit line borrowing capacity and EBITDA.

¹⁸See Carlstrom et al. (2017) and Lubello and Rouabah (2024).

follows:

$$\psi_t^L I_t(j) \leq \ell_t^f(j) \quad (36)$$

$$\psi_t^L = S_t^\psi \bar{\psi}^L \quad (37)$$

$$\log(S_t^\psi) = \rho_\psi \log(S_{t-1}^\psi) + u_t^\psi. \quad (38)$$

where $\ell_t^f(j) = u_t^f(j)\ell_t^{f,b}(j) + \tilde{\ell}_t^{f,cb}(j)$ is the total loan from both retail banks and the central bank. The shock process S_t^ψ captures exogenous changes in the ratio of borrowing and investment (ψ_t^L).

The last remark concerns the firm's financial distress cost. The cost function for intermediate good firm j is defined as:

$$\varphi_t = \kappa_f \left(\frac{\tilde{\ell}_t^{f,b}(j)}{\ell_t^f(j)} \right)^{\theta_u} = \kappa_f \left(1 + \frac{\tilde{\ell}_t^{f,cb}(j)}{u_t^f(j)\ell_t^{f,b}(j)} \right)^{-\theta_u}, \quad (39)$$

where κ_f and θ_u are the scale and elasticity of the function, respectively. I assume the cost increases when the share of loans from retail banks in total loans increases. Using the market clearing condition of firm loans, the cost function can be rewritten as the second expression in (39). The stylized function implies that the cost increases in the loan from the bank, but decreases in the loan from the central bank.¹⁹

In sum, intermediate good producer j maximizes expected lifetime profit considering the household's stochastic discount factor²⁰:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Lambda_t^h \Pi_t^F(j) \quad (40)$$

¹⁹As the committed credit ($\ell^{f,b}$) is a predetermined variable in the model, an increase in the utilization rate will increase the share of loans from the bank, resulting in the increase of distress.

²⁰The stochastic discount factor is defined by $\Lambda_{t,t+1}^h = \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}^h}{\lambda_t^h} \right]$. Here, λ_t^h is the nominal Lagrange multiplier for the household's budget constraint, and the real Lagrange multiplier is $\Lambda_t^h = \lambda_t^h / P_t$.

subject to (29), (31), and (36), and the final good firm's demand for intermediate goods:

$$y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\theta} y_t. \quad (41)$$

The nominal profit of intermediate good firm j is defined as follows:

$$\begin{aligned} \Pi_t^F(j) = & P_t(j)y_t(j) + P_t\ell_t^f(j) - P_t w_t h_t(j) - r_{t-1}^f P_{t-1} \ell_{t-1}^f(j) \\ & - P_t r_t^{k,n} K_t^n(j) - P_t I_t(j) - P_t G_{P,t}(j) - P_t G_{I,t}(j) - P_t \phi_t(j). \end{aligned} \quad (42)$$

3.5.2 Capital Good Producers

Given the initial capital stock, capital good producers produce additional capital goods for startups using final goods. There is an investment adjustment cost, which has an identical form to that of intermediate good producers. The details of the capital good producers' problem are provided in Appendix D.7.

3.5.3 Startups

The key feature of the startups' problem is the idiosyncratic return shock and the debt contract based on its realization. The shock process is defined as follows:

$$\omega_{t+1}^n(j) \stackrel{i.i.d.}{\sim} \text{lognormal}(1, \sigma_t^n) \quad (43)$$

$$\sigma_t^n = \bar{\sigma}^n S_t^n \quad (44)$$

$$\log S_t^n = \rho^n \log S_{t-1}^n + u_t^n, \quad (45)$$

where $\omega_{t+1}^n(j)$ is the idiosyncratic shock for startup j at the end of period $t + 1$. The parameter σ_t^n is the standard deviation of the shock process. The process captures exogenous changes in the uncertainty of startups' returns.

The realized real operating profit of startup j at the end of the period is $\omega_{t+1}^n(j) R_{t+1}^n q_t^n K_{t+1}^n(j)$,

where R_{t+1}^n is the gross return on capital for startups, q_t^n is the real market price of new capital goods, and $K_{t+1}^n(j)$ is the capital stock purchased by startup j .²¹ Based on the realized return, the debt contract is designed as follows:

$$\text{Debt payment} = \begin{cases} \bar{\omega}_{t+1}^n R_{t+1}^n q_t^n K_{t+1}^n(j) & \text{if } \omega_{t+1}^n(j) \geq \bar{\omega}_{t+1}^n \\ \omega_{t+1}^n(j) R_{t+1}^n q_t^n K_{t+1}^n(j) & \text{if } \omega_{t+1}^n(j) < \bar{\omega}_{t+1}^n. \end{cases} \quad (46)$$

In other words, startups pay the fixed debt payment if the realized $\omega_{t+1}^n(j)$ is above the threshold ($\bar{\omega}_{t+1}^n$), whereas they pay all of their operating profit to the shadow bank otherwise. In the latter case, shadow banks must pay a monitoring cost to receive the payment from the defaulted loans. The cost is assumed to be a fixed share (μ_n) of the default debt payment of firm j , $\mu_n \omega_{t+1}^n(j) R_{t+1}^n q_t^n K_{t+1}^n(j)$.

Given the setup above, shadow banks' participation constraint can be defined using the expected return from loans to startups. The startups maximize their expected wealth subject to the participation constraint of shadow bank. The details of the startups' problem are elaborated in Appendix D.8.

3.6 Central Bank and Macroeconomic Policies

There are four types of macroeconomic policies in the model: conventional monetary policy using the interest rate on reserves, asset purchase policy using the central bank's balance sheet, and two macroprudential policies using leverage regulation and capital regulation in the banking sector.

First, the central bank issues reserves (re_t) to the wholesale bank and pays the interest rate (R_t) on them. The framework behind the monetary policy follows Sims et al. (2023); however, I distinguish the interest rate on reserves from the short-term rates that affect the household's

²¹For simplicity, I omit the transformation process of new capital goods into startup capital and keep the notation of the new capital stock, K^n .

inter-temporal decisions. Thus, the change in the policy rate is transmitted to households through traditional banks.

The policy rate is determined by a typical Taylor-type rule responding to inflation and output deviations from the steady state:

$$R_t = \bar{R} \left(\frac{\Pi_t}{\bar{\Pi}} \right)^{m_\pi} \left(\frac{y_t}{\bar{y}} \right)^{m_y} S_t^i \quad (47)$$

$$\log S_t^i = \rho_i \log S_{t-1}^i + u_t^i \quad (48)$$

The specification of the monetary policy rule without interest rate smoothing follows Kumhof and Wang (2021). This can be interpreted as a forward-looking and continuous attempt to adjust inflation to its steady-state level. In the baseline model, the response to the output gap is muted ($m_y = 0$), and the response to inflation (m_π) and the standard deviation of the monetary policy shock ($\sigma_t^{u^i}$) are estimated.

Second, the asset purchase policy is implemented. The central bank lends to intermediate good producers using the newly issued reserves. The policy rule has an analogous form to the Taylor-type rule:

$$\tilde{\ell}_t^{f,cb} = re_t \quad (49)$$

$$re_t = \bar{r}e \left(\frac{\Pi_t}{\bar{\Pi}} \right)^{-q_\pi} \left(\frac{\mathbb{E}_t \ell_{t+1}^{f,b}}{\bar{\ell}^{f,b}} \right)^{-q_\ell} S_t^q \quad (50)$$

$$\log(S_t^q) = \rho_q \log(S_{t-1}^q) + u_t^q \quad (51)$$

where S_t^q is the shock process to the reserve policy. The parameters q_π and q_ℓ are the response coefficients to inflation and the amount of total committed credit line loans. The framework behind the asset purchase policy in Sims et al. (2023) was originally designed to boost aggregate demand in the economy by purchasing bonds issued by households when the output gap falls below the steady state. However, I assume that the central bank supports the production sector by providing

loans with loose terms and conditions. This is modeled after the Main Street Lending Program (MSLP) by the Federal Reserve in 2020, which is one of the unconventional monetary policies during the COVID-19 pandemic period.²² In the baseline model, the response to inflation is muted ($q_\pi = 0$), and the response to committed loan deviation from its steady state is estimated.

Third, I assume that the financial authority sets the capital-asset ratio for wholesale banks. The process for the ratio is the following:

$$\mathbf{v}_t^k = (1 - \rho_k) \bar{\mathbf{v}}^k + (1 - \rho_k) \left[\chi_k \left(\frac{\tilde{\ell}_t^{f,b} + b_t^{sb}}{y_t} - \frac{\bar{\ell}^{f,b} + \bar{b}^{sb}}{\bar{y}} \right) \right] + \rho_k \mathbf{v}_{t-1}^k + u_t^{v,k} \quad (52)$$

The process is designed after the macroprudential policy rule in Angelini et al. (2014). The steady state target ratio $\bar{\mathbf{v}}^k$ is set to 10.5% following the Basel III requirements for the capital adequacy ratio of 8% and the capital conservation buffer of 2.5%.²³ χ_k determines the cyclicity of the capital regulation. In the baseline model, $\chi_k > 0$, which means that the target ratio increases when the assets of banks expand relative to output, asking for more equity and liabilities to back up their balance sheets or shrinkage of assets. Additionally, χ_k is highly relevant to the determinacy of the equilibrium, and I discuss this in Section 5.3.4.

Finally, I assume that the financial authority sets the leverage ratio for bankers. The policy rule for the leverage ratio is as follows:

$$\mathbf{v}_t^{bhc} = (1 - \rho_{bhc}) \bar{\mathbf{v}}^{bhc} + (1 - \rho_{bhc}) \left[\chi_{bhc} \left(\frac{e_t + b_t^b}{y_t} - \frac{\bar{e} + \bar{b}^b}{\bar{y}} \right) \right] + \rho_{bhc} \mathbf{v}_{t-1}^{bhc} + u_t^{v,bhc} \quad (53)$$

The macroprudential policy is implemented to test the effect of controlling indebtedness within

²²See Board of Governors of the Federal Reserve System (2025). For eligible borrowers, the loans issued under the MSLP have a five-year maturity, deferral of principal payments for two years, and deferral of interest payments for one year.

²³See Basel Committee on Banking Supervision (2020a).

the banking sector. The form of the policy rule is analogous to the capital-asset ratio process for wholesale banks. In the baseline model, the adjustment channel is muted ($\chi_{bhc} = 0$) as the parameter is not well-identified through the model estimation. However, I conduct a sensitivity analysis for the parameter in Section 5.3.3 to show the effect of the leverage ratio regulation on the model dynamics. The target steady state leverage ratio for bankers ($\bar{v}^{bhc}$) is set to 1.2 based on the discussion in Federal Reserve Bank of Richmond (2020).²⁴

Following Gerali et al. (2010), the macroprudential policy shocks ($u_t^{v,k}$ and $u_t^{v,bhc}$) are added to capture unexpected changes in the target capital-asset ratio and the target leverage ratio, respectively. The shocks can be interpreted as exogenous changes in the regulatory environment.

3.7 Market Clearing Conditions and Exogenous Shocks

All markets clear in each period: labor (h_t), bankers' bonds (b_t^b), retail deposits (d_t), equity investment (e_t^b), retail loans to intermediate good producers ($\tilde{\ell}_t^{f,b}$) and shadow banks (b_t^{sb}), reserves (re_t), capital goods (K_t^n), loans to startups (b_t^n) markets, and internal market between retail banks and the wholesale bank. The loan market for intermediate good producers and the final good market are explicitly written here.

The loan market for intermediate good producers clears as follows:

$$\ell_t^f = \tilde{\ell}_t^{f,b} + \tilde{\ell}_t^{f,cb}. \quad (54)$$

The final good market clears as follows:

$$y_t = c_t^b + c_t^h + I_t + I_t^n + \text{Costs}, \quad (55)$$

²⁴The double leverage ratio refers to the ratio of a bank holding company when its total investment in subsidiary banks over its total equity is greater than 1. Following Federal Reserve Bank of Richmond (2020), a high double leverage ratio can harm subsidiary banks when their parent companies require large bank dividends to service those obligations.

where the costs in the economy are categorized as:

$$\text{Financial distress cost} = d_t^h \Phi(s_t) + \varphi_t$$

$$\text{Adjustment cost} = G_{P,t} + G_{I,t} + G_{I,t}^n$$

$$\text{Remuneration cost} = \left(\mathcal{C}_{t-1}^{bhc,b} + \mathcal{C}_{t-1}^{sb,n} \right) \Pi_t^{-1}$$

$$\text{Macroprudential cost} = \left(\mathcal{C}_{t-1}^{bhc} + \mathcal{C}_{t-1}^b \right) \Pi_t^{-1}$$

$$\text{Monitoring cost} = m_t^{sb} \Pi_t^{-1}.$$

The exogenous aggregate shocks in the model are the following: technology shock (u_t^a), loan-investment shock (u_t^ψ), monetary policy shock (u_t^i), asset purchase policy shock (u_t^q), household financial distress shock (u_t^ϑ) and macroprudential policy shocks ($u_t^{v,k}$ and $u_t^{v,bhc}$). Additionally, there is an idiosyncratic shock to startups' return dispersion ($u_t^n(j)$).

For macroprudential policies, one-time impulse shocks are considered. The shocks are designed to capture the unexpected changes in the variables, and dissipate according to the endogenous dynamics of the variables themselves. The rest of the shocks are designed to be persistent and forward-looking, and the standard AR(1) process is adopted to capture a sustained change in the variables as in Kumhof and Wang (2021).

4 Parameterization

I calibrate the model to match key interest spreads in the U.S. economy after the COVID-19 pandemic. Based on the fundamental parameters from the DSGE literature, I estimate the rest of the parameters using U.S. data after the Global Financial Crisis. Table 1 summarizes the targeted steady-state interest rates and returns. The full parameter set and the data justification for each target are reported in Appendix F, and the details of the estimation are provided in Appendix C.

Table 1: Targeted Steady-State Interest Rates and Returns

Target	Symbol	Value	Data counterpart
Retail deposit rate	$\bar{r}^d$	2.00%	3-month CD rate, 1.68% (2024–2025) ^a
Policy rate	$\bar{R}$	3.00%	Longer-run FOMC SEP ^b , 2.4–3.0% (2019–2025) ^a
Banker’s bond spread	$\bar{R}^b - \bar{R}$	−0.40%	Recent negative corporate–FFR ^c spread ^d
Firm (C&I) loan spread	$\bar{r}^f - \bar{R}$	1.75%	C&I loan spread (Pandolfo, 2025)
Shadow-bank loan spread	$\bar{r}^{sb} - \bar{R}$	1.92%	Bank loans to private debt funds (Berrospide et al., 2025)
Shadow-bank lending spread	$\bar{R}^{fn} - \bar{R}$	6.00%	Shadow-bank loan spread (Pandolfo, 2025)
Startup return spread	$\bar{R}^n - \bar{R}$	10.00%	Adjusted market ROE, 6.8–15.3% (2016–2025) (Damodaran, 2025)

^a Retrieved from the FRED database. All figures are in annual terms.

^b The Federal Open Market Committee (FOMC) Summary of Economic Projections (SEP).

^c Federal Funds Rate (FFR).

^d Retrieved from the FRED database. See Appendix F for details.

5 Analysis of the Model

I analyze the model along two dimensions. First, I trace the impulse responses to the household financial distress shock to show how it propagates from the household’s portfolio to the real economy. Second, I estimate the model on U.S. data and quantify each structural shock’s contribution to the observed fluctuations. The two exercises deliver a consistent message: the household financial distress shock is a *funding-mix* shock that reshapes the *composition* of household savings, bank funding, and credit across borrower types, while moving the *level* of aggregate credit only modestly. By contrast, the fluctuations in credit volumes are driven mainly by demand-side shocks in the credit market.

5.1 Impulse Response Analysis

In this section, I analyze the impulse response of the economy to the financial distress shock affecting the representative household. The key finding is that a negative shock to the elasticity process variable (ϑ_t) triggers a reallocation of household portfolios away from deposits toward bankers' bonds, which subsequently propagates through the financial and production sectors and affects the real economy. The mechanism operates through several interconnected channels involving the household, banker, traditional banks, shadow banks, intermediate good producers, and startups.

Figure 6 presents the impulse responses of key macroeconomic variables to a negative one standard deviation shock to the household's financial distress cost process (S_t^ϑ).

Portfolio reallocation and funding contraction. The shock lowers the elasticity of the household's financial distress cost, making bankers' bonds relatively more attractive than deposits. The deposit share drops by 2.14 pp on impact (d^h : -3.63% , b^b : $+6.47\%$), and the additional bond funding is channeled into wholesale banks as bankers' equity investment (e^b : $+6.48\%$ at impact). Unlike standard credit or financial shocks studied in the macroprudential literature, this shock is a *funding-mix* shock rather than a funding-level shock: it rotates the bank's liabilities from deposits toward equity while contracting total assets only modestly, so bank assets and bank leverage move in opposite directions. The substitution toward bond funding still leaves traditional banks with a net contraction of their balance sheet (-0.15% at impact, -0.35% by horizon 5) because the deposit outflow dominates. The capital-asset ratio simultaneously improves and opens a gap of up to 0.37 pp above the target by horizon 5 (the target is nearly fixed given $\rho_k = 0.98$). Three forces push in the same direction at short horizons: the equity inflow, the deleveraging of total assets, and a falling risk-weighted-asset (RWA) density driven by a rising reserve share in assets and endogenously declining risk weights. The density channel reverses around horizon 13, when the asset purchase rule pulls reserves below steady state, but by then the equity-side improvement is already decaying. A formal log decomposition of the ratio is provided in Appendix G.

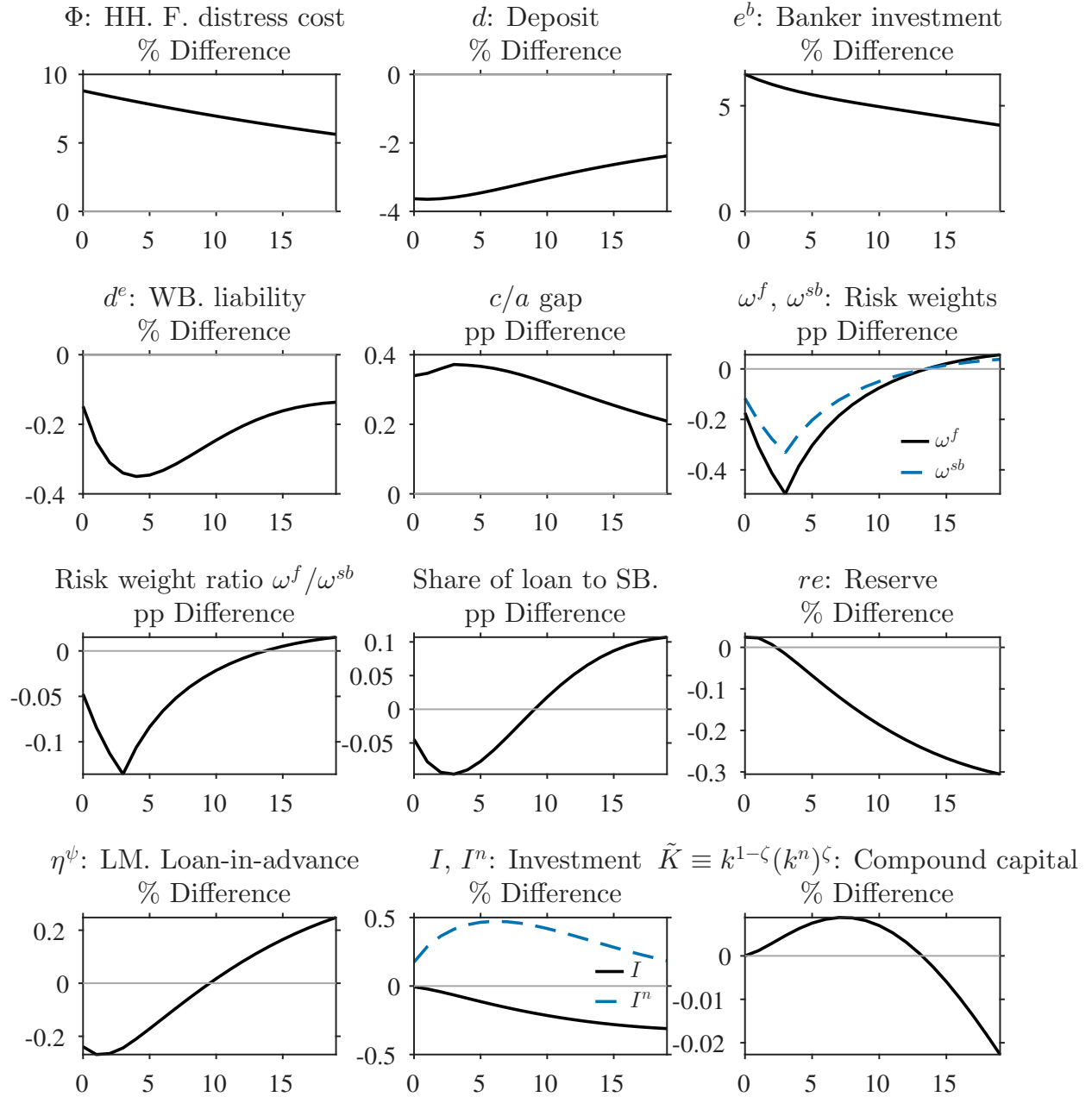


Figure 6: IRFs to a negative shock to the household financial distress shock
Notes: The figure shows the IRFs to one standard deviation negative shock to the household financial distress cost process, S_t^{ϑ} . A blue dashed line is employed to represent two series in one panel. The posterior modes of the parameters are used for the IRF analysis. A one standard deviation shock drops the deposit share by 2.14 pp at the impact of the shock. The variables are selected to show the key channels of the propagation mechanism. The full set of IRFs is provided in Appendix H.

Credit reallocation by traditional banks. The widened CA-ratio gap and the higher marginal cost of lending push banks to reprice and to reallocate the smaller pool of credit. The estimated risk-weight process implies that the risk weight on firm loans is more sensitive to output than that on shadow-bank loans ($(1 - \rho_f)\chi_f < (1 - \rho_{sb})\chi_{sb}$, both negative), so an output expansion lowers the marginal regulatory cost of firm loans by more than that of shadow-bank loans. The retail rate on shadow-bank loans therefore rises slightly more, and lending to shadow banks (b^{sb}) falls more than lending to intermediate-good firms: b^{sb} drops by 0.48% on impact, troughs at -0.98% around horizon 4, and overshoots to $+0.27\%$ by horizon 20.

The central bank's asset purchase rule shapes this reallocation through two related channels. First, since the committed credit line rises with firms' EBITDA, the counter-cyclical rule in equation (49) cuts central bank loans to firms by up to 0.31% over twenty quarters. Second, the resulting decline in $\tilde{\ell}^{f,cb}$ raises the share of bank loans in firms' total borrowing, which feeds back into a higher firm's financial distress cost (39). Firms respond by trimming investment via the loan-in-advance constraint (36); the implied marginal cost η^ψ rises by 0.25% by horizon 20. The contraction in firms' own investment is partly absorbed by an offsetting substitution: firms rent more capital from startups (K : -0.07% , K^n : $+0.18\%$ at horizon 20), which lifts startup investment ($+0.47\%$ at its peak) and net worth, and lowers the default rate by up to 0.021 pp. Bank credit to shadow banks gradually recovers as bank capacity improves, and exceeds steady state from about horizon 13 onward.

Aggregate response. The induced reallocation generates a modest but persistent expansion. Real GDP rises by 0.41% on impact and remains $+0.16\%$ above steady state at horizon 20. Aggregate consumption rises ($+0.50\%$ at impact) because bankers' consumption increases sharply with the bond inflow (c^b : $+1.97\%$), even though household consumption falls slightly (-0.05% at impact, -0.17% by horizon 20). Inflation and the policy rate peak modestly at $+0.04$ pp and $+0.08$ pp respectively around horizon 4. Two general lessons emerge. First, the credit policies in the model are far from neutral once households shift away from deposits: the asset purchase rule

together with the committed-credit and financial-distress mechanisms amplifies the reallocation across borrower types. The risk-weight channel lessens the reallocation by adjusting the marginal cost of lending, but the capital regulation channel amplifies it as the financial authority counts only the deterioration of the asset side of the bank's balance sheet. Second, the substitutability between firms' internal capital and rented capital from startups is what allows the credit reallocation to transmit to the real economy without generating large aggregate disruption.

5.2 Estimation

I estimate the model using Bayesian techniques with U.S. data from 2009:Q2 to 2025:Q4. The steady-state deviations of eight endogenous variables in the model are matched to their data counterparts: output (y), inflation (Π), policy rate (R), investment (I), deposit (d), share of deposits in total household savings (s), wholesale bank capital ($X^b + e^b$), and default rate of startups (F^n). I estimate 29 parameters, including the standard deviations of 8 shocks. The technical details of the estimation procedure, together with the prior distributions and posterior estimates, are provided in Appendix C.2.

Using the posterior estimates of the parameters, I conduct forecast error variance decomposition (FEVD) and historical shock-share decomposition of observables and the credit flow to intermediate good producers and shadow bank (ℓ^f and b^{sb}) to analyze the contribution of each structural shock. Table 2 summarizes the dominant shock for each variable on impact, while the full FEVD across horizons (Table 9) and the historical shock-decomposition summary over 2022:Q1–2025:Q4 (Table 10) are reported in Appendix C.2.4. The graphical representation of the variance decomposition for observables is presented in Figure 7.

First, the FEVD results delineate the scope of the household financial distress shock: it governs the composition of household savings and bank funding rather than the volume of credit. The shock explains all unexpected fluctuations in the share of deposits in total household savings at any horizon implied by the model structure, and it has substantial explanatory power for bank capital

and household deposits at the impact horizon (39.3% and 26.5% for each observable, respectively). By contrast, the unanticipated changes in the credit flows to intermediate good producers and the shadow bank are driven mainly by demand-side shocks originating with borrowers, namely the technology and investment shocks. This pattern is economically sensible under the financial intermediation and regulatory environment of the model: shocks that directly move the firm's investment and borrowing decisions have more explanatory power for the fluctuations in credit supply than a shock that reshapes the household's funding mix.

Table 2: Dominant Structural Shock at the Impact and One-Year Horizons

Variable	Impact ($h = 1$)		One year ($h = 4$)	
	Dominant shock	Share (%)	Dominant shock	Share (%)
Output	Technology	59.8	HH. F.D.	36.2
Policy rate	Monetary policy	55.7	Startup return	28.0
Inflation	Technology	45.7	Startup return	27.0
Deposit share	HH. F.D.	100.0	HH. F.D.	100.0
Bank capital	HH. F.D.	39.3	Technology	65.3
Investment	Investment	60.6	Startup return	27.6
Deposit	HH. F.D.	26.5	Technology	59.6
Default rate	Startup return	91.0	Startup return	28.2
Credit to bank (ℓ^f)	Investment	49.0	Investment	24.0 [†]
Loan to SB (b^{sb})	Technology	79.8	Technology	61.9

Notes: For each variable and horizon, the “Dominant shock” column reports the structural shock with the largest share of the h -step-ahead forecast error variance, evaluated at the posterior mode; “Share (%)” is that shock’s contribution. At $h = 4$, startup return shocks dominate several variables (policy rate, inflation, investment, and default rate) as their propagation through credit and real sectors accumulates, while technology shocks dominate more persistent aggregate variables (bank capital, deposit, and loan to SB), and the HH. F.D. shock becomes the dominant driver of output. The full horizon profile is reported in Table 9. [†]Closely followed by startup return (23.7%).

Second, the historical shock decomposition results show that the financial distress shock to the household explains a substantial portion of the fluctuations in the share of deposits in total household savings, bank capital, and deposits from 2022:Q1 to 2025:Q4. For instance, the shock explains 100% of the fluctuations in the share of deposits, and 21.1 % and 13.5% of the fluctuations in bank capital and deposits, respectively. However, consistent with the FEVD results, the shock

explains only a small portion of the fluctuations in credit flows to intermediate good producers and shadow banks (0.7% and 1.0%, respectively).

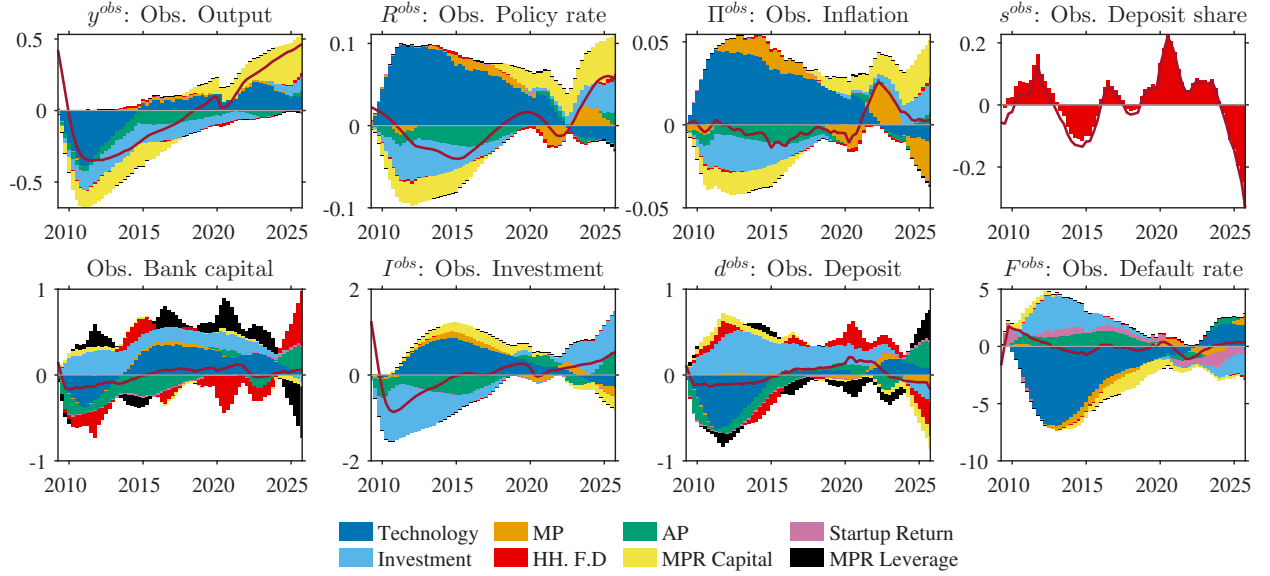


Figure 7: Variance decomposition of selected variables and shocks

Note: The red line represents the smoothed series for the selected variables, and the stacked bar represents the contribution of each shock to the fluctuations in the variables. The abbreviations are as follows: MP denotes monetary policy shock, AP denotes asset purchase policy shock, HH. F.D denotes financial distress shock to the household, MPR capital denotes macroprudential policy shock for capital regulation, and MPR Leverage denotes macroprudential policy shock for leverage ratio regulation.

Although the household financial distress shock has a limited direct impact on the volume of credit flows, the propagation analysis in the previous sections shows why the mechanism still matters: it reshapes the composition of credit across borrower types even when it moves the level little. Since the credit flow to intermediate good producers is tied to their investment decisions as specified in the loan-in-advance constraint, equation (36), it is not surprising that the unanticipated changes in the production environment drive the fluctuation in credit supply. Additionally, intermediate good producers' investment decision affects the demand for startup capital as in the production function, equation (29); this, in turn, affects startups' credit demand from the shadow bank.

5.3 Robustness Checks

This section presents robustness checks for decisions in the model setup and parameterization. First, I conducted the sensitivity analysis for the parameters in the asset purchase policy and the macroprudential policies. Second, the household and banker's welfare under the financial distress shock are analyzed with different scenarios of policy adjustment. Third, the role of leverage ratio regulation for the banker is examined. Finally, I verify the determinacy of the equilibrium for two policies activated simultaneously: the monetary policy and the macroprudential policy for capital regulation.

5.3.1 The Role of Credit Policies for Intermediate Good Producers

In this section, I conduct a sensitivity analysis for the parameters in the asset purchase policy and the macroprudential policies to show their role in the propagation mechanism of the financial distress shock. I compare the IRFs to a one standard deviation negative shock to the household financial distress cost process, S_t^ϑ , with and without the policy adjustment. Also, I amplify the policy parameters to show these effects more clearly. The results are summarized in Figure 8, 9, and 10.

First, the asset purchase policy plays a central role in the amplification of the reallocation mechanism. When the policy adjustment is muted, the reallocation of credit supply from traditional banks is dampened, and investment by intermediate good producers becomes flat, with a stable marginal cost of investment from borrowing that is lower than its steady-state level. However, capital accumulation by startups is still required to meet the increase in aggregate demand. In this case, intermediate good producers choose to borrow less from traditional banks because it is cheaper to rent capital from startups. In the case of the amplified policy parameter, the reallocation is stronger as the responses of share of loans to shadow bank become more volatile.

Second, the macroprudential policy for capital regulation is weaker, but with enough amplification, the policy strengthens the reallocation mechanism. Since the estimated persistence of

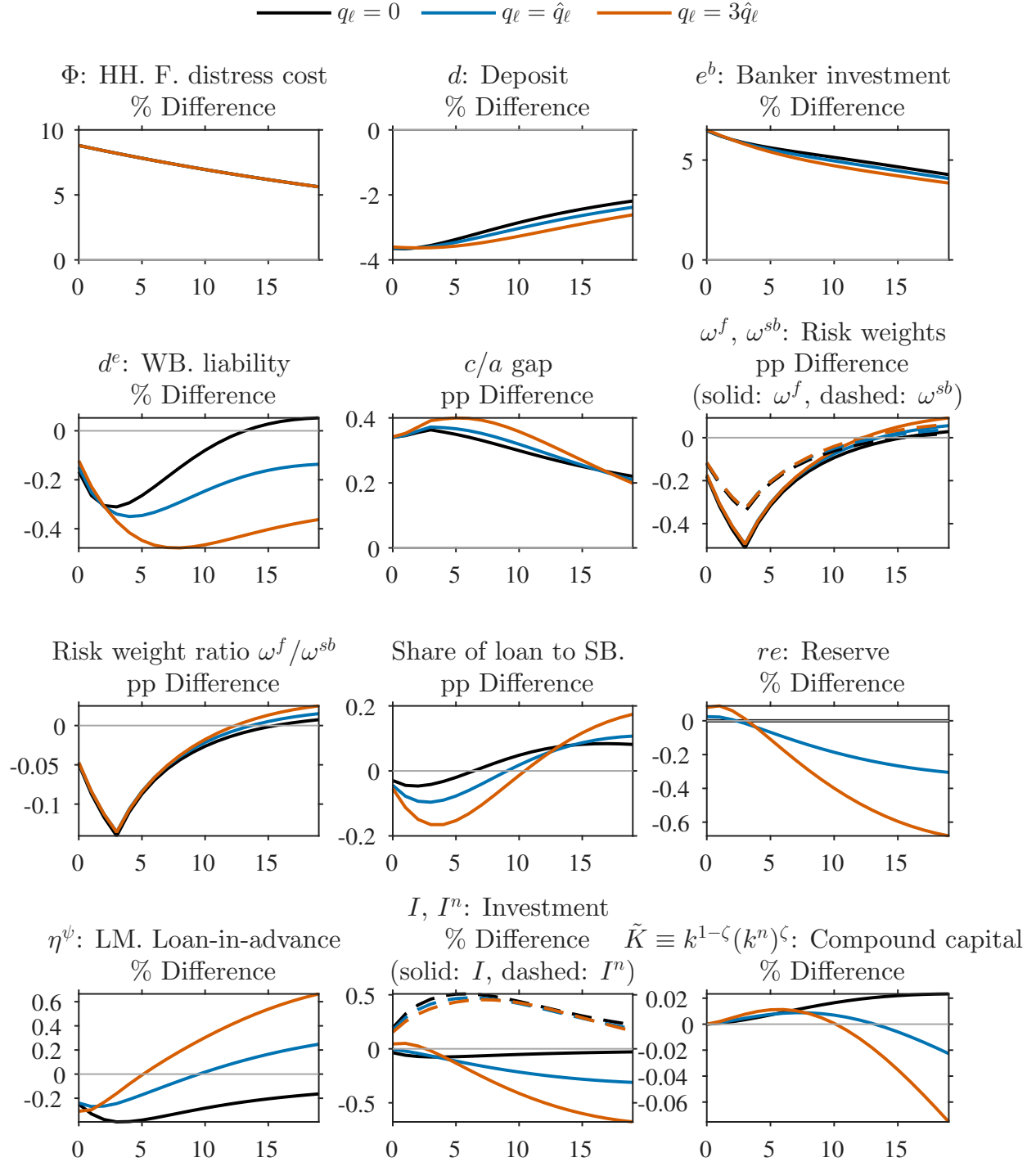


Figure 8: IRFs for asset purchase policies

Notes: The figure shows the IRFs to one standard deviation negative shock to the household financial distress cost process, S_t^ψ , with and without the asset purchase policy. The black line represents when the policy is muted. The blue line represents when the policy parameter is set to the posterior mode. The orange line represents when the policy parameter is amplified to be 3 times the posterior mode. All other parameters are set to the baseline values. The identical variables are selected to the main analysis to show the key differences in the propagation mechanism.

the policy is high ($\rho_k = 0.98$), it has a long-lasting effect on the reallocation, while the counter-cyclicality of the policy is weak even with the amplified parameter. Following equation (52), the target ratio decreases with a positive response parameter χ_k as the balance sheet-to-output ratio decreases. A larger value of χ_k widens the gap between the target and actual capital-asset ratio, which creates a stronger incentive for banks to adjust the price of loans and reallocate credit supply across different borrowers. As expected, the discrepancy between the risk weight ratios for retail loans increases with the strong counter-cyclical policy, which further accelerates the accumulation of the startups' capital consequently.

Third, a strong cyclical risk weight adjustment policy for loans to shadow banks and intermediate good producers dampens the reallocation mechanism. By enlarging the parameters, the difference between the risk weights increases, which makes the relative cost of loans to shadow banks higher than before. This caps the output growth by limiting the accumulation of the startup capital, which is the key propagation channel. An interesting point is that the dampened output shrinks the EBITDA gap and hence firms' committed credit line (34); because the asset purchase rule (49) responds counter-cyclically to that line through the negative exponent $-q_\ell$, the central bank raises its own loans to intermediate good producers. This lowers the bank-loan share in firms' total borrowing and their reliance on bank credit, which, as shown in the IRFs, further dampens the reallocation.

5.3.2 Welfare Analysis

To verify the role of the policies for the household and banker's welfare, I calculate the linear-quadratic welfare under the household financial distress shock u_{vt} . Three counterfactual regimes are compared: the regime without capital regulation adjustment ($\chi_k = 0$), the regime without risk weight adjustment ($\chi_f = \chi_{sb} = 0$), the regime without asset purchase policy adjustment ($q_\ell = 0$). The consumption-equivalent variation (CEV) is calculated for regimes deviated from the baseline, and the results are summarized in Table 3.

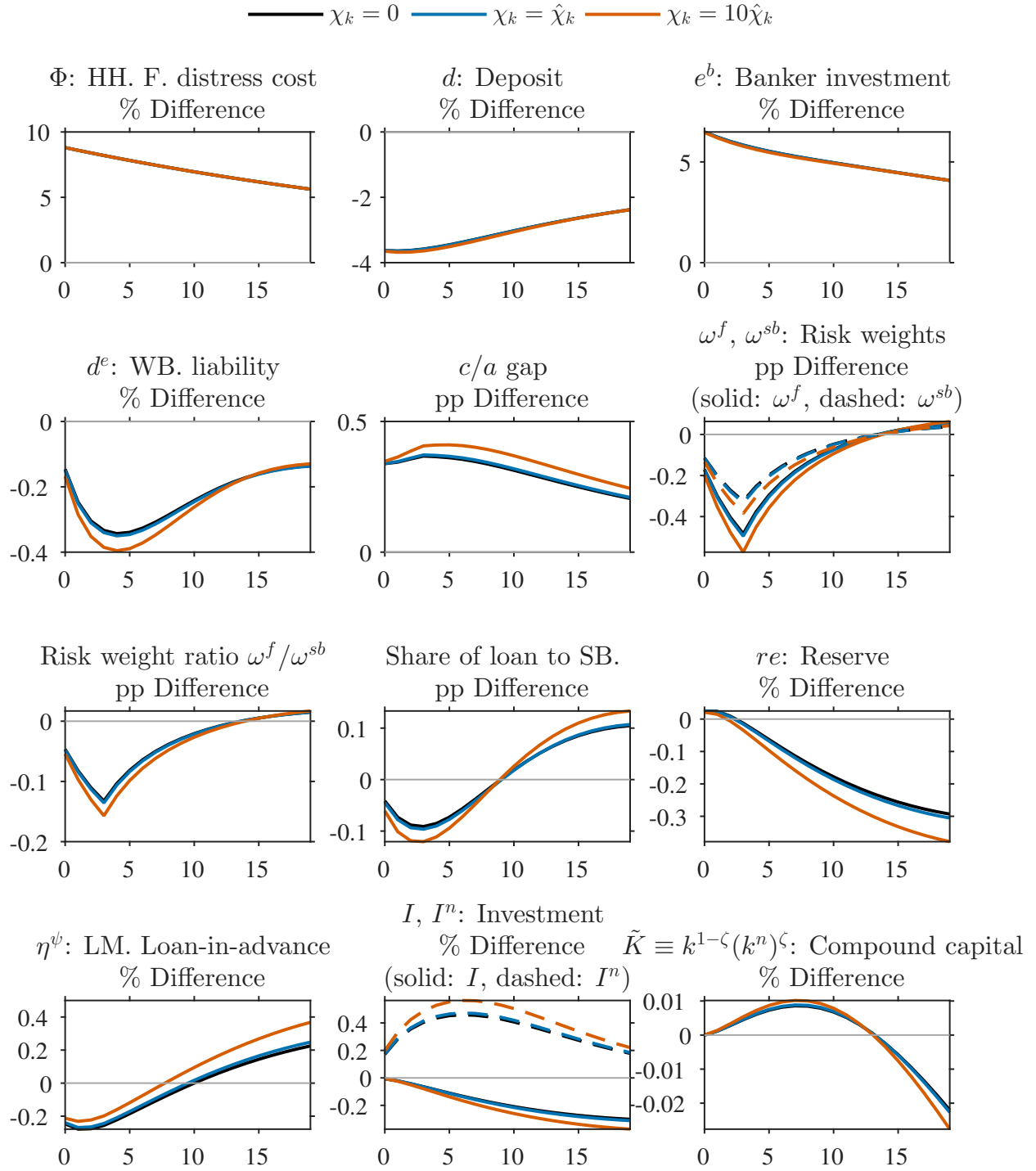


Figure 9: IRFs for macroprudential policies

Notes: The figure shows the IRFs to a one standard deviation negative shock to S_t^ϑ , with and without the macroprudential policy for capital regulation. Line conventions and selected variables follow Figure 8, except that the orange line amplifies the policy parameter to 10 times the posterior mode.

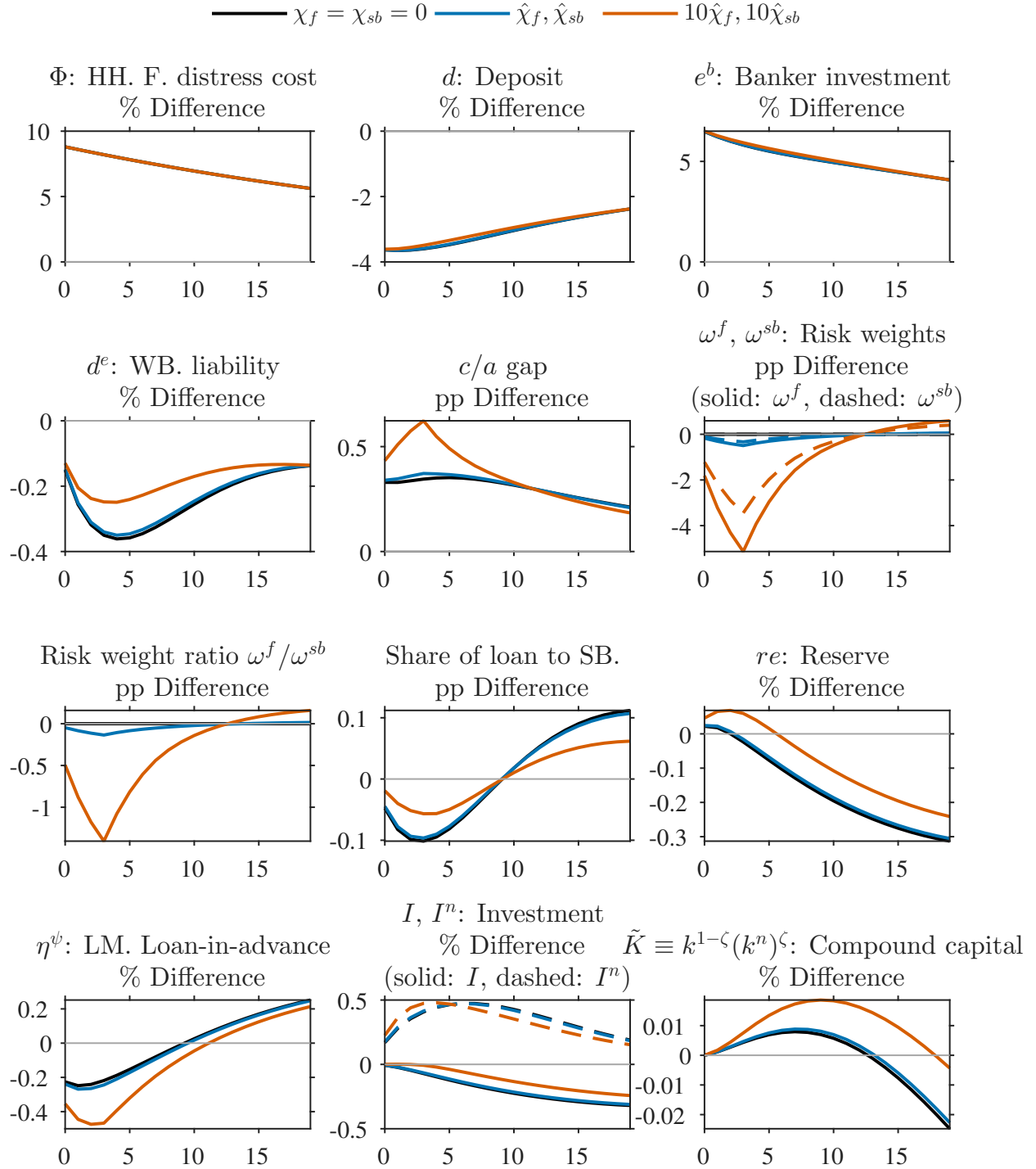


Figure 10: IRFs for risk weight adjustment policies

Notes: The figure shows the IRFs to a one standard deviation negative shock to S_t^ϑ , with and without the risk weight adjustment policy. Line conventions and selected variables follow Figure 8, with the orange line amplifying the policy parameter to 10 times the posterior mode.

The results show that capital regulation reduces both the household's and the banker's welfare, while the risk-weight adjustment and the asset purchase policy improve welfare. Quantitatively, switching off the capital-regulation adjustment ($\chi_k = 0$) raises household welfare by 0.26 bp and banker welfare by 2.84 bp of permanent consumption, whereas shutting down the risk-weight adjustment ($\chi_f = \chi_{sb} = 0$) costs the household 0.07 bp and the banker 0.86 bp, and shutting down the asset purchase adjustment ($q_\ell = 0$) is the most damaging counterfactual at -0.58 bp for the household and -7.71 bp for the banker.

The welfare effect of the capital-regulation adjustment becomes intuitive once it is read together with the IRF decomposition in Appendix G. Under the household financial-distress shock, the actual capital-asset ratio rises endogenously through the equity-inflow, deleveraging, and falling-risk-density channels. With $\chi_k > 0$ active, the policy mechanically lowers the target ratio in response, since bank assets contract relative to output, which widens the gap between the rising actual ratio and the falling target and forces banks to keep retail loan rates elevated. Switching the adjustment off ($\chi_k = 0$) leaves the target fixed and lets the gap stay narrower, lowers the borrowing cost for the production sector, and spills over to the real economy as a welfare improvement. The risk weight adjustment works in the opposite direction: by lowering risk weights when output expands, it directly reduces the banks' regulatory cost of lending, so muting it raises borrowing costs and lowers welfare. The asset purchase adjustment operates differently. As shown in the sensitivity analysis in Section 5.3.1, an active rule ($q_\ell > 0$) actually amplifies the credit reallocation across borrower types: when committed credit expands with firm EBITDA, the central bank withdraws its own loans to firms counter-cyclically, which forces firms to rely more on bank funding and propagates the shock more strongly through the reallocation toward shadow-bank lending and the substitution toward startup capital. Muting the rule ($q_\ell = 0$) keeps central bank credit support to firms in place, dampens the reallocation, and lowers the expansionary response of GDP and bankers' equity. The resulting welfare loss is the largest among the three counterfactuals because the very mechanism that drives the bankers' consumption gain (and the modest aggregate expansion) under the baseline

shock is shut down.

One caveat of analysis in this section is that the welfare comparison cannot be connected to the IRF analysis since IRFs are computed from a first-order perturbation and therefore convey only the conditional mean responses. The CEV, however, is a second-order welfare object that depends on the variance of consumption induced by the shock. Also, the value of CEV is not valid by itself since the parameters are fixed for different regimes without re-estimating them. The welfare comparison is only used to verify the role of the policies for the household and banker under the shock, and it should be read as a stand-alone exercise.

Table 3: Welfare Comparison under the Household Financial Distress Shock u_{vt}

Regime	$CEV_{u_{vt}}^h$ (%)	$CEV_{u_{vt}}^b$ (%)	ω_μ^*
$\chi_k = 0$	+0.00257	+0.02839	1.809
$\chi_f = \chi_{sb} = 0$	-0.00069	-0.00855	1.668
$q_\ell = 0$	-0.00581	-0.07707	1.592

Notes: The table reports the consumption-equivalent variation (CEV) for the household ($CEV_{u_{vt}}^h$) and the banker ($CEV_{u_{vt}}^b$) when the economy is switched from the baseline regime to each counterfactual regime, conditional on the household financial distress shock u_{vt} being the only active shock. A positive CEV indicates that the agent prefers the counterfactual to the baseline; a negative value indicates the opposite. ω_μ^* is the planner weight on the household that equalizes the two agents' CEVs; $\omega_\mu^* > 1$ in every row indicates the absence of a Pareto conflict, so the sign of either CEV suffices to read off the welfare-preferred regime.

5.3.3 The Role of Leverage Ratio Regulation

One of the macroprudential policies in the model is the regulation of the banker's leverage ratio. A positive value of χ_{bhc} in equation (53) accelerates the banker's investment to the wholesale bank. Specifically, the target ratio increases when the balance sheet of the banker expands relative to output. A larger target ratio allows the banker to hold more assets for a given level of debt. To minimize the adjustment costs, the banker can either increase equity investment or decrease borrowing.

To illustrate the role of leverage ratio regulation, I conducted a sensitivity analysis by varying

the value of χ_{bhc} . Figure 11 summarizes the IRFs to a one standard deviation negative household financial distress shock when χ_{bhc} is set to -5 , 0 , and 5 while keeping other parameters unchanged. The shock decreases household's deposit holding but increases banker's bond holding. Relative to the baseline ($\chi_{bhc} = 0$), a positive value $\chi_{bhc} = +5$ raises the target leverage ratio by $+1.25$ pp at horizon 20 and shifts banker financing toward equity: at impact b^b is 0.04 pp lower and e^b is 0.05 pp higher than the baseline, with $\chi_{bhc} = -5$ producing symmetric differences in the opposite direction.

While more investment from the banker to the wholesale bank improves the capital-asset ratio of the wholesale bank, there is more adjustment in the wholesale bank's balance sheet to minimize the adjustment cost. The share of loan to shadow bank out of total bank loan varies more as the target leverage ratio steadily increases: at horizon 20, the shadow-bank share $b^{sb}/(\tilde{\ell}^{f,b} + b^{sb})$ is 0.02 pp above the baseline when $\chi_{bhc} = +5$ and 0.03 pp below it when $\chi_{bhc} = -5$. This implies that regulation of the banker's investment in the wholesale bank can affect the reallocation of credit across different types of borrowers.

While the channel of leverage ratio regulation is muted in the baseline model, the double-leverage funding structure of holding companies is an important feature for understanding financial stability in recent years, as highlighted in Federal Reserve Bank of Richmond (2020) and Bressan (2018). Identifying and investigating the role of this policy in the model deserves further attention in future research.

5.3.4 Determinacy of the Model as a Function of Monetary and Macroprudential Policy Parameters

In this section, I analyze the determinacy of the equilibrium when both macroprudential policy and monetary policy are activated. The baseline model with other parameters fixed, allows strong pro-cyclical capital regulation for the wholesale bank only when the monetary policy is sufficiently

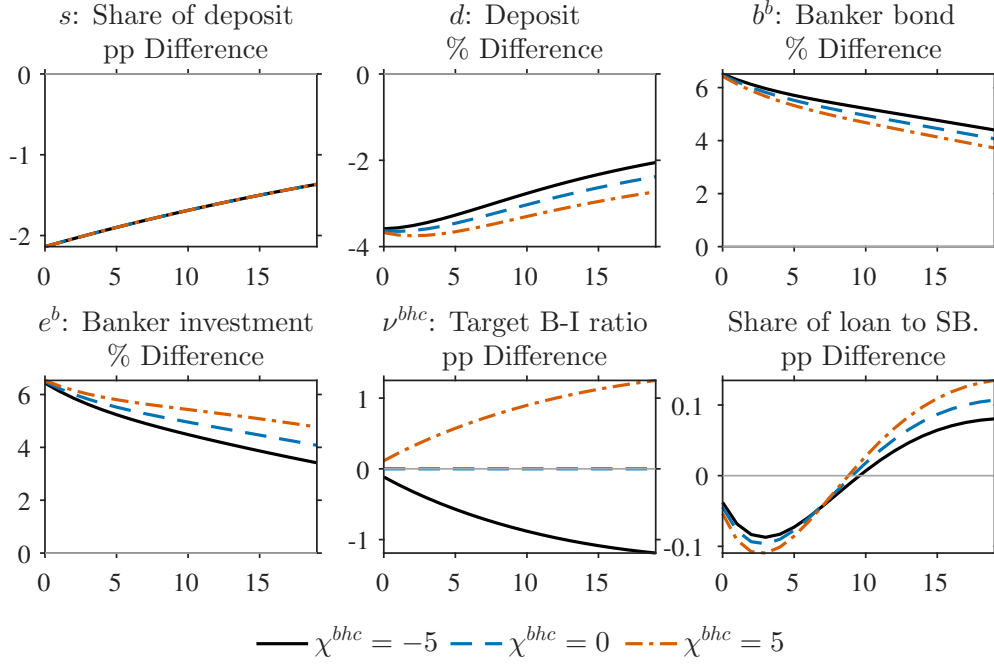


Figure 11: IRFs to Household financial distress shock with different χ_{bhc}

weak in response to the inflation deviation from its steady state.²⁵

Figure 12 shows the determinacy region of the model equilibrium according to the policy parameters m^π and χ_k , given the baseline calibration. There are two stable determinacy regions in the parameter space. The first region represents a policy combination $m^\pi > 1.4$ (i.e., the Taylor principle is satisfied) and $\chi_k > -2.5$ (i.e., the capital regulation is not strongly pro-cyclical). The second region represents a policy combination with $1 < m^\pi < 1.3$ and $\chi_k < -2.5$, which allows a strong capital regulation that is pro-cyclical to the business cycle.

In the baseline model, I employ a policy combination of strong monetary policy response to inflation deviations ($m^\pi = 2.07$) and the weak counter-cyclical capital regulation with high persistence ($\chi_k = 0.37$ and $\rho_k = 0.98$) from the estimation.²⁶ The policy mix is supported by Angelini

²⁵The pro-cyclicality of capital regulation is defined by the sign of χ_k in equation (52). A negative value of χ_k implies that the target capital-asset ratio decreases when the balance sheet-to-output ratio increases. So, the bank can expand its balance sheet with lower capital requirements when the economy is doing well.

²⁶In the baseline estimation, the prior for χ_k is assumed to follow a Gamma distribution, reflecting the belief in counter-cyclical capital regulation. The neutral prior, for instance, a Normal distribution with enough dispersion, $N(0, 2)$ does not alter the counter-cyclical nature of the estimated policy.

et al. (2014) as the counter-cyclicality of capital regulation can obtain further macroeconomic stability. I leave the investigation of a different policy mix for future research.

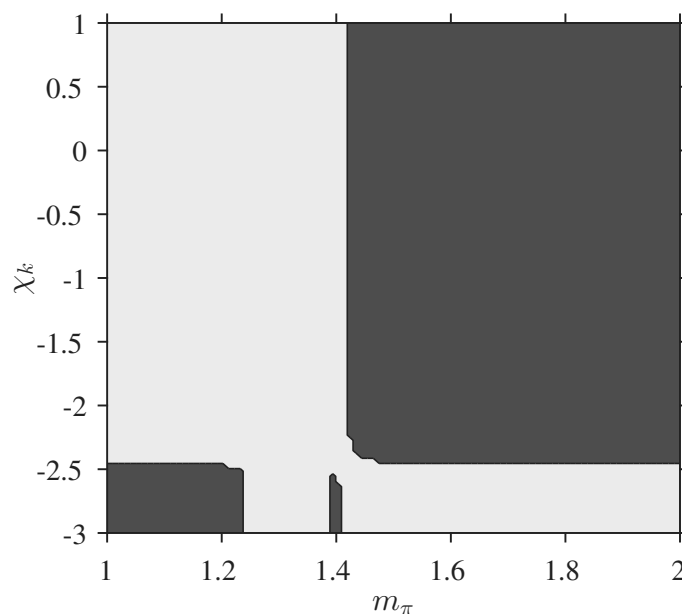


Figure 12: Determinacy as a function m^π and χ_k

Notes: The figure shows the determinacy of the equilibrium according to changes in the response coefficient to the inflation deviation in the monetary policy rule, m^π , and the response coefficient to the credit supply relative to output in the capital regulation, χ_k , given the baseline calibration. The shaded area represents the indeterminacy of the equilibrium.

6 Conclusion

This paper develops a tractable DSGE model that links the decline in U.S. deposit shares to a reallocation of bank funding and credit across borrower types, which I estimate on U.S. data from 2009:Q2 to 2025:Q4. I show that a household-driven shift out of deposits acts as a funding-mix shock: it reshapes the composition of bank funding and credit—toward equity-financed intermediation, shadow banks, and startup capital—while moving the level of aggregate credit only modestly.

This mechanism also sharpens a caution for macroprudential design. A counter-cyclical capital rule that keys off asset-to-GDP presumes that assets, leverage, and funding move together, as

in a classic credit boom, but a funding-mix shock breaks that co-movement—deposits fall while bank equity and the capital-asset ratio rise—so the rule loosens the target precisely when the balance sheet is strengthening and turns welfare-reducing. Conditioning the trigger on a leverage- or risk-weighted measure would preserve its intent without this perverse response. This result is subject to caveats that also mark the agenda ahead: the deposit share is exogenous, the firm funding structure abstracts from bond and equity issuance, and the first-order solution understates the nonlinear amplification suggested by a second-order comparison. Endogenizing the deposit share and developing a leverage-based capital rule are the natural next steps.

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Appendix A. Additional Figures and Tables

Figures in this section present additional data for balance sheets, loans, financial literacy, stock market participation, and mobile banking adoption in the U.S. The data span for balance sheets and loans is from 2000:Q1 to 2025:Q4. For other data, the span is specified in each subsection. I describe the data sources and construction for each figure in the corresponding subsection.

A.1 Liabilities of U.S. Chartered Depository Institutions

Figure 13 presents the real equities and liabilities of U.S. chartered depository institutions. The data are from L.111 of Z.1 Financial Accounts of the United States, adjusted using the Personal Consumption Expenditures (PCE) deflator (2017=100). The data were retrieved from the Federal Reserve Economic Data (FRED) database.

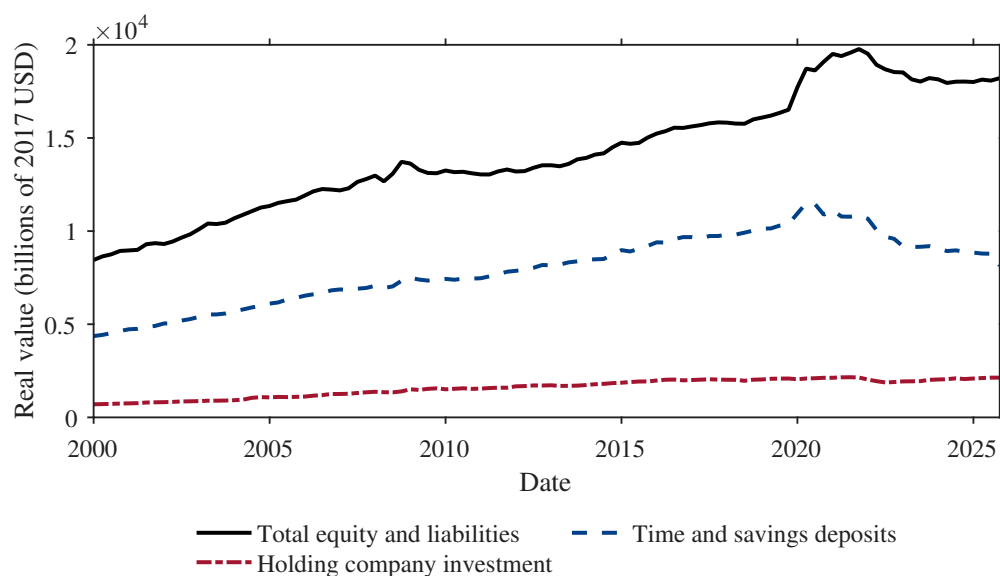


Figure 13: Liabilities of U.S. Chartered Depository Institutions

A.2 Loans from U.S. Chartered Depository Institutions

Figure 14 presents the real loans from U.S. chartered depository institutions to nonfinancial corporates and NBFIs. The data are from L.215 of Z.1 Financial Accounts of the United States,

adjusted using the Personal Consumption Expenditures (PCE) deflator (2017=100). The data were retrieved from the Federal Reserve Economic Data (FRED) database. The loan to NBFIs covers the loan to “Domestic financial sectors” (FL793168005), which includes private depository institutions, life insurance companies, government-sponsored enterprises (GSEs), issuers of asset-backed securities, and mortgage real estate investment trusts (mREITs), following Board of Governors of the Federal Reserve System (2024a).

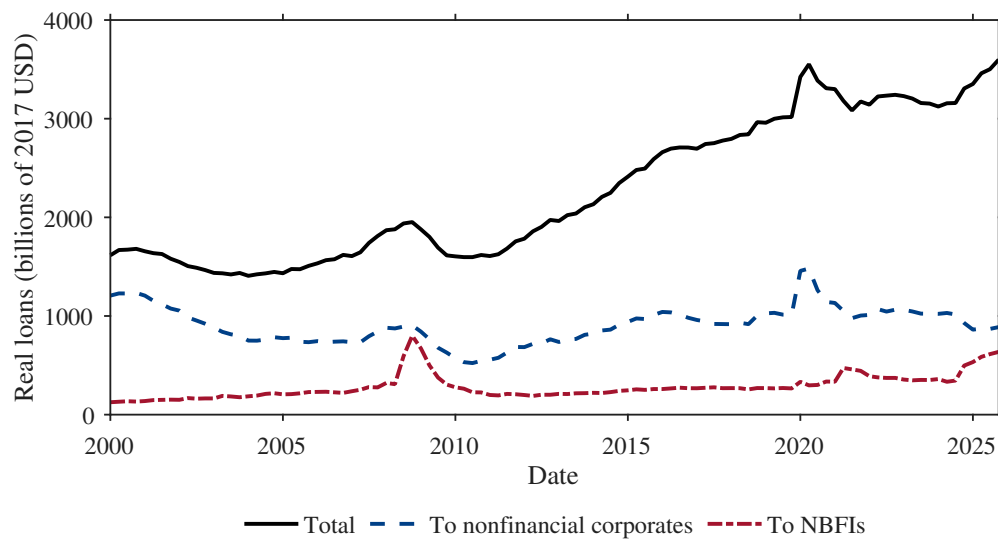


Figure 14: Loans from U.S. Chartered Depository Institutions

A.3 Loans to Nonfinancial Corporates

Figure 15 presents the real loans to nonfinancial corporates from depository institutions and NBFIs. The data are from L.103 and L.216 of Z.1 Financial Accounts of the United States, adjusted using the Personal Consumption Expenditures (PCE) deflator (2017=100). The data were retrieved from the Federal Reserve Economic Data (FRED) database.

To derive the loans from NBFIs, I select different sorts of loans in L.216. Specifically, I include loans from life insurance companies, mutual funds, ABS issuers, brokers, and other financial businesses. I excluded loans from the household sector (FL153069803). Additionally, I include loans from “Finance company loans to business” (FL103169535) in the loans from NBFIs. Fol-

lowing Board of Governors of the Federal Reserve System (2024a), “Finance companies” are defined as companies where 50 percent or more of their assets are held in loans and lease assets. U.S.-chartered depository institutions, cooperative banks, credit unions, investment banks, and industrial loan corporations are not included among them.

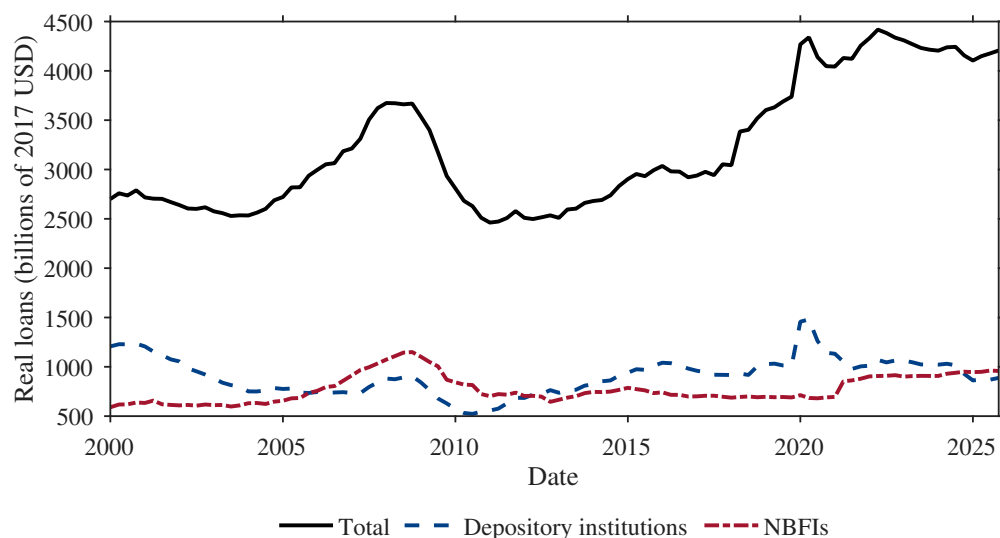


Figure 15: Loans to Nonfinancial Corporates

A.4 Financial Literacy of U.S. Adults

The Teachers Insurance and Annuity Association of America (TIAA) Personal Finance Index (P-Fin Index) measures the financial literacy of U.S. adults through 28 questions covering eight key areas: earning, consuming, saving, investing, borrowing and managing debt, insuring, comprehending risk, and go-to information sources.²⁷ The survey is conducted online with a representative sample of U.S. adults aged 18 and older, with sample sizes increasing from approximately 1,000 respondents before 2021 to around 3,000 thereafter. TIAA weights the responses to ensure representativeness of the U.S. adult population.

The data show little variation in financial literacy scores over the nine-year period, with the percentage of correct responses ranging between 48% and 52%. The average score across all years

²⁷Yakoboski et al. (2025).

Table 4: Financial Literacy of U.S. Adults: P-Fin Index Results

Year	Percentage Correct
2017	49%
2018	50%
2019	51%
2020	52%
2021	50%
2022	50%
2023	48%
2024	48%
2025	49%
Average	49.7%

Source: TIAA Institute Personal Finance Index (P-Fin Index), various years.

Note: Sample sizes are approximately 1,000 respondents (2017-2020) and 3,000 respondents (2021-2025). The P-Fin Index measures financial literacy through 28 questions covering eight key areas of personal finance.

is 49.7%, indicating that only half of questions have been answered correctly each year. Notably, there is no discernible upward trend that would support the hypothesis of improving financial sophistication among households during the period of interest.

A.5 Stock Market Participation Rate of U.S. Adults

Gallup (2025) reports the annual stock market participation rate of U.S. adults from 1998 to 2025. The survey is conducted with U.S. adults aged 18 and older. During the telephone interview, respondents answer the following question: “Do you, personally, or jointly with a spouse, have any money invested in the stock market right now—either in an individual stock, a stock mutual fund, or in a self-directed 401(k) or IRA?” Figure 16 presents the stock market participation rate of U.S. adults over the period. The data show an upward trend in stock market participation after the COVID-19 pandemic up to 62%, and the rate reached its highest level of 63% in 2004 before the Global Financial Crisis.

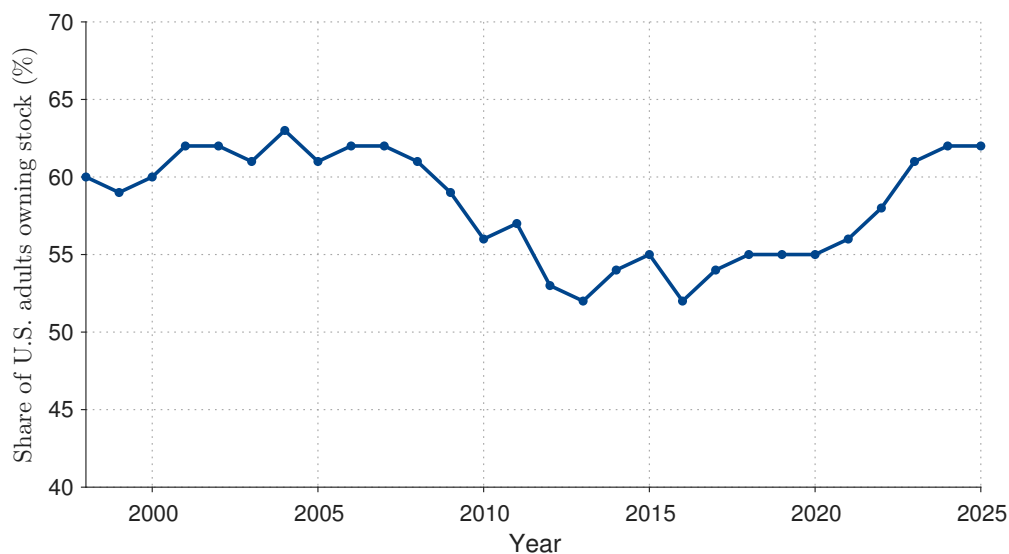


Figure 16: Stock Market Participation Rate of U.S. Adults

A.6 Number of Mobile Banking Customers

Figure 17 presents the number of mobile banking customers of JPMorgan Chase and Bank of America from 2013 to 2025. The data were retrieved from *Statista* and banks' annual reports.²⁸ Here, active mobile banking customers are defined similarly for both banks. JPMorgan Chase (2025b) defines active mobile customers as those who have logged in to any mobile platform within the last 90 days. Bank of America (2025a) defines active mobile banking users as active users over the past 90 days.

The data show a steady increase in the number of mobile banking customers for both banks over the period. The number of mobile banking customers for JPMorgan Chase increased from approximately 15.6 million in 2013 to 61.7 million in 2025. Similarly, Bank of America saw an increase from around 14.4 million in 2013 to 41.4 million in 2025. This trend reflects the growing adoption of mobile banking services among consumers in the U.S.

²⁸See JPMorgan Chase (2025a) and Bank of America (2025b) for the data from *Statista*. Observations for the two most recent years are obtained from the banks' annual reports published in 2025 and 2026: JPMorgan Chase (2025b), JPMorgan Chase (2026), Bank of America (2025a), Bank of America (2026).

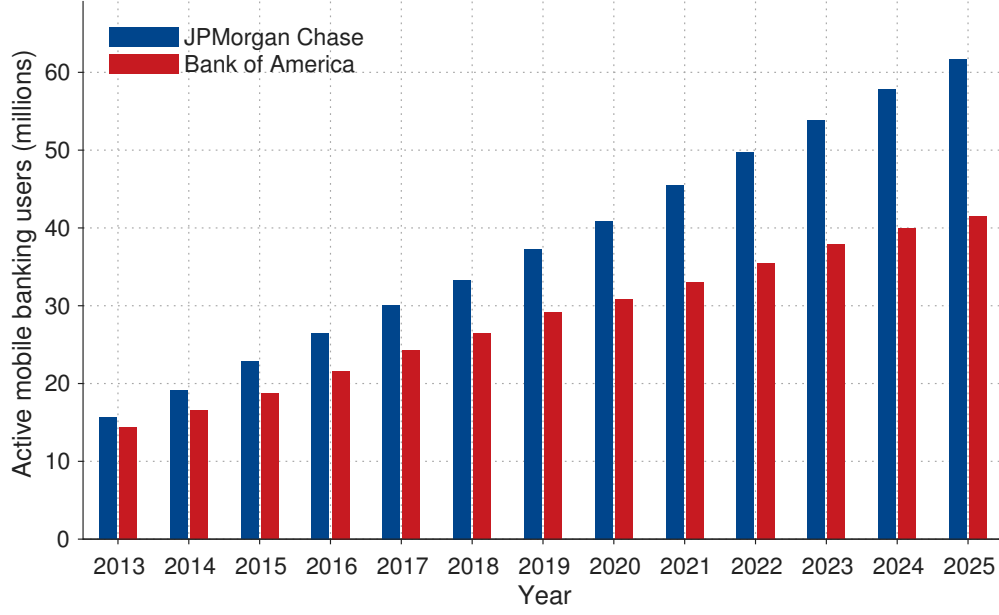


Figure 17: Number of Mobile Banking Customers

Appendix B. Proofs and Derivations

In this section, I provide the proofs and derivations for the steady state share of deposits and the capital weight of equity investment in the bank's capital regulation. The variable with bar denotes the steady state value of the variable.

B.1 Proof of Steady State Share of Deposits

The first-order conditions of the household's optimization problem with respect to the total savings d_t^h and the share of deposits s_t in steady state are as follows:

$$\begin{aligned}
 d_t^h : \quad & \beta_h \bar{\Pi} \left(\bar{R}^b (1 - \bar{s}) + \bar{r}^d \bar{s} \right) = 1 + \Phi(\bar{s}) \\
 s_t : \quad & \beta_h \bar{\Pi} \left(\bar{R}^b - \bar{r}^d \right) = -\Phi'(\bar{s}).
 \end{aligned}$$

Using $\beta^h \bar{r}^d = 1$ from the steady state of the household's inter-temporal Euler equation, the problem simplifies to:

$$\begin{aligned}\beta_h \bar{\Pi} \bar{R}^b (1 - \bar{s}) + \bar{\Pi} \bar{s} &= 1 + \Phi(\bar{s}) \\ \beta_h \bar{\Pi} \bar{R}^b - \bar{\Pi} &= -\Phi'(\bar{s}).\end{aligned}$$

I assume that there is no trend in inflation, or $\bar{\Pi} = 1$. Then, the two equations above become:

$$\begin{aligned}\beta_h \bar{R}^b (1 - \bar{s}) + \bar{s} &= 1 + \Phi(\bar{s}) \\ \beta_h \bar{R}^b - 1 &= -\Phi'(\bar{s}).\end{aligned}$$

Combining the two equations above, I obtain:

$$(1 - \Phi'(\bar{s}))(1 - \bar{s}) + \bar{s} = 1 + \Phi(\bar{s})$$

Rearranging the equation gives:

$$1 - \Phi'(\bar{s}) - \bar{s} + \Phi'(\bar{s})\bar{s} + \bar{s} = 1 + \Phi(\bar{s})$$

which simplifies to:

$$\Phi'(\bar{s})(1 - \bar{s}) + \Phi(\bar{s}) = 0.$$

or equivalently,

$$\bar{s} = F(\bar{s}) \equiv \frac{\Phi(\bar{s})}{\Phi'(\bar{s})} + 1.$$

When the function Φ is defined as

$$\Phi(s_t) = \chi_1 \left(\frac{1 - s_t}{s_t} \right)^{\vartheta_t},$$

$F(s_t)$ is calculated as follows:

$$F(s_t) = \frac{\chi_1 \left(\frac{1-s_t}{s_t} \right)^{\vartheta_t}}{-\chi_1 \vartheta_t \left(\frac{1-s_t}{s_t} \right)^{\vartheta_t-1} \left(\frac{1}{s_t^2} \right)} + 1 = -\frac{s_t(1-s_t)}{\vartheta_t} + 1.$$

Finally, I have

$$\bar{s} = -\frac{\bar{s}(1-\bar{s})}{\bar{\vartheta}} + 1 \implies \bar{s} = \bar{\vartheta}.$$

B.2 Derivation of the Capital Weight of Equity Investment

Basel III requires banks to maintain a minimum of 6% of tier 1 capital and 8% of total capital against their risk-weighted assets. Additionally, the capital conservation buffer requires banks to hold an additional 2.5% of risk-weighted assets for all banks. Given these requirements, the target capital-asset ratio v^k in steady state should satisfy the following two conditions:

$$\begin{aligned} v^{k1} &= \frac{\bar{X}^b}{\bar{r}e + \bar{\omega}^f \bar{\ell}^{f,b} + \bar{\omega}^{sb} \bar{b}^{sb}} = 6\% \\ v^{k2} &= \frac{\bar{X}^b + \rho_e \bar{e}^b}{\bar{r}e + \bar{\omega}^f \bar{\ell}^{f,b} + \bar{\omega}^{sb} \bar{b}^{sb}} = 8\% + 2.5\% \end{aligned}$$

The ratio between the two conditions is as follows:

$$\frac{v^{k1}}{v^{k2}} = \frac{\bar{X}^b}{\bar{X}^b + \rho_e \bar{e}^b} = \frac{6\%}{10.5\%},$$

and rearranging the equation with respect to ρ_e gives

$$\rho_e = 0.75 \times \frac{\bar{X}^b}{\bar{e}^b}.$$

The value of ρ_e is endogenously determined by the steady state equilibrium of the model.

B.3 Derivation of the Representative Household's Problem

In this section, I derive the representative household's problem by aggregating the optimization problems of household member $i \in [0, 1]$.

A member of the representative household, $i \in [0, 1]$ maximizes the expected lifetime utility:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_h^t \left[\frac{1}{1 - \frac{1}{\gamma_h}} \left\{ c_t^h(i) \right\}^{1 - \frac{1}{\gamma_h}} - \frac{\chi_h}{1 + \frac{1}{\zeta_h}} \left\{ h_t^h(i) \right\}^{1 + \frac{1}{\zeta_h}} \right]$$

subject to the budget constraint:

$$\begin{aligned} R_{t-1}^b P_{t-1} b_{t-1}^b(i) + r_{t-1}^d P_{t-1} d_{r,t-1}(i) + P_t w_t h_t^h(i) + P_t Z_t^{total} \\ \geq P_t c_t^h(i) + P_t b_t^b(i) + P_t d_{r,t}(i) + P_t b_t^b(i) \phi(i) \end{aligned}$$

where the transfer $Z_t^{total} = D_t^f + D_t^k + (1 - \iota) \chi_n \mathcal{W}_t^n + T_t^{cb} - X_0^{bhc}$, is identical across members.

Assume that a member $i \leq s_t$ saves through deposits, while a member $s_t < i \leq 1$ saves through bank bonds. Additionally, assume that the amount of saving is the same for all members, i.e. $b_t^b(i) = d_{r,t}(i) = d_t^h$. Then, the aggregated budget constraint can be rewritten as follows:

$$\begin{aligned} R_{t-1}^b P_{t-1} \int_{s_t}^1 d_{t-1}^h di + r_{t-1}^d P_{t-1} \int_0^{s_t} d_{t-1}^h di + P_t w_t \int_0^1 h_t^h(i) di + P_t Z_t^{total} \\ \geq P_t \int_0^1 c_t^h(i) di + P_t \int_{s_t}^1 d_t^h di + P_t \int_0^{s_t} d_t^h di + P_t \int_{s_t}^1 d_t^h \phi(i) di \end{aligned}$$

Rearranging the equation above gives:

$$\begin{aligned} R_{t-1}^b P_{t-1} (1 - s_t) d_{t-1}^h + r_{t-1}^d P_{t-1} s_t d_{t-1}^h + P_t w_t h_t^h + P_t Z_t^{total} \\ \geq P_t c_t^h + P_t (1 - s_t) d_t^h + P_t s_t d_t^h + P_t d_t^h \int_{s_t}^1 \phi(i) di \end{aligned}$$

using the symmetry of the household members, where $c_t^h(i) = c_t^h(j) = c_t^h$ and $h_t^h(i) = h_t^h(j) = h_t^h$ for all $i, j \in [0, 1]$.

Finally, I define the aggregate household financial distress cost, $\Phi(s_t)$, as follows:

$$\Phi(s_t) \equiv \int_{s_t}^1 \phi(i) di,$$

From the results above, one can derive the representative household's lifetime utility (1) and its budget constraint (2).

Appendix C. Estimation of Parameters

C.1 Parameterization of Risk Weights Parameters

In this section, I elaborate on the parameterization of the risk weights process parameters, χ_f , χ_{sb} , ρ_f and ρ_{sb} in equation (16). First, the parameters in the risk weight process for firm loans are estimated following Angelini et al. (2010). The regression equation is as follows:

$$\hat{\omega}_t^f = \rho_f \hat{\omega}_{t-1}^f + (1 - \rho_f) \chi_{f,ols} (\log(\hat{y}_t) - \log(\hat{y}_{t-4})) + \varepsilon_t^f,$$

where $\hat{\omega}_t^f$ is the percent deviation of the risk weight on firm loans from its steady state, and $\hat{y}_t$ is the percent deviation of real GDP from its steady state. For the proxy of the risk weight on firm loans, I use the delinquency rate on business loans data (DRBLACBS) and employ the real GDP data (GDPC1) from 1987:Q1 to 2025:Q4. The data are retrieved from FRED.

The cyclical component of the data is extracted using the same filter as in the estimation of the model parameters, which is provided by Kamber et al. (2025). Then, I run the OLS with Newey-West HAC errors on the post-Global Financial Crisis data from 2009Q3 to 2025Q4, which is a subset of the data most relevant for the model estimation. The estimates of $\chi_{f,ols}$ and ρ_f are -2.82

and 0.87 respectively. Then, the parameter of the DSGE model, χ_f , is recovered as follows:

$$\begin{aligned} \left(\frac{\omega_t^f - \bar{\omega}^f}{\bar{\omega}^f} \right) &= \rho_f \left(\frac{\omega_{t-1}^f - \bar{\omega}^f}{\bar{\omega}^f} \right) + (1 - \rho_f) \chi_f^{ols}(\dots) + \varepsilon_t^f, \\ \iff \omega_t^f - \bar{\omega}^f &= \rho_f (\omega_{t-1}^f - \bar{\omega}^f) + (1 - \rho_f) \chi_{f,ols} \bar{\omega}^f (\dots) + \bar{\omega}^f \varepsilon_t^f, \\ \implies \chi_f &= \chi_{f,ols} \bar{\omega}^f. \end{aligned}$$

I set the value of $\bar{\omega}^f = \bar{\omega}^{sb} = 1.2$ parsimoniously.²⁹ The estimated strength of response to the output gap does not change with the level of the risk weight as χ_f increases proportionally to $\bar{\omega}^f$.

For the parameters in the risk weight process of shadow bank loans, I calibrate χ_{sb} to be $\chi_f/1.5$ and estimate the persistence parameter ρ_{sb} in the model estimation. As Krainer et al. (2024) point out, the NBFIs have a better and more homogenous credit quality than the general borrowers from banks, which motivates the smaller value of χ_{sb} compared to χ_f . Also, the estimates of ρ_{sb} is 0.87, and the parameter is well-identified through the model estimation. However, as stressed by Vines and Wills (2018), a separate estimation of χ_{sb} and ρ_{sb} is necessary to improve the reliability of the DSGE model analysis. Unfortunately, accessible data for the proxy of the risk weight on shadow bank loans are rare, so I leave it for future research.

C.2 Bayesian Estimation of the Model

The parameters are estimated using Bayesian techniques. In this section, I summarize the data treatment, prior distributions of parameters, and estimation results.

C.2.1 Data Treatment

I employ the U.S. data of real GDP per capita, inflation, policy rate, real household deposits per capita, equipment investment, holding companies' equities and debt, U.S. chartered depository

²⁹I presume a fictitious asset without additional risk with $\omega = 1$ as the benchmark, and add a risk premium of 20% for loans in the model referring to the risk premium of AAA corporate bonds in Basel Committee on Banking Supervision (2020b).

institutions' equities, investment from holding companies, and charge-off rates on business loans from 2009:Q2 to 2025:Q4. The data are retrieved from FRED, and details are summarized in Table 5. The data treatment is as follows. First, level data are transformed into real per capita terms using inflation and population data. If the data are not seasonally adjusted, I seasonally adjust them using the X-13ARIMA-SEATS method. Then I log-transform the data and detrend them using the filter provided by Kamber et al. (2025).

Based on the Beveridge-Nelson filter, Kamber et al. (2025) provide a method to filter out the trend component from the data with large shocks such as the COVID-19 Pandemic. Since the model in this paper does not have a specific trend component, I use the cyclical component of the data for estimation, and match them to the corresponding variables' log of deviation from the steady state in the model. While volatile, the cyclical component of the data captures the dynamics before and after the COVID-19 pandemic, which is considered to be an inflection point for the change in the flow of funds in the U.S. economy.

Second, the 3-month Treasury bill rates and shadow rates are transformed into quarterly gross rates. Similarly, I detrend the log of the gross rates using the same filter. Third, I generate an empirical counterpart of the share of deposits using household deposits and holding companies' debt and equities. Aligning with other observables, I take the log of the shares and detrend them.³⁰ Additionally, the steady state value of the share of deposits is calculated as the average of the empirical counterpart over the period.

Finally, the charge-off rate on business loans is used to match the default rate on shadow bank loans, F_t^n , in the model. Because of the availability of data, the charge-off rates are used as a proxy for F_t^n based on the assumption that the cyclical component of the charge-off rates on business loans comoves with the cyclical component of the default rate of NBFI loans. As with other variables, I detrend the log of the charge-off rates using the identical filter and demean the cyclical

³⁰To obtain the empirical counterpart of the deposit share in the model, I match bankers' bonds to the debt and equity of holding companies. Following Board of Governors of the Federal Reserve System (2024b), holding companies are "Parent-only bank holding companies, savings and loan holding companies and security holding companies that file Federal Reserve Board form FR Y-9LP, FR Y-9SP, or FR 2320".

component. Figure 18 plots the observed variables.

C.2.2 Prior Distributions of Parameters

The estimated parameters and their prior distributions are summarized in Table 6. There are a few remarks regarding the prior distributions. First, the priors for persistence parameters and the standard deviations of shocks follow Gerali et al. (2010). There are two exceptions. First, the prior distribution for ρ_{sb} is set to be more narrowly concentrated around 0.80 with a standard deviation of 0.05. Also, the prior distribution for ρ_k is set to be more persistent around 0.85 with a standard deviation of 0.05. Without hurting the parsimony of the estimation, narrower distributions are employed for the stability of the estimation results.

Second, I set the mean of prior distributions of parameters following literature and allow a relatively wide dispersion to the distribution following Gerali et al. (2010). The prior distribution for coefficients of adjustment costs, κ_{bhc} , κ_{Kb} and κ_f are explored in a parsimonious way as there is limited guidance in the literature. As noted in Gerali et al. (2010), these type of parameters are hard to be observed in the data, and one way to circumvent this issue is to set the prior distribution with a wide dispersion.

The prior distribution for policy parameters, m_π , q_ℓ , χ_{bhc} and χ_k are set to fulfill theories and match the observed data. m_π and q_ℓ are assumed to be positive and represent central bank's capability to stabilize inflation and credit supply, respectively. The prior distribution of χ_k is set to be a Gamma distribution to capture the counter-cyclical of capital regulation, as Angelini et al. (2014) show the effectiveness of $\chi_k > 0$ in stabilizing the economy in response to technology and financial shocks.

For other parameters, the rationale for setting the prior distribution is as follows. First, The prior distribution of the dividend of shadow bank, v_{sb} is centered around 0.10 referring to the value, 11.4% in Brunnermeier and Koby (2018).

Second, the prior distribution of the household consumption-to-output ratio is centered around

0.7 with a dispersion of 0.05. As in Sims et al. (2023), when there are two types of households in the model, calibrating the ratio of one type of household's consumption to output is a useful way to pin down the steady state of the model. Instead of fixing the ratio parsimoniously as in Sims et al. (2023), I estimate the ratio with a wide dispersion to allow for more flexibility in the estimation. The mean of the prior distribution is obtained from the average of the ratio of U.S. PCE to GDP from 2009:Q2 to 2025:Q4, which is around 0.677. Given the posterior mode of the ratio, the total consumption-to-output ratio is 0.811 in steady state, which is in the acceptable range compared to the data.

Third, the steady state reserve-to-output ratio is estimated with an informative prior distribution. The mean of the distribution is set to 0.1, the value calibrated in Sims et al. (2023), and the standard deviation is set to 0.01, to allow for limited variation around the calibrated value. Because of the stylized assumptions, the model does not allow a large variation in the reserve-to-output ratio, which makes it difficult to target the reserve-to-output ratio using the balance sheet data of Federal Reserve.

Fourth, the prior distribution of the habit formation parameter, h_c is set to be Beta distribution with a mean of 0.7 and a standard deviation of 0.1, which is in line with Smets and Wouters (2007).

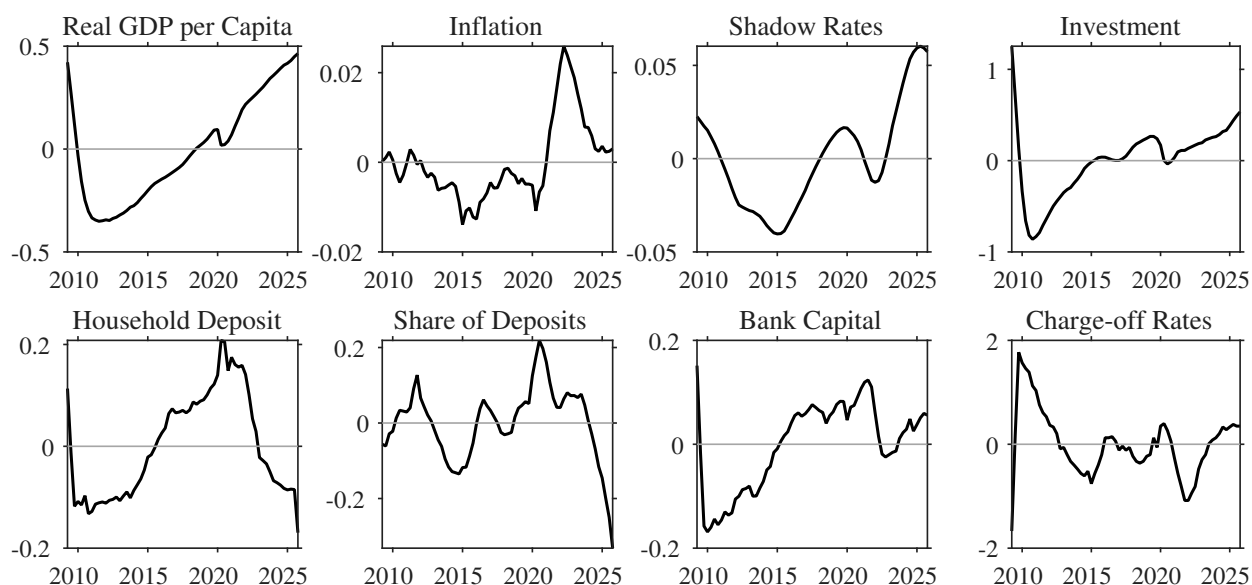


Figure 18: U.S. Observable Variables

Table 5: Data Sources and Descriptions

Variable Name	Unit	FRED Code
Population	Thousands	POPTHM
Personal consumption expenditures (implicit price deflator)	Index 2017=100	DPCERD3Q086SBEA
Real GDP per capita	Dollars	A939RX0Q048SBEA
3-Month Treasury Bill Secondary Market Rate	Percent	TB3MS
Shadow rates (Wu and Xia (2016))	Percent	—
Real Gross Private Domestic Investment: Fixed Investment: Nonresidential: Equipment	Billions of Dollars	Y033RX1Q020SBEA
Households; Other Deposits Including Time and Savings Deposits	Millions of Dollars	BOGZ1FL193030205Q
Holding Companies; Equity and Investment Fund Shares Excluding Mutual Fund Shares and Money Market Fund Shares	Millions of Dollars	BOGZ1LM733181105Q
Holding Companies; Debt Securities	Millions of Dollars	HCDSL
U.S.-Chartered Depository Institutions; Equity and Investment Fund Shares Excluding Mutual Fund Shares and Money Market Fund Shares	Millions of Dollars	BOGZ1FL763181005Q
U.S.-Chartered Depository Institutions; Other Investment by Holding Company Parent	Million of Dollars	BOGZ1FL763194705Q
Charge-Off Rate on Business Loans, All Commercial Banks	Percent	CORBLACBS

All data are quarterly from 2009:Q2 to 2025:Q4.

Source: Shadow rates are retrieved from Federal Reserve Bank of Atlanta. Others are from Federal Reserve Economic Data (FRED), Federal Reserve Bank of St. Louis.

C.2.3 Details of Estimation Process and Results

The technical details of the estimation process are the following. First, the observation equations are specified following Pfeifer (2014). The observation equations are the steady state deviation of log of corresponding variables as follows:

$$\begin{aligned}
y_t^{obs} &= \log(y_t) - \log(\bar{y}) \\
\Pi_t^{obs} &= \log(\Pi_t) - \log(\bar{\Pi}) \\
R_t^{obs} &= \log(R_t) - \log(\bar{R}) \\
I_t^{obs} &= \log(I_t) - \log(\bar{I}) \\
d_t^{obs} &= \log(d_t) - \log(\bar{d}) \\
s_t^{obs} &= \log(s_t) - \log(\bar{s}) \\
\text{bank capital}_t^{obs} &= \log(X_t^b + e_t^b) - \log(\bar{X}^b + \bar{e}^b) \\
F_t^{n,obs} &= \log(F_t^n) - \log(\bar{F}^n)
\end{aligned}$$

The estimation is conducted using the Bayesian approach with the Metropolis-Hastings algorithm. Five blocks of 1,000,000 posterior draws are obtained after a 50% burn-in. The scale parameter of the proposal distribution's covariance matrix is adjusted to achieve an acceptance rate of approximately 20~30%.³¹

To assess the convergence of estimation, I rely on the Monte Carlo Markov Chain (MCMC) univariate and multivariate convergence diagnostics (Brooks and Gelman (1998)). The convergence diagnostic compares the 80% interval of pooled and within-chain draws with respect to the first, second, and third central moments. The univariate convergence diagnostic assesses the convergence of each parameter separately, while the multivariate convergence diagnostic evaluates the posterior likelihood function. Figure 21 shows the convergence is satisfactory for the mean, vari-

³¹The acceptance ratios for the 5 chains are 24.4%, 24.4%, 24.2%, 24.3%, 24.3%.

ance, and skewness of the log posterior likelihood function. Details of the convergence diagnostic are provided in the figure notes.

Additionally, the potential scale reduction factor (PSRF) from Brooks and Gelman (1998) is checked for all individual parameters, and the values are below 1.01, which indicates that the estimation has converged satisfactorily. Table 7 reports the PSRF values for all parameters along with their posterior central tendencies and dispersion.

Three different methods are employed to assess the identification of parameters. First, the difference between the prior and posterior mean are checked following Gerali et al. (2010). The results are summarized in the second column in Table 8. The significant difference between the prior and posterior mean for most parameters indicates that the data are informative about the parameters. Second, the Bayesian learning rate, which is the ratio of the posterior to the prior standard deviation, is calculated for each parameter and summarized in the third column of Table 8. Referring to An and Schorfheide (2007), the values below 1 indicate that the data are informative about the parameters. Most parameters have a learning rate below 1; however, a few parameters are near or above 1. Third, I assess local identification using the rank-based test of Iskrev (2010), as implemented in Dynare's `identification` command. The Jacobian of the first two theoretical moments with respect to the estimated parameters has full column rank at the posterior mode, indicating that all 29 estimated parameters are locally identified from the model's moments. Overall, the results from these three methods suggest that the parameters are well-identified in the estimation.

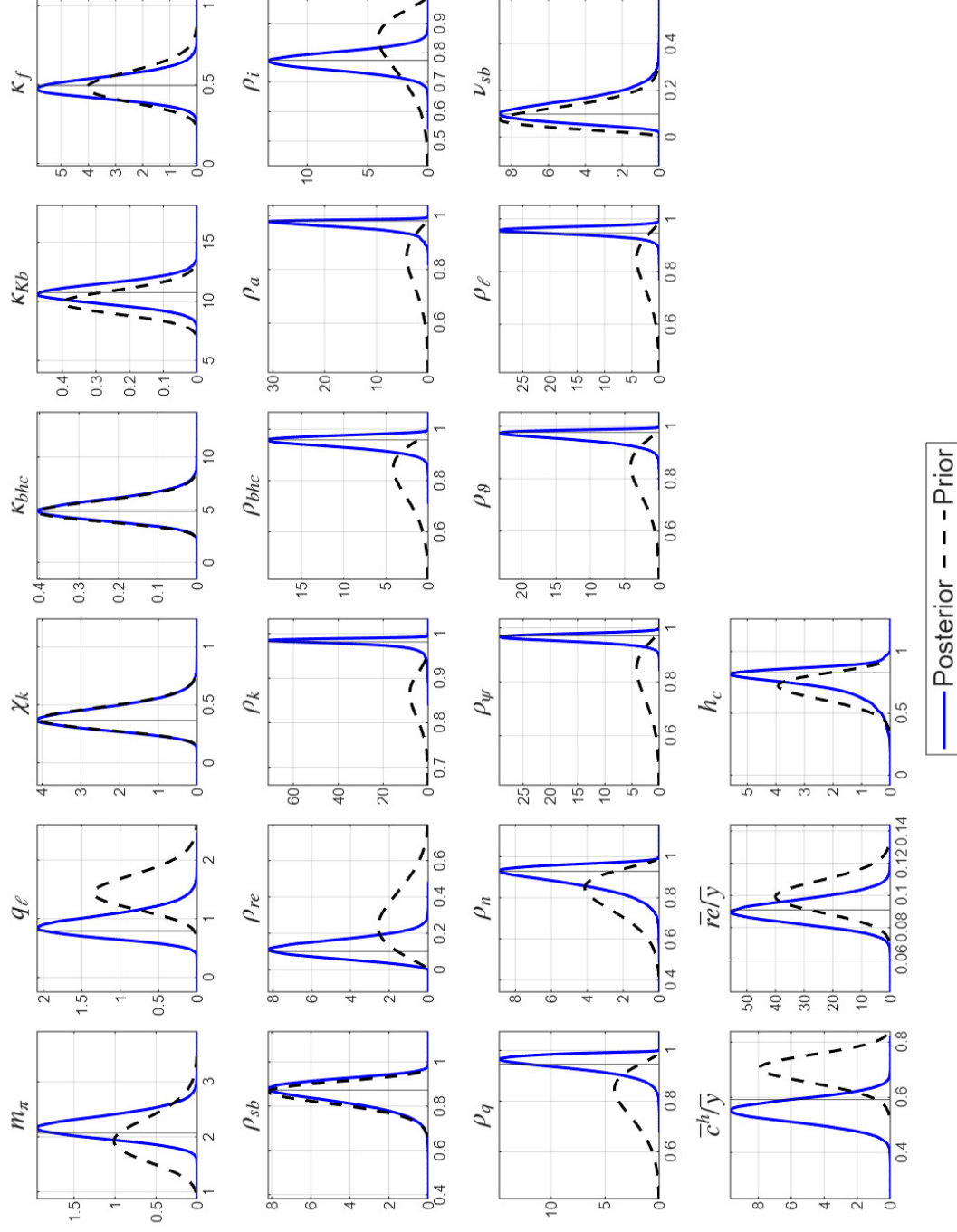


Figure 19: Prior (dashed) vs. posterior (solid) densities of the structural parameters; vertical line marks the posterior mode.

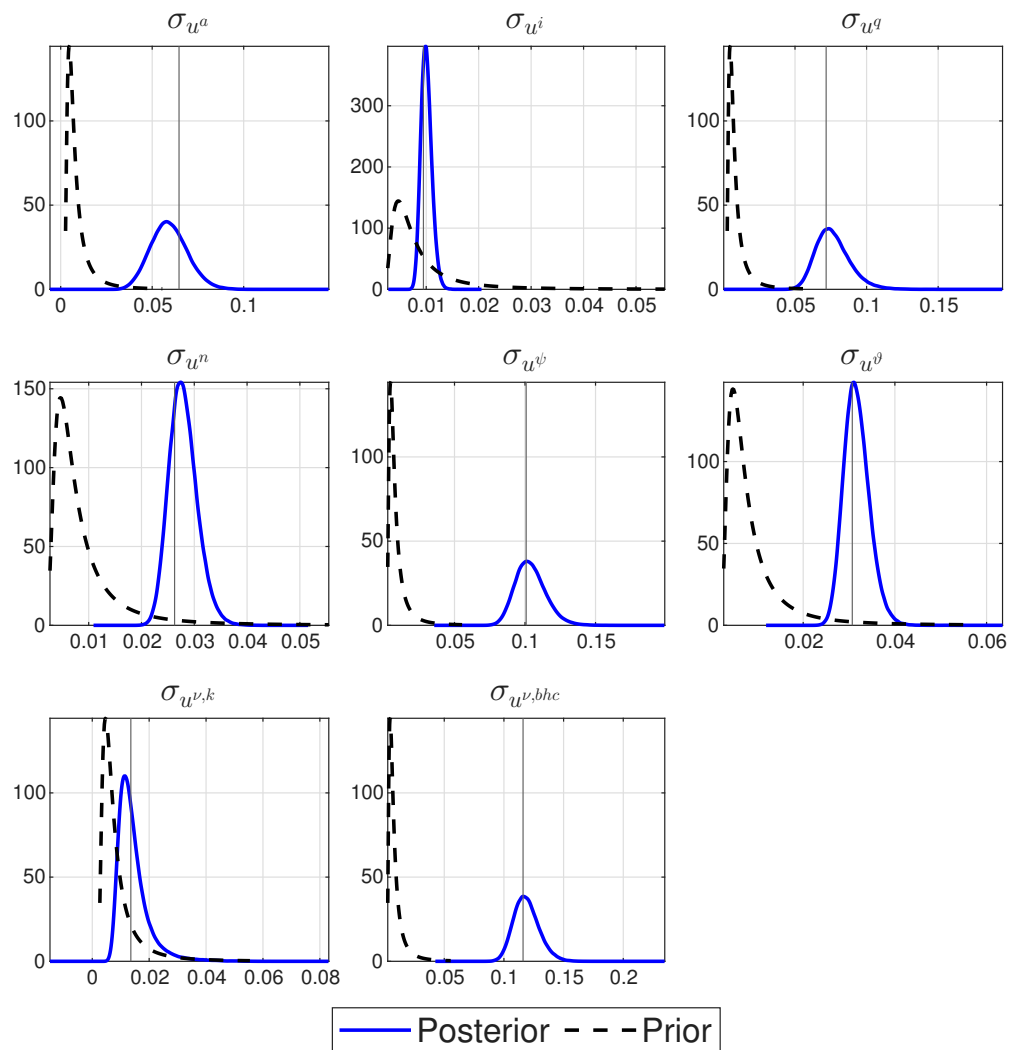


Figure 20: Priors and Posteriors of Estimated Shock Standard Deviations

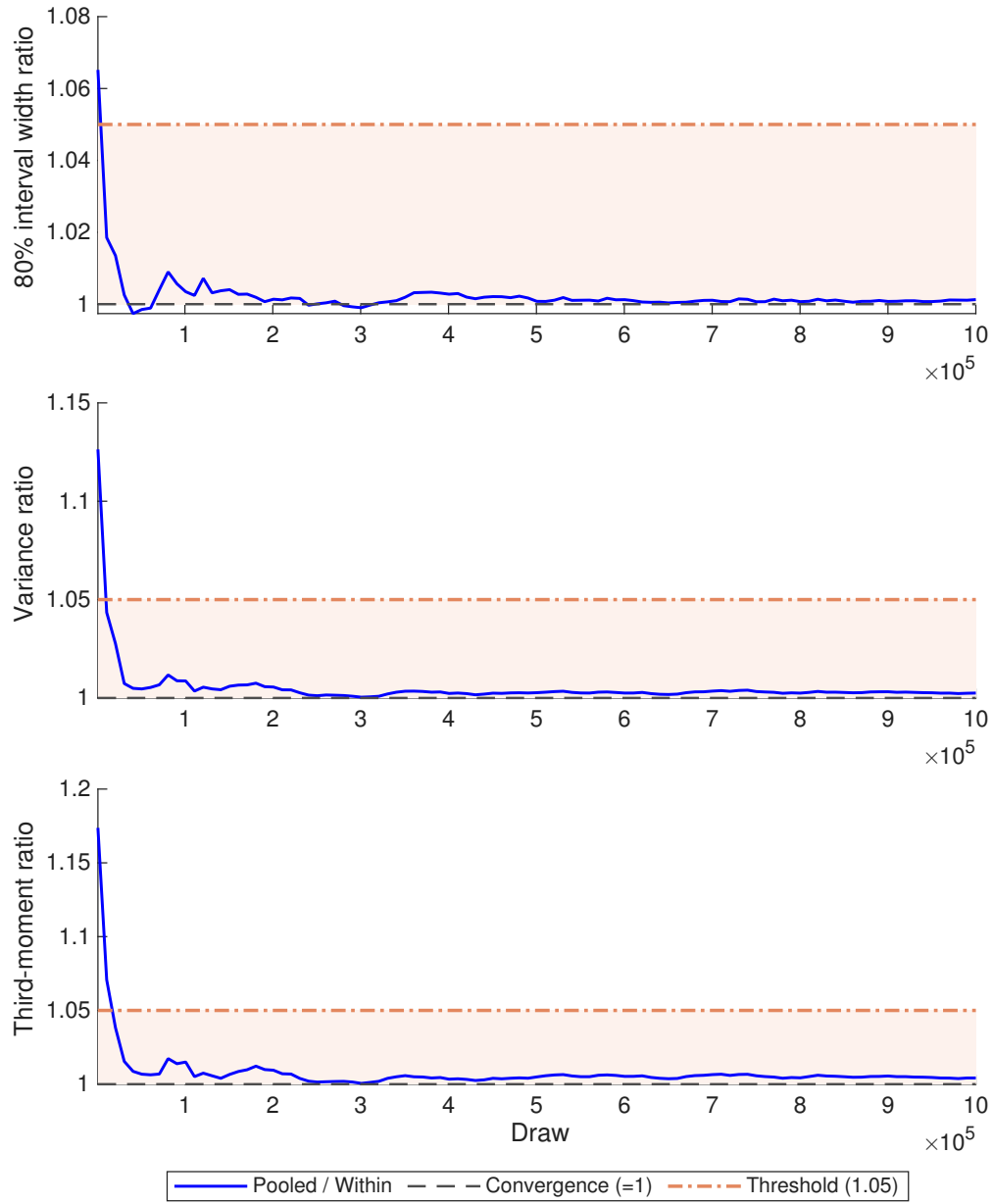


Figure 21: Multivariate Convergence Diagnostic

Note: Multivariate Brooks–Gelman (1998) convergence diagnostic. For 80% of the interval of the pooled and within-chain draws, the mean, variance, and skewness of the log posterior are calculated with widening windows. The step size is set to a ceiling of $(\text{Total draws} - 1000) / 100$. The window i ranges from $(1000 + (i - 1) \times \text{step size}) / 2$ to $(1000 + i \times \text{step size})$. The blue line represents the ratio of the pooled and within-chain statistics, while the red dotted line represents the reference threshold of 1.05. In the final window, which includes all draws after the burn-in, the ratios are 0.999, 1.001, and 1.002 for the mean, variance, and skewness, respectively, indicating satisfactory convergence of the estimation process.

Table 6: Prior Distributions for Bayesian Estimation

Parameter	Description	Dist.	Mean	Std.
Policy Parameters				
m_{π}	Strength of MP response to inflation gap	Gamma	2	0.4
q_{ℓ}	Strength of AP response to credit gap	Gamma	1.5	0.3
χ_k	Cyclicality of macroprudential policy for bank	Gamma	0.4	0.1
Adjustment Cost Parameters				
κ_{bhc}	Banker leverage	Gamma	5	1
κ_{Kb}	Bank capital	Gamma	10	1
κ_f	Firm financial distress	Gamma	0.5	0.1
Risk Weight Parameters				
ρ_{sb}	Persistence of loan to shadow bank	Beta	0.8	0.05
ρ_{re}	Risk weight for reserve holding	Beta	0.3	0.15
Macroprudential Policy Persistence				
ρ_k	Policy for wholesale bank	Beta	0.85	0.05
ρ_{bhc}	Policy for banker	Beta	0.8	0.1
Shock Persistence Parameters				
ρ_a	Technology shock	Beta	0.8	0.1
ρ_i	Monetary policy shock	Beta	0.8	0.1
ρ_q	Asset purchase policy	Beta	0.8	0.1
ρ_n	Return shock	Beta	0.8	0.1
ρ_{ψ}	Debt-Investment shock	Beta	0.8	0.1
ρ_{ϑ}	HH financial distress cost shock	Beta	0.8	0.1
Shock Standard Deviations				

Continued on next page

Table 6 (*continued*)

Parameter	Description	Dist.	Mean	Std.
σ_{u^a}	Technology shock	Inv. Gamma	0.01	0.05
σ_{u^i}	Monetary policy shock	Inv. Gamma	0.01	0.05
σ_{u^q}	Asset purchase policy	Inv. Gamma	0.01	0.05
σ_{u^n}	Return shock	Inv. Gamma	0.01	0.05
σ_{u^ψ}	Debt-Investment shock	Inv. Gamma	0.01	0.05
σ_{u^ϑ}	HH financial distress cost shock	Inv. Gamma	0.01	0.05
$\sigma_{u^{v,k}}$	Macroprudential policy shock for wholesale bank	Inv. Gamma	0.01	0.05
$\sigma_{u^{v,bhc}}$	Macroprudential policy shock for banker	Inv. Gamma	0.01	0.05
Other Parameters				
ρ_ℓ	Persistence of committed credit process	Beta	0.8	0.1
v_{sb}	Dividend payout ratio of shadow bank	Beta	0.1	0.05
$\bar{c}^h/\bar{y}$	SS consumption-to-output ratio	Beta	0.70	0.05
$\bar{r}e/\bar{y}$	SS reserve-to-GDP ratio	Beta	0.1	0.01
h_c	Consumption habit persistence	Beta	0.7	0.1

Note: Prior distributions for Bayesian estimation using U.S. quarterly data from 2009:Q2 to 2025:Q4. MP = Monetary Policy; AP = Asset Purchase; HH = Household.

Table 7: Posterior Statistics for Bayesian Estimation

Parameter	Mode	Mean	5%	Median	95%	$\hat{R}$
Policy Parameters						
m_π	2.071	2.194	1.875	2.182	2.555	1.000
q_ℓ	0.790	0.900	0.610	0.881	1.257	1.001
χ_k	0.365	0.395	0.247	0.386	0.573	1.000
Adjustment Cost Parameters						
κ_{bhc}	4.848	5.064	3.539	4.992	6.830	1.000
κ_{Kb}	10.742	10.667	9.302	10.648	12.090	1.000
κ_f	0.494	0.486	0.379	0.482	0.606	1.000
Risk Weight Parameters						
ρ_{sb}	0.873	0.859	0.765	0.866	0.932	1.000
ρ_{re}	0.101	0.118	0.045	0.115	0.200	1.000
Policy Persistence Parameters						
ρ_k	0.982	0.980	0.962	0.982	0.989	1.000
ρ_{bhc}	0.958	0.947	0.906	0.950	0.977	1.000
Shock Persistence Parameters						
ρ_a	0.980	0.963	0.927	0.967	0.983	1.006
ρ_i	0.774	0.770	0.721	0.770	0.819	1.000
ρ_q	0.946	0.947	0.888	0.952	0.986	1.000
ρ_n	0.929	0.894	0.791	0.905	0.961	1.000
ρ_ψ	0.971	0.962	0.936	0.963	0.982	1.000
ρ_ϑ	0.977	0.959	0.920	0.964	0.985	1.000
Shock Standard Deviations						

Continued on next page

Table 7 (*continued*)

Parameter	Mode	Mean	5%	Median	95%	$\hat{R}$
σ_{u^a}	0.065	0.059	0.044	0.059	0.076	1.002
σ_{u^i}	0.009	0.010	0.008	0.010	0.012	1.000
σ_{u^q}	0.072	0.077	0.059	0.075	0.098	1.000
σ_{u^n}	0.026	0.028	0.024	0.028	0.033	1.001
σ_{u^ψ}	0.101	0.104	0.088	0.103	0.123	1.000
σ_{u^ϑ}	0.031	0.032	0.027	0.031	0.037	1.000
$\sigma_{u^{v,k}}$	0.014	0.014	0.008	0.013	0.024	1.000
$\sigma_{u^{v,bhc}}$	0.116	0.119	0.103	0.118	0.137	1.000
Other Parameters						
ρ_ℓ	0.946	0.952	0.926	0.954	0.973	1.001
v_{sb}	0.098	0.121	0.054	0.114	0.214	1.000
$\bar{c}^h/\bar{y}$	0.592	0.549	0.481	0.550	0.615	1.000
$\bar{r}e/\bar{y}$	0.091	0.090	0.079	0.090	0.102	1.000
h_c	0.828	0.759	0.550	0.782	0.879	1.002

Note: Posterior statistics from Bayesian estimation using U.S. quarterly data from 2009:Q2 to 2025:Q4.

All values are rounded to three decimal places. $\hat{R}$ is the Brooks–Gelman (1998) univariate potential scale reduction factor computed across 5 chains after discarding the first 50% of draws; values below 1.05 indicate convergence.

Table 8: Prior versus Posterior: Bayesian Learning Rate

Well identified ($\sigma_{\text{post}}/\sigma_{\text{prior}} \leq 0.5$)			Weakly identified ($\sigma_{\text{post}}/\sigma_{\text{prior}} > 0.5$)		
Param.	Diff.	$\sigma_{\text{post}}/\sigma_{\text{prior}}$	Param.	Diff.	$\sigma_{\text{post}}/\sigma_{\text{prior}}$
σ_{u^i}	0.000	0.02	ρ_{sb}	0.059	0.51
$\sigma_{u^{\pi}}$	0.018	0.05	m_{π}	0.194	0.52
$\sigma_{u^{\vartheta}}$	0.022	0.06	ρ_n	0.094	0.54
$\sigma_{u^{v,k}}$	0.004	0.11	q_{ℓ}	-0.600	0.66
ρ_{ψ}	0.162	0.14	κ_f	-0.014	0.70
ρ_{ℓ}	0.152	0.15	$\bar{r}e/\bar{y}$	-0.010	0.71
ρ_a	0.163	0.18	$\bar{c}^h/\bar{y}$	-0.151	0.81
ρ_k	0.130	0.20	κ_{Kb}	0.667	0.85
σ_{u^a}	0.049	0.20	v_{sb}	0.021	0.99
ρ_{ϑ}	0.159	0.20	χ_k	-0.005	1.00
$\sigma_{u^{v,bhc}}$	0.109	0.21	κ_{bhc}	0.064	1.01
$\sigma_{u^{\psi}}$	0.094	0.22	h_c	0.059	1.03
ρ_{bhc}	0.147	0.22			
σ_{u^q}	0.067	0.24			
ρ_i	-0.030	0.30			
ρ_q	0.147	0.30			
ρ_{re}	-0.382	0.47			

Note: Parameters sorted in ascending order of the Bayesian learning rate $\sigma_{\text{post}}/\sigma_{\text{prior}}$, the ratio of posterior to prior standard deviations Poirier (1998); Koop et al. (2013). A small ratio indicates that the data sharpen the prior substantially; a ratio near or above one indicates that the data are essentially uninformative for the parameter, so the marginal posterior coincides with the marginal prior (see An and Schorfheide (2007)). The left panel collects the 17 parameters for which the data at least halve the prior standard deviation; the right panel collects the remaining 12 parameters. The Diff. column is the posterior mean minus the prior mean; prior and posterior means are reported in Tables 6 and 7. For κ_{bhc} the ratio is marginally above one and for χ_k the ratio rounds to one; together with $|\text{Diff.}|$ values below 0.1 prior standard deviations, this indicates that the marginal posteriors of these two parameters remain close to their marginal priors in this sample. The local rank condition is still satisfied at the posterior mode (see Section C.2).

C.2.4 Shock Decomposition Results

This appendix reports the full shock decomposition tables referenced in Section 5. Table 9 shows the forecast error variance decomposition (FEVD) at selected horizons. Table 10 summarizes the historical contribution of each shock to the smoothed series over 2022:Q1–2025:Q4, reported in both signed-mean (Panel A) and absolute-mean (Panel B) terms.

Table 9: Forecast Error Variance Decomposition at Selected Horizons (%)

Variable	Technology	MP	AP	Startup Return	Investment	HH. F.D.	MPR Capital	MPR Leverage
<i>Horizon: 1 quarter(s)</i>								
Output	59.8	3.7	0.7	0.0	0.4	0.8	34.5	0.1
Policy rate	36.6	55.7	0.2	0.0	0.0	0.3	7.1	0.1
Inflation	45.7	44.7	0.2	0.0	0.1	0.3	8.9	0.1
Deposit share	0.0	0.0	0.0	0.0	0.0	100.0	0.0	0.0
Bank capital	6.0	0.4	8.2	0.0	8.8	39.3	0.4	36.8
Investment	3.0	3.9	32.5	0.0	60.6	0.0	0.0	0.0
Deposit	12.3	0.7	17.8	0.0	19.5	26.5	0.9	22.4
Default rate	3.6	4.3	0.0	91.0	0.0	0.0	1.0	0.0
Credit to bank (ℓ^f)	3.8	5.1	42.1	0.0	49.0	0.0	0.0	0.0
Loan to SB (b^{sb})	79.8	5.5	0.0	0.2	0.4	0.5	13.6	0.0
<i>Horizon: 4 quarter(s)</i>								
Output	25.8	3.7	2.5	15.0	11.7	36.2	2.0	3.0
Policy rate	17.1	3.4	6.2	28.0	17.3	20.2	3.5	4.3
Inflation	16.4	7.0	5.9	27.0	16.7	19.4	3.4	4.1
Deposit share	0.0	0.0	0.0	0.0	0.0	100.0	0.0	0.0
Bank capital	65.3	17.6	6.5	0.2	5.8	0.5	0.5	3.6
Investment	12.2	14.0	7.5	27.6	11.5	20.7	1.0	5.4
Deposit	59.6	23.1	9.7	0.3	3.9	0.5	0.8	2.1
Default rate	25.1	13.9	1.5	28.2	4.8	22.2	0.6	3.6
Credit to bank (ℓ^f)	10.5	12.0	6.5	23.7	24.0	17.8	0.9	4.6
Loan to SB (b^{sb})	61.9	21.0	8.8	0.4	3.9	0.4	0.8	2.9
<i>Horizon: 8 quarter(s)</i>								
Output	58.7	21.6	5.8	1.6	6.0	5.5	0.3	0.5
Policy rate	56.0	22.8	5.7	2.9	5.8	6.4	0.3	0.2
Inflation	55.9	22.8	5.7	2.9	5.7	6.4	0.3	0.2
Deposit share	0.0	0.0	0.0	0.0	0.0	100.0	0.0	0.0
Bank capital	62.7	19.4	6.9	0.0	5.9	2.3	0.3	2.5
Investment	56.2	25.2	8.3	1.8	3.6	3.7	0.7	0.5
Deposit	62.7	19.4	7.0	0.0	5.8	2.1	0.4	2.6
Default rate	60.4	22.6	8.0	0.3	4.5	2.3	0.6	1.3
Credit to bank (ℓ^f)	55.9	25.1	8.3	1.8	4.1	3.7	0.7	0.5
Loan to SB (b^{sb})	62.8	19.4	7.0	0.0	5.8	2.2	0.4	2.6
<i>Horizon: 16 quarter(s)</i>								
Output	62.6	19.6	6.9	0.0	5.8	2.3	0.3	2.4
Policy rate	62.6	19.6	6.9	0.0	5.8	2.3	0.3	2.4
Inflation	62.6	19.6	6.9	0.0	5.8	2.3	0.3	2.4
Deposit share	0.0	0.0	0.0	0.0	0.0	100.0	0.0	0.0
Bank capital	62.6	19.6	6.9	0.0	5.9	2.4	0.3	2.4
Investment	62.6	19.5	6.9	0.0	5.9	2.3	0.3	2.4
Deposit	62.6	19.6	6.9	0.0	5.9	2.4	0.3	2.4
Default rate	62.6	19.5	6.9	0.0	5.9	2.4	0.3	2.4
Credit to bank (ℓ^f)	62.6	19.5	6.9	0.0	5.9	2.3	0.3	2.4
Loan to SB (b^{sb})	62.6	19.6	6.9	0.0	5.9	2.4	0.3	2.4

Notes: Each entry is the percentage of the h -step-ahead forecast error variance of the row variable attributable to the column shock, computed from the first-order decision rule $y_t = g_x x_{t-1} + g_u e_t$ evaluated at the posterior mode using Σ_e . Rows sum to 100 by construction. Shocks are assumed mutually orthogonal, consistent with estimation.

Table 10: Average Magnitude Share of Each Structural Shock, 2022:Q1–2025:Q4 (%)

Variable	Technology	MP	AP	Startup Return	Investment	HH. F.D.	MPR Capital	MPR Leverage	Dominant
<i>Panel A. Mean signed share (net direction)</i>									
Output	32.4	1.4	-3.9	-0.1	5.8	0.8	47.4	-0.3	MPR Capital
Policy rate	-10.7	6.6	1.8	-0.4	22.3	0.1	39.4	-0.1	MPR Capital
Inflation	-9.3	8.7	0.8	-0.3	18.9	0.2	32.1	-0.1	MPR Capital
Deposit share	0.0	-0.0	-0.0	0.0	-0.0	12.5	-0.0	-0.0	HH. F.D.
Bank capital	23.7	-0.5	4.1	2.3	8.3	2.8	-2.6	1.0	Technology
Investment	-2.3	6.1	1.0	-0.1	48.5	0.6	-9.8	-0.0	Investment
Deposit	20.1	0.0	1.2	2.3	1.1	-0.1	-12.5	-1.1	Technology
Default rate	26.6	-9.0	-2.4	-21.3	-17.8	-0.5	-4.5	1.0	Technology
Credit to bank (ℓ^f)	-7.7	6.7	2.3	-0.1	27.4	0.7	-15.3	-0.0	Investment
Loan to SB (b^{sb})	40.5	-8.1	-4.7	3.1	-25.0	-0.9	4.9	1.0	Technology
<i>Panel B. Mean absolute share (average magnitude). Daggers mark “two-sided” cells.</i>									
Output	32.4	2.0	6.4	0.1	9.2	1.8	47.4	0.6	MPR Capital
Policy rate	13.6	15.6	5.7	0.4	22.3	2.1	39.4	0.9	MPR Capital
Inflation	11.5	30.2 [†]	4.7	0.3	18.9	1.6	32.1	0.7	MPR Capital
Deposit share	0.0	0.0	0.0	0.0	0.0	100.0 [†]	0.0	0.0	HH. F.D.
Bank capital	23.7	0.7	15.7 [†]	2.3	9.6	21.1 [†]	4.5	22.5 [†]	Technology
Investment	13.3 [†]	11.9	15.8 [†]	0.1	48.5	0.6	9.8	0.0	Investment
Deposit	20.1	8.6	18.2 [†]	2.3	11.9 [†]	13.5 [†]	12.5	13.0 [†]	Technology
Default rate	29.6	13.6	10.6 [†]	21.3	17.8	1.2	4.9	1.0	Technology
Credit to bank (ℓ^f)	17.9	15.7	22.9 [†]	0.1	27.4	0.7	15.3	0.0	Investment
Loan to SB (b^{sb})	40.5	12.9	10.9	3.1	25.0	1.0	5.6	1.0	Technology

Notes: Within each quarter, the share of shock k in the smoothed deviation of variable i from its steady state is computed as $\text{share}_{i,k,t} = 100 \times \text{contrib}_{i,k,t} / \sum_k |\text{contrib}_{i,k,t}|$, so that $\sum_k |\text{share}_{i,k,t}| = 100$. Panel A reports the time average of $\text{share}_{i,k,t}$ over 2022:Q1–2025:Q4: a positive (negative) value means the shock, on net, pushed the variable above (below) its steady state during the sub-period. Panel B reports the time average of $|\text{share}_{i,k,t}|$, which measures the average magnitude of the shock’s effect and is invariant to directional cancellation. A [†] flags “two-sided” cells where the magnitude exceeds 10 pp but the absolute value of the signed mean is below 40% of the magnitude, i.e. the shock exerts large but offsetting upward and downward pressure over the sub-period. The presample initial-values effect is excluded.

Appendix D. Full Nonlinear Model

Lower case letters denote real variables. The parameters are elaborated in the calibration section.

D.1 Households

List of variables:

- c_t = consumption.
- h_t = labor supply.
- b_t^b = banker bond demand.
- d_t = deposit demand.
- D_t^f = dividend from intermediate good firm.
- D_t^k = dividend from capital good firm.
- Z_t = operational profit of central bank.
- $\mathcal{W}_t^n$ = wealth of startups.
- $\Phi(s_t)$ = household financial distress cost.
- s_t = share of depositors.
- P_t = nominal price of final good.
- w_t = labor income.
- R_t^b = interest rate on banker's bond.
- r_t^d = retail interest rate on deposit.
- X_0^{bhc} = fixed transfer to banker.

D.1.1 Optimization Problem

- Objective function:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_h^t \left[\frac{1}{1 - \frac{1}{\gamma_h}} \left\{ c_t^h - h_c c_{t-1}^h \right\}^{1 - \frac{1}{\gamma_h}} - \frac{\chi_h}{1 + \frac{1}{\zeta_h}} \left\{ h_t^h \right\}^{1 + \frac{1}{\zeta_h}} \right]$$

- Budget constraint:

$$\begin{aligned} R_{t-1}^b P_{t-1} b_{t-1}^b + r_{t-1}^d P_{t-1} d_{r,t-1} + P_t w_t h_t^h + P_t Z_t^{total} \\ \geq P_t c_t^h + P_t b_t^b + P_t d_{r,t} + P_t d_t^h \Phi(s_t) \end{aligned}$$

where $Z_t^{total} = D_t^f + D_t^k + (1 - \iota)\chi_n \mathcal{W}_t^n + T_t^{cb} - X_0^{bhc}$.

- Lubello and Rouabah (2024) Financial distress cost:

$$\begin{aligned}\Phi(s_t) &\equiv \chi_1 \left(\frac{1 - s_t}{s_t} \right)^{\vartheta_t} \\ \vartheta_t &= S_t^\vartheta \bar{\vartheta} \\ \log(S_t^\vartheta) &= \rho_\vartheta \log(S_{t-1}^\vartheta) + u_t^\vartheta\end{aligned}$$

- Lubello and Rouabah (2024) Deposits and banker's debt:

$$d_{r,t} = s_t d_t^h, \quad b_t^b = (1 - s_t) d_t^h$$

D.1.2 Lagrangean

$$\begin{aligned}\max_{c_t, h_t, d_t^h, s_t,} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_h^t &\left[\frac{1}{1 - \frac{1}{\gamma_h}} \left\{ c_t^h - h_c c_{t-1}^h \right\}^{1 - \frac{1}{\gamma_h}} - \frac{\chi_h}{1 + \frac{1}{\zeta_h}} \left\{ h_t^h \right\}^{1 + \frac{1}{\zeta_h}} \right. \\ &+ \lambda_t^h \left(R_{t-1}^b (1 - s_{t-1}) \Pi_t^{-1} d_{t-1}^h + r_{t-1}^d s_{t-1} \Pi_t^{-1} d_{t-1}^h + w_t^h h_t^h + Z_t^{total} \right. \\ &\left. \left. - \left(c_t^h + d_t^h + d_t^h \Phi(s_t) \right) \right) \right]\end{aligned}$$

where $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$.

D.1.3 First-Order Conditions

- FOC for consumption c_t^h :

$$\lambda_t^h = \left(c_t^h - h_c c_{t-1}^h \right)^{-\frac{1}{\gamma_h}} - \beta_h h_c \left(c_{t+1}^h - h_c c_t^h \right)^{-\frac{1}{\gamma_h}}$$

- FOC for labor supply h_t^h :

$$h_t^h = \left(\lambda_t^h \frac{w_t^h}{\chi_h} \right)^{\zeta_h}$$

- FOC for saving d_t^h :

$$\mathbb{E}_t \left[\Lambda_{t,t+1}^h \Pi_{t+1}^{-1} \left(R_t^b (1 - s_t) + r_t^d s_t \right) \right] = 1 + \Phi(s_t)$$

where $\Lambda_{t,t+1}^h = \beta_h \frac{\lambda_{t+1}^h}{\lambda_t^h}$.

- FOC for share of deposit s_t :

$$\mathbb{E}_t \left[\Lambda_{t,t+1}^h \Pi_{t+1}^{-1} \left(R_t^b - r_t^d \right) \right] = -\Phi'(s_t)$$

- FOC for deposit demand $d_{r,t}$:

$$\mathbb{E}_t \left[\Lambda_{t,t+1}^h \Pi_{t+1}^{-1} r_t^d \right] = 1$$

D.2 Banker

List of variables:

c_t^b = consumption by banker.

b_t^b = banker debt supply.

e_t^b = equity investment in wholesale bank.

D_t^b = dividend from wholesale and retail banks.

D_t^{sb} = dividend from shadow bank.

$\mathcal{C}_t^{bhc,b}$ = cost of borrowing.

$\mathcal{C}_t^{bhc}$ = double-leverage adjustment cost.

R_t^{eb} = interest rate on equity investment.

X_0^{bhc} = fixed transfer from the household.

D.2.1 Optimization Problem

- Objective function:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_b^t \log c_t^b$$

- Budget constraint:

$$\begin{aligned}
P_t b_t^b + R_{t-1}^{eb} P_{t-1} e_{t-1}^b + P_t D_t^b + P_t D_t^{sb} + P_t X_0^{bhc} \\
\geq P_t c_t^b + R_{t-1}^b P_{t-1} b_{t-1}^b + P_t e_t^b + P_{t-1} \mathcal{C}_{t-1}^{bhc,b} + P_{t-1} \mathcal{C}_{t-1}^{bhc}
\end{aligned}$$

- Kumhof and Wang (2021) The cost of issuing debt:

$$\mathcal{C}_t^{bhc,b} = \frac{\kappa_{bhc,b}}{1 + \frac{1}{\theta_{bhc,b}}} \bar{b}^b \left(\frac{b_t^b}{\bar{b}^b} \right)^{1 + \frac{1}{\theta_{bhc,b}}}$$

- Angelini et al. (2010) The cost of leverage ratio adjustment:

$$\mathcal{C}_t^{bhc} = \frac{\kappa_{bhc}}{2} \left(\frac{e_t^b}{b_t^b} - v_t^{bhc} \right)^2 e_t^b$$

D.2.2 Lagrangean

$$\begin{aligned}
\max_{c_t^b, b_t^b, e_t^b} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_t^i \left[\log c_t^b + \lambda_t^b \left(b_t^b + R_{t-1}^{eb} \Pi_t^{-1} e_{t-1}^b + D_t^b + P_t X_0^{bhc} \right. \right. \\
\left. \left. - \left(c_t^b + R_{t-1}^b \Pi_t^{-1} b_{t-1}^b + e_t^b + \Pi_t^{-1} \mathcal{C}_{t-1}^{bhc,b} + \Pi_t^{-1} \mathcal{C}_{t-1}^{bhc} \right) \right) \right]
\end{aligned}$$

D.2.3 First-Order Conditions

- FOC for consumption c_t^b :

$$\lambda_t^b = \frac{1}{c_t^b}$$

- FOC for debt supply b_t^b :

$$\mathbb{E}_t \left[\Lambda_{t,t+1}^b \Pi_{t+1}^{-1} \left(R_t^b + \frac{\partial \mathcal{C}_t^{bhc,b}}{\partial b_t^b} + \frac{\partial \mathcal{C}_t^{bhc}}{\partial b_t^b} \right) \right] = 1$$

where $\Lambda_{t,t+1}^b = \beta_b \frac{\lambda_{t+1}^b}{\lambda_t^b}$.

- FOC for equity investment in wholesale bank e_t^b :

$$\mathbb{E}_t \left[\Lambda_{t,t+1}^b \Pi_{t+1}^{-1} \left(R_t^{eb} - \frac{\partial \mathcal{C}_t^{bhc}}{\partial e_t^b} \right) \right] = 1$$

D.3 Wholesale Bank

D.3.1 Optimization Problem

List of variables:

N_t^b = net worth.

$\tilde{\ell}_t^{f,b}$ = wholesale lending supply to retail banks (intermediate good firm loan).

b_t^{sb} = wholesale lending supply to retail banks (shadow bank loan).

re_t = reserve demand.

d_t = wholesale deposit supply to retail banks (household's deposit).

e_t^b = equity demand.

$\mathcal{C}_t^b$ = capital-asset ratio adjustment cost.

X_t^b = internal capital.

R_t^f = wholesale interest rate on firm lending.

R_t^{sb} = wholesale interest rate on shadow bank lending.

R_t = policy interest rate.

R_t^d = wholesale interest rate on deposit.

- Net worth:

$$\Pi_{t+1} N_{t+1}^b = R_t^f \tilde{\ell}_t^{f,b} + R_t^{sb} b_t^{sb} + R_t re_t - R_t^d d_t - R_t^{eb} e_t^b - \mathcal{C}_t^b$$

- Angelini et al. (2014) The cost of capital-asset ratio adjustment:

$$\mathcal{C}_t^b = \frac{\kappa_K b}{2} \left(\frac{X_t^b + \rho_e e_t^b}{\rho_{re} re_t + \omega_t^f \tilde{\ell}_t^{f,b} + \omega_t^{sb} b_t^{sb}} - v_t^k \right)^2 (X_t^b + \rho_e e_t^b)$$

- Risk weight process:

$$\omega_t^s = (1 - \rho_s) \bar{\omega}^s + (1 - \rho_s) \chi_s (y_t - y_{t-4}) + \rho_s \omega_{t-1}^s$$

where $s \in \{f, sb\}$.

- Balance sheet condition:

$$\tilde{\ell}_t^{f,b} + b_t^{sb} + re_t = d_t + e_t^b + X_t^b$$

- Brunnermeier and Koby (2018) Capital accumulation formula:

$$X_{t+1}^b = (1 - v_b)N_{t+1}^{b,total}$$

where $N_{t+1}^{b,total} = N_{t+1}^b + N_{t+1}^r$, the sum of wholesale and retail bank net worth.

- Objective function:

$$\max \mathbb{E}_t \left[\Lambda_{t,t+1}^b N_{t+1}^b \right]$$

D.3.2 Lagrangean

$$\begin{aligned} \max_{\tilde{\ell}_t^{f,b}, b_t^{sb}, re_t, e_t^b} \mathbb{E}_t \Lambda_{t,t+1}^b \Pi_{t+1}^{-1} & \left(\left(R_t^f - R_t^d \right) \tilde{\ell}_t^{f,b} + \left(R_t^{sb} - R_t^d \right) b_t^{sb} + \left(R_t - R_t^d \right) re_t \right. \\ & \left. - \left(R_t^{eb} - R_t^d \right) e_t^b + R_t^d X_t^b - \mathcal{C}_t^b \right) \end{aligned}$$

D.3.3 First-Order Conditions

- FOC for firm lending supply to retail banks:

$$\mathbb{E}_t \left[\Lambda_{t,t+1}^b \Pi_{t+1}^{-1} \left(R_t^f - R_t^d - \frac{\partial \mathcal{C}_t^b}{\partial \tilde{\ell}_t^{f,b}} \right) \right] = 0$$

- FOC for shadow bank debt supply to retail banks:

$$\mathbb{E}_t \left[\Lambda_{t,t+1}^b \Pi_{t+1}^{-1} \left(R_t^{sb} - R_t^d - \frac{\partial \mathcal{C}_t^b}{\partial b_t^{sb}} \right) \right] = 0$$

- FOC for reserve demand:

$$\mathbb{E}_t \left[\Lambda_{t,t+1}^b \Pi_{t+1}^{-1} \left(R_t - R_t^d - \frac{\partial \mathcal{C}_t^b}{\partial r e_t} \right) \right] = 0$$

- FOC for equity investment demand:

$$\mathbb{E}_t \left[\Lambda_{t,t+1}^b \Pi_{t+1}^{-1} \left(-R_t^{eb} + R_t^d - \frac{\partial \mathcal{C}_t^b}{\partial e_t^b} \right) \right] = 0$$

D.4 Retail Bank

List of variables:

N_t^r = net worth of retail bank.

$\tilde{\ell}_{r,t}^{f,b}$ = retail lending to intermediate good firms.

$b_{r,t}^{sb}$ = retail lending to shadow bank.

$d_{r,t}$ = retail deposit demand from the household.

r_t^f = retail interest rate on firm lending.

r_t^{sb} = retail interest rate on shadow bank lending.

r_t^d = retail interest rate on deposits.

$\varepsilon_{b,f}$ = elasticity of demand for firm loans.

$\varepsilon_{b,sb}$ = elasticity of demand for shadow bank loans.

$\varepsilon_{b,d}$ = elasticity of supply for deposits.

D.4.1 Optimization Problem

- The net worth of retail banks:

$$N_{t+1}^r = \Pi_{t+1}^{-1} \left(r_t^f \tilde{\ell}_{r,t}^{f,b} + r_t^{sb} b_{r,t}^{sb} - r_t^d d_{r,t} - \left(R_t^f \tilde{\ell}_t^{f,b} + R_t^{sb} b_t^{sb} - R_t^d d_t \right) \right)$$

- The net worth of depository banks in total:

$$\begin{aligned}
\Pi_{t+1}N_{t+1}^{total} &= \Pi_{t+1} \left(N_{t+1}^b + N_{t+1}^r \right) \\
&= \left(r_t^f - r_t^d \right) \tilde{\ell}_{r,t}^{f,b} + \left(r_t^{sb} - r_t^d \right) b_{r,t}^{sb} + \left(R_t - r_t^d \right) r e_t \\
&\quad - \left(R_t^{eb} - r_t^d \right) e_t^b + r_t^d X_t^b - \mathcal{C}_t^b
\end{aligned}$$

- Gambacorta and Karmakar (2018) The retail loans and deposit demand schedules:

$$\begin{aligned}
\tilde{\ell}_{r,t}^{f,b}(j) &= \left(\frac{r_t^f(j) - 1}{r_t^f - 1} \right)^{-\varepsilon_{b,f}} \tilde{\ell}_{r,t}^{f,b} \\
b_{r,t}^{sb}(j) &= \left(\frac{r_t^{sb}(j) - 1}{r_t^{sb} - 1} \right)^{-\varepsilon_{b,sb}} b_{r,t}^{sb} \\
d_{r,t}(j) &= \left(\frac{r_t^d(j) - 1}{r_t^d - 1} \right)^{-\varepsilon_{b,d}} d_{r,t}
\end{aligned}$$

- Objective function for loan branches:

$$\begin{aligned}
\mathbb{E}_0 \sum_{t=0}^{\infty} \beta_b^t \lambda_t^b &\left[\left(r_t^f(j) - 1 \right) \tilde{\ell}_{r,t}^{f,b}(j) + \left(r_t^{sb} - 1 \right) (j) b_{r,t}^{sb}(j) \right. \\
&\quad \left. - \left(\left(R_t^f - 1 \right) \tilde{\ell}_t^{f,b}(j) + \left(R_t^{sb} - 1 \right) b_t^{sb}(j) \right) \right]
\end{aligned}$$

- Objective function for deposit branches:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta_b^t \lambda_t^b \left[- \left(r_t^d(j) - 1 \right) d_{r,t}(j) + \left(R_t^d - 1 \right) d_t(j) \right]$$

D.4.2 Lagrangean

- Loan branches

$$\begin{aligned} \max_{r_t^f(j), r_t^{sb}(j)} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_b^t \lambda_t^b & \left[\left(r_t^f(j) - 1 \right) \left(\frac{r_t^f(j) - 1}{r_t^f - 1} \right)^{-\varepsilon_{b,f}} \tilde{\ell}_{r,t}^{f,b} \right. \\ & + \left(r_t^{sb}(j) - 1 \right) \left(\frac{r_t^{sb}(j) - 1}{r_t^{sb} - 1} \right)^{-\varepsilon_{b,sb}} b_{r,t}^{sb} \\ & \left. - \left((R_t^f - 1) \tilde{\ell}_t^{f,b}(j) + (R_t^{sb} - 1) b_t^{sb}(j) \right) \right] \end{aligned}$$

- Deposit branches

$$\max_{r_t^d(j)} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_b^t \lambda_t^b \left[- \left(r_t^d(j) - 1 \right) \left(\frac{r_t^d(j) - 1}{r_t^d - 1} \right)^{-\varepsilon_{b,d}} d_{r,t} + (R_t^d - 1) d_t(j) \right]$$

D.4.3 First-Order Conditions

- FOC for lending supply to intermediate good producers:

$$1 - \varepsilon_{b,f} + \varepsilon_{b,f} \frac{R_t^f - 1}{r_t^f - 1} = 0$$

- FOC for lending supply to shadow bank:

$$1 - \varepsilon_{b,sb} + \varepsilon_{b,sb} \frac{R_t^{sb} - 1}{r_t^{sb} - 1} = 0$$

- FOC for deposit demand from household:

$$-1 + \varepsilon_{b,d} - \varepsilon_{b,d} \frac{R_t^d - 1}{r_t^d - 1} = 0$$

D.5 Shadow Bank

List of variables:

N_t^{sb} = net worth of shadow bank.

b_t^n = lending supply to startups.

b_t^{sb} = borrowing demand from retail banks.

X_t^{sb} = internal capital.

$\mathcal{C}_t^{sb,n}$ = cost of lending to startups.

m_t^{sb} = monitoring cost for default loans.

R_{t+1}^{fn} = return on fictitious firm lending.

D.5.1 Optimization Problem

- The balance sheet:

$$b_t^n = b_t^{sb} + X_t^{sb}$$

- Net worth:

$$\Pi_{t+1} N_{t+1}^{sb} = \left(R_{t+1}^{fn} b_t^n - r_t^{sb} b_t^{sb} - \mathcal{C}_t^{sb,n} - m_t^{sb} \right)$$

- The cost of lending to startups:

$$\mathcal{C}_t^{sb,n} = \frac{\kappa^{sb,n}}{1 + \frac{1}{\theta_{sb,n}}} \bar{b}^n \left(\frac{b_t^n}{\bar{b}^n} \right)^{1 + \frac{1}{\theta_{sb,n}}}$$

- Brunnermeier and Koby (2018) Capital accumulation formula:

$$X_{t+1}^{sb} = (1 - v_{sb}) N_{t+1}^{sb}$$

- Objective function:

$$\max \mathbb{E}_t \left[\Lambda_{t,t+1}^b N_{t+1}^{sb} \right]$$

D.5.2 Lagrangean

$$\max_{b_t^n} \mathbb{E}_t \Lambda_{t,t+1}^b \Pi_{t+1}^{-1} \left((R_{t+1}^{fn} - r_t^{sb}) b_t^n - \mathcal{C}_t^{sb,n} - m_t^{sb} \right)$$

D.5.3 First-Order Conditions

$$\mathbb{E}_t \Lambda_{t,t+1}^b \Pi_{t+1}^{-1} \left[(R_{t+1}^{fn} - r_t^{sb}) - \frac{\partial \mathcal{C}_t^{sb,n}}{\partial b_t^n} \right] = 0$$

D.6 Intermediate Good Producers

List of variables for intermediate good firm j :

$y_t(j)$ = output of intermediate good.

$h_t(j)$ = labor demand.

$I_t(j)$ = investment.

$K_t(j)$ = owned capital stock.

$K_t^n(j)$ = rented capital from startups.

$\ell_t^f(j)$ = total credit.

$\ell_t^{f,b}(j)$ = committed credit line.

$\ell_t^{f,cb}(j)$ = central bank loan to firm j .

$u_t^f(j)$ = credit line utilization rate.

$P_t(j)$ = price of intermediate good.

$r_t^{k,n}$ = rental rate on capital from startups.

mc_t^{kl} = marginal cost of production.

q_t = Tobin's Q (price of capital).

$G_{P,t}(j)$ = Price adjustment cost.

$G_{I,t}(j)$ = Investment adjustment cost.

$\varphi_t(j)$ = Financial distress cost.

D.6.1 Optimization Problem

- Nominal profit of firm j :

$$\begin{aligned}\Pi_t^F(j) = & P_t(j)y_t(j) + P_t\ell_t^f(j) - P_t w_t h_t(j) - r_{t-1}^f P_{t-1}\ell_{t-1}^f(j) \\ & - P_t r_t^{k,n} K_t^n(j) - P_t I_t(j) - P_t G_{P,t}(j) - P_t G_{I,t}(j) - P_t \varphi_t(j)\end{aligned}$$

where $\ell_t^f(j) = u_t^f(j)\ell_t^{f,b}(j) + \ell_t^{f,cb}(j)$.

- Kumhof and Wang (2021), Gasparini et al. (2024) Inflation adjustment cost:

$$G_{P,t}(j) = \frac{\phi_P}{2} y_t \left(\frac{\frac{P_t(j)}{\bar{P}_{t-1}(j)}}{\Pi_{t-1}} - 1 \right)^2$$

- Christiano et al. (2005) Investment adjustment cost:

$$G_{I,t}(j) = \frac{\phi_I}{2} I_t \left(\frac{I_t(j)}{I_{t-1}(j)} - 1 \right)^2$$

- Financial distress cost:

$$\varphi_t = \kappa_f \left(\frac{\tilde{\ell}_t^{f,b}}{\ell_t^f} \right)^{\theta_u}$$

- Loan in advance constraint:

$$\begin{aligned}\psi_t^L P_t I_t(j) & \leq P_t \ell_t^f(j) \\ \psi_t^L & = S_t^\Psi \bar{\psi}^L\end{aligned}$$

- The shock process for S^Ψ :

$$\log(S_t^\Psi) = \rho_\Psi \log(S_{t-1}^\Psi) + u_t^\Psi$$

- Production technology:

$$y_t(j) = S_t^a (h_t(j))^{1-\alpha} \left[(K_t(j))^{1-\zeta} (K_t^n(j))^\zeta \right]^\alpha$$

- The technology shock process S_t^a :

$$\log(S_t^a) = \rho_a \log(S_{t-1}^a) + u_t^a$$

- Final good firm's demand for intermediate goods:

$$y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\theta} y_t$$

- Dixit–Stiglitz aggregation of intermediate goods:

$$y_t = \left(\int_0^1 y_t(j)^{\frac{\theta-1}{\theta}} dj \right)^{\frac{\theta}{\theta-1}}$$

- Aggregate price index:

$$P_t = \left(\int_0^1 P_t(j)^{1-\theta} dj \right)^{\frac{1}{1-\theta}}$$

- Owned capital accumulation formula:

$$K_{t+1}(j) = (1 - \delta)K_t(j) + I_t(j)$$

- Committed credit line process:

$$\begin{aligned} \Pi_{t+1} \ell_{t+1}^{f,b}(j) = & \rho_\ell \ell_t^{f,b}(j) + \\ & (1 - \rho_\ell) \bar{\ell}^{f,b}(j) + (1 - \rho_\ell) \theta_\pi \left(\Pi_t^{EBITDA}(j) - \bar{\Pi}^{EBITDA} \right) \end{aligned}$$

where $\Pi_t^{EBITDA}(j) = P_t y_t(j) - P_t w_t h_t(j) - P_t r_t^{k,n} K_t^n(j)$.

D.6.2 Lagrangean

$$\begin{aligned}
& \max_{\{P_t(j), \ell_{t+1}^{f,b}(j), u_t^f(j), h_t(j), I_t(j), K_{t+1}(j), K_{t+1}^n(j)\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_h^t \lambda_t^h \left[(P_t(j))^{1-\theta} (P_t)^{\theta-1} y_t \right. \\
& + u_t^f(j) \ell_t^{f,b}(j) + \tilde{\ell}_t^{f,cb}(j) - w_t h_t(j) - r_{t-1}^f \left(u_{t-1}^f(j) \ell_{t-1}^{f,b}(j) + \tilde{\ell}_{t-1}^{f,cb}(j) \right) \Pi_t^{-1} \\
& - r_t^{k,n} K_t^n(j) - I_t(j) - G_{P,t}(j) - G_{I,t}(j) - \varphi_t \\
& - mc_t^{kl} \left((P_t(j))^{-\theta} (P_t)^{\theta} y_t - S_t^a h_t(j)^{1-\alpha} \left[(K_t(j))^{1-\zeta} (K_t^n(j))^{\zeta} \right]^{\alpha} \right) \\
& + q_t^k ((1-\delta)K_t(j) + I_t(j) - K_{t+1}(j)) \\
& \left. + \eta_t^{\psi} \left(u_t^f(j) \ell_t^{f,b}(j) + \tilde{\ell}_t^{f,cb}(j) - \psi_t^L I_t(j) \right) \right]
\end{aligned}$$

D.6.3 First-Order Conditions

- Using the symmetry across intermediate good firms, I drop the index j in the following FOCs.
- FOC for price setting $P_t(j)$:

$$\begin{aligned}
\mu mc_t^{kl} - 1 &= \phi_p(\mu - 1) \left(\frac{\Pi_t}{\Pi_{t-1}} - 1 \right) \frac{\Pi_t}{\Pi_{t-1}} \\
&\quad - \mathbb{E}_t M_t \frac{y_{t+1}}{y_t} \phi_p(\mu - 1) \left(\frac{\Pi_{t+1}}{\Pi_t} - 1 \right) \frac{\Pi_{t+1}}{\Pi_t}
\end{aligned}$$

where $\mu = \frac{\theta}{\theta-1}$.

- FOC for credit utilization $u_t^f(j)$:

$$\eta_t^{\psi} = \mathbb{E}_t \Lambda_{t,t+1}^h \left[r_t^f \Pi_{t+1}^{-1} \right] - 1 + \frac{\partial \varphi_t}{\partial u_t^f} \frac{1}{\ell_t^{f,b}}$$

- FOC for labor demand $h_t(j)$:

$$w_t h_t = (1 - \alpha) m c_t^{kl} y_t$$

- FOC for investment $I_t(j)$:

$$q_t = 1 + \eta_t^\psi \psi_t^L + \phi_I \left(\frac{I_t}{I_{t-1}} - 1 \right) \frac{I_t}{I_{t-1}} - \mathbb{E}_t \Lambda_{t,t+1}^h \left(\phi_I \left(\frac{I_{t+1}}{I_t} - 1 \right) \left(\frac{I_{t+1}}{I_t} \right)^2 \right)$$

- FOC for owned capital $K_{t+1}(j)$:

$$q_t = \mathbb{E}_t \Lambda_{t,t+1}^h \left[\alpha (1 - \zeta) m c_{t+1}^{kl} \frac{y_{t+1}}{K_{t+1}} + q_{t+1} (1 - \delta) \right]$$

- FOC for rented capital from startups $K_{t+1}^n(j)$:

$$r_{t+1}^{k,n} = \alpha \zeta m c_{t+1}^{kl} \frac{y_{t+1}}{K_{t+1}^n}$$

D.7 Capital Good Producers

List of variables:

Q_t^n = nominal price of capital.

I_t^n = investment.

$G_{I,t}^n$ = investment adjustment cost.

q_t^n = Tobin's Q.

D.7.1 Optimization Problem

- Nominal profit of capital good producers:

$$\Pi_t^K = Q_t^n I_t^n - P_t I_t^n - P_t G_{I,t}^n$$

- Christiano et al. (2005) Investment adjustment cost:

$$G_{I,t}^n = \frac{\phi_I^n}{2} I_t^n \left(\frac{I_t^n}{I_{t-1}^n} - 1 \right)^2$$

- Capital accumulation formula:

$$K_{t+1}^n = (1 - \delta_n) K_t^n + I_t^n$$

D.7.2 Lagrangean

$$\max_{I_t^n} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_h^t \lambda_t^h (q_t^n I_t^n - I_t^n - G_{I,t}^n)$$

D.7.3 First-Order Conditions

$$\begin{aligned} q_t^n = & 1 + \frac{\phi_I^n}{2} \left(\frac{I_t^n}{I_{t-1}^n} - 1 \right)^2 + \phi_I^n \left(\frac{I_t^n}{I_{t-1}^n} - 1 \right) \frac{I_t^n}{I_{t-1}^n} \\ & - \mathbb{E}_t \Lambda_{t,t+1}^h \left(\phi_I^n \left(\frac{I_{t+1}^n}{I_t^n} - 1 \right) \left(\frac{I_{t+1}^n}{I_t^n} \right)^2 \right) \end{aligned}$$

D.8 Startups

- The startups are identical to “entrepreneurs” in Gasparini et al. (2024) who are exposed to idiosyncratic return shock.
- A crucial difference is that startups borrow from shadow bank instead of traditional banks.
- Also, the capital rented from startups is additional inputs for intermediate good producers.

D.8.1 Optimization Problem

- Net worth of startup j , $n_t^n(j)$:

$$n_t^n(j) = q_t^n K_{t+1}^n(j) - b_t^n(j)$$

- Rate of return on capital R_t^n :

$$R_t^n = \frac{r_t^{k,n} + (1 - \delta^n) q_t^n}{q_{t-1}^n} \Pi_t$$

- Wealth of startup j , $\mathcal{W}_{t+1}^n(j)$:

$$P_{t+1} \mathcal{W}_{t+1}^n(j) = (1 - \Gamma_{t+1}^n) R_{t+1}^n P_t q_t^n K_{t+1}^n(j)$$

where Γ_{t+1}^n is the fraction of return that goes to shadow bank as a debt payment.

- The relationship between wealth and net worth in aggregate.
 - A portion of startups exit and bring their wealth $\mathcal{W}_{t+1}^n(j)$ to the household. The aggregate amount of their wealth is equivalent to a portion of total startups' wealth, which is $\chi_n \mathcal{W}_{t+1}^n$.
 - The household transfers a fraction of ι of exiting startups' wealth to new startups.
 - Then the aggregate net worth of startups, n_{t+1}^n :

$$n_{t+1}^n = (1 - \chi_n + \iota \chi_n) \mathcal{W}_{t+1}^n$$

- The capital-to-net worth ratio of startups

$$\rho_t = \frac{q_t^n K_t^n}{n_t^n};$$

- Idiosyncratic return shock process, $\omega_{t+1}^n(j)$:

$$\omega_{t+1}^n(j) \stackrel{i.i.d.}{\sim} \text{lognormal}(1, \sigma_t^n)$$

where

$$\sigma_t^n = \bar{\sigma}^n S_t^n$$

$$\log S_t^n = \rho_n \log S_{t-1}^n + u_t^n$$

- Financial contract with shadow bank.
 - A startup j borrows $b_t^n(j)$ from the shadow bank.
 - The startup j faces the idiosyncratic return shock $\omega_{t+1}^n(j)$ at $t + 1$.
 - The startup j 's payment to the shadow bank depends on the realization of $\omega_{t+1}^n(j)$.
 - For a threshold $\bar{\omega}_{t+1}^n(j)$, a startup j with $\omega_{t+1}^n(j) \geq \bar{\omega}_{t+1}^n(j)$ will pay $\bar{\omega}_{t+1}^n(j) R_{t+1}^n q_t^n K_{t+1}^n(j)$.
 - A startup j with $\omega_{t+1}^n(j) < \bar{\omega}_{t+1}^n(j)$ will default and will pay $\omega_{t+1}^n(j) R_{t+1}^n q_t^n K_{t+1}^n(j)$
 - To receive the payment from default loans, the shadow bank should pay the real monitoring cost, $m_t^{sb} = \mu^n \omega_t^n(j) R_t^n q_{t-1}^n K_t^n(j)$, which is the fraction μ^n of the return from default startups.

- Given the contract, the payment fraction Γ_{t+1}^n is determined as follows:

$$\begin{aligned}
\Gamma_{t+1}^n &= \Gamma^n(\bar{\omega}^n(j)) \\
&= \int_0^{\bar{\omega}_{t+1}^n(j)} \omega_{t+1}^n(j) f^n(\omega_{t+1}^n(j)) d\omega_{t+1}^n(j) \\
&\quad + \int_{\bar{\omega}_{t+1}^n(j)}^{\infty} \bar{\omega}_{t+1}^n(j) f^n(\omega_{t+1}^n(j)) d\omega_{t+1}^n(j) \\
&= \int_0^{\bar{\omega}_{t+1}^n(j)} \omega_{t+1}^n(j) f^n(\omega_{t+1}^n(j)) d\omega_{t+1}^n(j) + [1 - F^n(\bar{\omega}_{t+1}^n(j))] \bar{\omega}_{t+1}^n(j)
\end{aligned}$$

where the default rate F_{t+1}^n is defined as:

$$F_{t+1}^n = F^n(\bar{\omega}_{t+1}^n(j)) = \int_0^{\bar{\omega}_{t+1}^n(j)} f^n(\omega_{t+1}^n(j)) d\omega_{t+1}^n(j)$$

- Participation constraint of shadow bank.
 - Considering the chance of default and monitoring costs accompanied with lending to startups, the shadow bank requires the expected return to be the same as the return, R_{t+1}^{fn} , from the same amount of riskless loans to fictitious firms.

$$\mathbb{E}_t[(\Gamma_{t+1}^n - \mu^n G_{t+1}^n) R_{t+1}^n q_t^n K_{t+1}^n(j)] \geq \mathbb{E}_t R_{t+1}^{fn} b_t^n$$

where G_{t+1}^n is the expectation of the shock process $\omega_{t+1}^n(j)$ of default startups:

$$G_{t+1}^n = G_{t+1}^n(\bar{\omega}^n(j)) \equiv \int_0^{\bar{\omega}_{t+1}^n(j)} \omega_{t+1}^n(j) f^n(\omega_{t+1}^n(j)) d\omega_{t+1}^n(j)$$

- In sum, the startup j maximizes its wealth $\mathcal{W}_{t+1}^n(j)$ subject to the participation constraint of the shadow bank.
 - Consider the contractual repayment rate, $Z_{t+1}^n(j)$. For the startups who do not default,

the following should hold:

$$\bar{\omega}_{t+1}^n(j) R_{t+1}^n q_t^n K_{t+1}^n(j) = Z_{t+1}^n(j) b_t^n(j)$$

- Define the loan-to-value ratio $x_t^n(j)$ as follows:

$$x_t^n(j) \equiv \frac{Z_t^n(j) b_t^n(j)}{q_t^n K_{t+1}^n(j)}$$

- From the contractual obligation identity, we have:

$$\bar{\omega}_{t+1}^n(j) = \frac{x_t^n(j)}{R_{t+1}^n}$$

- In the optimization problem, a startup j chooses $K_{t+1}^n(j)$ and, instead of $b_t^n(j)$, it chooses $x_t^n(j)$ to maximize its wealth.

D.8.2 Lagrangean

$$\begin{aligned} \max_{x_t^n(j), K_{t+1}^n(j)} \mathbb{E}_t \Lambda_{t,t+1}^h \Bigg\{ & \left[1 - \Gamma^n \left(\frac{x_t^n(j)}{R_{t+1}^n} \right) \right] R_{t+1}^n q_t^n K_{t+1}^n(j) \\ & + \xi_t^n(j) \left[\left(\Gamma^n \left(\frac{x_t^n(j)}{R_{t+1}^n} \right) - \mu^n G^n \left(\frac{x_t^n(j)}{R_{t+1}^n} \right) \right) R_{t+1}^n q_t^n K_{t+1}^n(j) \right. \\ & \left. - R_{t+1}^{fn} \left(q_t^n \int_0^1 K_{t+1}^n(j) dj - n_t^n \right) \right] \Bigg\} \end{aligned}$$

D.8.3 First-Order Conditions

- I drop the index j in the following FOCs.

- FOC for loan-to-value ratio $x_t^n(j)$:

$$\mathbb{E}_t \Lambda_{t,t+1}^h \left\{ \left[-\Gamma_{t+1}' + \xi_t^n \left(\Gamma_{t+1}' - \mu^n G_{t+1}' \right) \right] \right\} = 0$$

- FOC for capital demand $K_{t+1}^n(j)$:

$$\mathbb{E}_t \Lambda_{t,t+1}^h \left\{ \left[1 - \Gamma_{t+1}^n \right] R_{t+1}^n + \xi_t^n \left[\Gamma_{t+1}^n - \mu^n G_{t+1}^n R_{t+1}^n - R_{t+1}^{fn} \right] \right\} = 0$$

D.9 Central Bank

D.9.1 Monetary Policy

$$R_t = \bar{R} \left(\frac{\Pi_t}{\bar{\Pi}} \right)^{m_\pi} \left(\frac{y_t}{\bar{y}} \right)^{m_y} S_t^i$$

where

$$\log(S_t^i) = \rho_i \log(S_{t-1}^i) + u_t^i$$

D.9.2 Asset Purchases Policy

- The central bank supplies reserves to the wholesale bank and lends to intermediate good firms' debt:

$$\tilde{\ell}_t^{f,cb} = re_t$$

- The central bank transfers its operating profit to the household:

$$\Pi_t T_t^{cb} = r_{t-1}^f \tilde{\ell}_{t-1}^{f,cb} - R_{t-1} re_{t-1}$$

- The reserve policy responds to deviations in inflation and credit line limit from their steady

states respectively:

$$re_t = \bar{r}e \left(\frac{\Pi_t}{\bar{\Pi}} \right)^{-q\pi} \left(\frac{\mathbb{E}_t \ell_{t+1}^{f,b}}{\bar{\ell}^{f,b}} \right)^{-q\ell} S_t^q$$

where

$$\log(S_t^q) = \rho_q \log(S_{t-1}^q) + u_t^q$$

D.9.3 Macroprudential Policies

- Macroprudential policy for banker's leverage ratio:

$$\begin{aligned} v_t^{bhc} = \\ (1 - \rho_{bhc}) \bar{v}^{bhc} + (1 - \rho_{bhc}) \left[\chi_{bhc} \left(\frac{e_t + b_t^b}{y_t} - \frac{\bar{e} + \bar{b}^b}{\bar{y}} \right) \right] + \rho_{bhc} v_{t-1}^{bhc} + u_t^{v,bhc} \end{aligned}$$

- Angelini et al. (2014) Macroprudential policy for wholesale bank's capital-asset ratio:

$$\begin{aligned} v_t^k = \\ (1 - \rho_k) \bar{v}^k + (1 - \rho_k) \left[\chi_k \left(\frac{\tilde{\ell}_t^{f,b} + b_t^{sb}}{y_t} - \frac{\tilde{\ell}^{f,b} + \bar{b}^{sb}}{\bar{y}} \right) \right] + \rho_k v_{t-1}^k + u_t^{v,k} \end{aligned}$$

D.10 Resource Constraints and GDP

$$\begin{aligned} y_t = c_t^b + c_t^h + I_t + I_t^n + d_t^h \Phi(s_t) + G_{P,t} + G_{I,t} + G_{I,t}^n + \varphi_t \\ + \left(\mathcal{C}_{t-1}^{bhc} + \mathcal{C}_{t-1}^{bhc,b} + \mathcal{C}_{t-1}^b + \mathcal{C}_{t-1}^{sb,n} + m_t^{sb} \right) \Pi_t^{-1} \end{aligned}$$

Appendix E. Methodology

E.1 Data Sources

- The following data were retrieved from the Federal Reserve Economic Data (FRED) database:
 - Z.1 Financial Accounts of the United States of Federal Reserve Board of Board from Governors of the Federal Reserve System.
 - Personal consumption expenditures (implicit price deflator) from U.S. Bureau of Economic Analysis.
 - National rates from Federal Deposit Insurance Corporation.
 - Treasury bill rates of U.S. from Board of Governors of the Federal Reserve System.
- The Wu-Xia shadow rates are obtained from Federal Reserve Bank of Atlanta.
- The P-Fin index is obtained from the Teachers Insurance and Annuity Association of America (TIAA).
- The following data is retrieved from Statista.
 - Stock market participation rate from Gallup.
 - Number of active mobile banking customers from JPMorgan Chase and Bank of America's annual reports.

E.2 Software

I used Matlab 2025a and Dynare 7.0 for solving and estimating the model. To eliminate the seasonality in the data for estimation, I used the X-13ARIMA-SEATS Seasonal Adjustment Program Ver. 1.1 provided by the U.S. Census Bureau. The Matlab tool box for the X-13ARIMA-SEATS program was developed by Lengwiler (2026).

E.3 Solution Methods

I solved the model using the first-order perturbation method around the non-stochastic steady state. I compare the impulse response functions of variables to the household financial distress shock from the first-order perturbation method with those from the second-order perturbation method. For the latter, the pruning algorithm provided by Kim et al. (2008) was employed, and the IRFs are expressed as deviations from the stochastic steady state. Figure 22 summarizes the comparison.

The results from the second-order perturbation method suggests the nonlinear amplification effects of the household financial distress shock through various remuneration costs in the model. The explosive responses of remuneration costs such as capital adjustment cost are strong enough to change the credit supply response to the shock. However, since the estimates of the model parameters are based on the Kalman filter with the first-order solution method, the responses from the second order perturbation method should be interpreted with caution. The non-linear estimation method, such as the particle filter estimation, would suggest different parameter estimates, which may lead to different responses.

In this paper, I will leave this result as a caveat and use the first-order perturbation method for the main analysis in the paper for computational efficiency. The investigation of the nonlinear amplification effects of the household financial distress shock is left for future research.

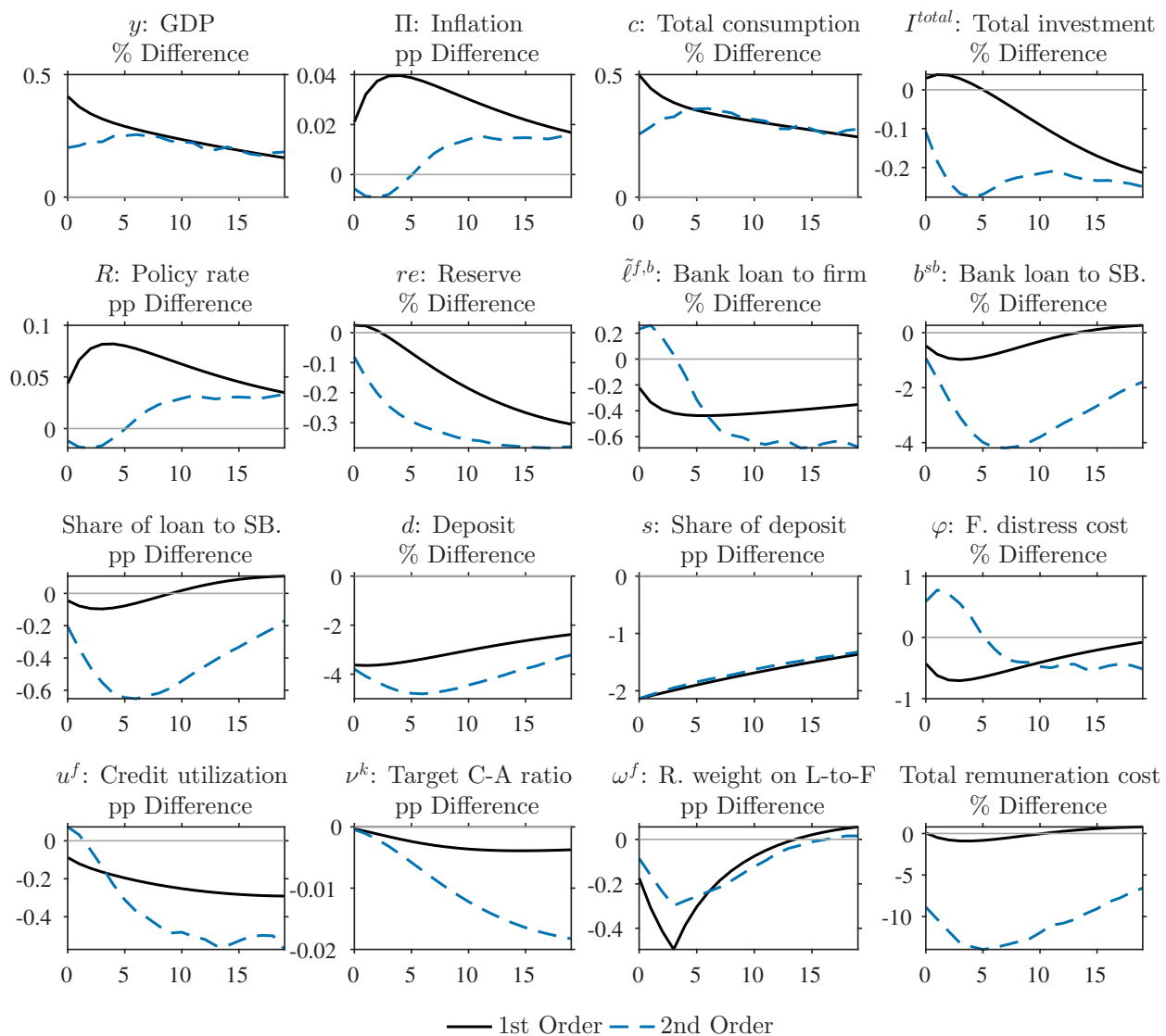


Figure 22: Financial distress shock 1

Appendix F. Calibration of Parameters

The calibration of fundamental parameters from the literature is summarized in Table 11. The prior distributions, posterior estimates, and details of the Bayesian procedure for the remaining parameters are reported in Appendix C.2.

Table 11: Calibration Parameters

Parameter	Value	Target	Description	Source
Households				
β_h	0.995	SS retail deposit rate, $\bar{r}^d = 1 + 0.02/4$	Time preference	KW (2021)
γ_h	0.5		Intertemporal elasticity of consumption	KW (2021)
χ_h	5.051	SS Labor supply, $\bar{h}^h = 1$	Coefficient of labor disutility	KW (2021)
ζ_h	1		Elasticity of labor supply	KW (2021)
$\bar{s}$	0.696		Steady state share of deposits	FRED, Z.1
$\bar{\vartheta}$	0.696	Appendix B.1	Elasticity of household financial distress cost	
χ_1	0.0007	$\bar{R}^b - \bar{r}^d = 0.0015$	Coefficient of household financial distress cost	
Banker				
β_b	0.993	SS policy rate, $\bar{R} = 1 + 0.03/4$	Time preference	
$\kappa_{bhc,b}$	0.001	Annual net interest= 0.004%	Coefficient of borrowing adjustment cost	
$\theta_{bhc,b}$	2		Elasticity of borrowing adjustment cost	KW (2021)
Wholesale Bank				
v_b	0.103	Tier 1 & Total capital ratio (6%, 10.5%)	Fixed dividend rate	

Continued on next page

Table 11 (*continued*)

Parameter	Value	Target	Description	Source
ρ_e	0.071	Appendix B.2	Weight for equity investment in capital	Basel III
$\bar{\omega}^f$	1.2		SS risk weight of loans to firms in asset	Basel III
$\bar{\omega}^{sb}$	1.2		SS risk weight of loans to shadow bank in as-set	Basel III
χ_f	-3.384		Risk weight adjustment of loans to firms	AENPQ (2010)
ρ_f	0.874		Persistence of risk weight adjustment of loans to firms	AENPQ (2010)
χ_{sb}	-2.256		Risk weight adjustment of loans to shadow bank	
Retail Bank				
$\varepsilon_{b,f}$	2.714	Annual C&I loan spread = 175bp	SS markup for firm lending	Pandolfo (2025)
$\varepsilon_{b,sb}$	2.563	Annual Private Debt Funds spread = 192bp	SS markup for SB lending	BCLZ (2025)
$\varepsilon_{b,d}$	-2.030	SS retail deposit rate, $\bar{r}^d = 1 + 0.02/4$	SS markdown for HH deposit	
Final & Intermediate Good Producer				
δ	0.025		Capital depreciation rate	GNSS (2010)
θ	11	Markup in the SS, $\mu = \theta / (\theta - 1) = 1.1$	Final good aggregation technology	KW (2021)

Continued on next page

Table 11 (continued)

Parameter	Value	Target	Description	Source
α	0.333		Share of capital in intermediate good production	KW (2021)
ζ	0.20	Total investment-to-output in SS= 0.198	Share of startups' capital in total capital	
$\bar{u}^f$	0.4		SS utilization rate of credit line	GKP (2020)
ϕ_p	200		Degree of stickiness in inflation	KW (2007)
ϕ_I	2.5		Degree of stickiness in investment	KW (2021)
θ_π	0.5		Responsiveness of committed credit to EBITDA gap	
θ_u	2		Firm curvature of financial distress cost	
$\bar{\psi}^L$	0.7		SS multiplier of debt-investment ratio	LR (2024)
Capital Good Producer & Startups				
δ_h	0.0375		Capital depreciation rate for startups	LH (2020)
χ_n	0.0245	SS capital-to-net worth ratio of startup= 1.1	Startups exit rate	GLMV (2024)
ι	0.002		Startups portion of starting funds	GK (2011)
ϕ_I''	2.5		Degree of stickiness in investment	KW (2021)

Continued on next page

Table 11 (continued)

Parameter	Value	Target	Description	Source
$\bar{R}^n$	1.025	Share of monitoring cost= 0.125	SS startup return	GLMV (2024)
$\bar{\sigma}^n$	1.033	SS startups default rate $\bar{F} = 0.03$	SS dispersion in the dist. of return shock	GKP (2020)
Shadow Banking				
$\kappa_{sb,n}$	0.010	Annual spread of private credit loan= 600bp	SB coeff. of borrowing adj. cost	Pandolfo (2025)
$\theta_{sb,n}$	2		Elasticity of borrowing adjustment cost	KW (2021)
Monetary Policy				
m_y	0		Strength of MP response to output gap	SWZ (2023)
q_π	0		Strength of AP response to inflation gap	SWZ (2023)
Macroprudential Policy				
χ_{bhc}	0		Cyclicality of macroprudential policy for banker	
$\bar{v}^k$	0.105		SS target capital-asset ratio for wholesale bank	Basel III
$\bar{v}^{bhc}$	1.2		SS target leverage ratio for banker	RF (2020)
Steady State				

Continued on next page

Table 11 (*continued*)

Parameter	Value	Target	Description	Source
$\bar{\Pi}$	1		SS inflation rate	

Note: SS refers to steady state.

Table 12: Sources and Abbreviations

Abbreviation	Full Reference
AENPQ (2010)	Angelini et al. (2010)
ANP (2014)	Angelini et al. (2014)
BCLZ (2025)	Berrospide et al. (2025)
FRED	Federal Reserve Economic Data
GK (2011)	Gertler and Karadi (2011)
GK (2018)	Gambacorta and Karmakar (2018)
GKP (2020)	Greenwald et al. (2020)
GLMV (2024)	Gasparini et al. (2024)
GNSS (2010)	Gerali et al. (2010)
KW (2007)	Keen and Wang (2007)
KW (2021)	Kumhof and Wang (2021)
LH (2020)	Li and Hall (2020)
LR (2024)	Lubello and Rouabah (2024)
RF (2020)	Federal Reserve Bank of Richmond (2020)
SW (2007)	Smets and Wouters (2007)
SWZ (2023)	Sims et al. (2023)
Z.1	Financial Accounts of the United States - Z.1

F.1 Data Targets for Interest Rates and Returns

This appendix provides the data justification for the steady-state interest rates and returns summarized in Table 1 and documents the remaining calibration choices.

First, the household's discount factor (β^h) is set to 0.995. This implies a steady state annual retail deposit interest rate ($\bar{r}^d$) of 2%. As the weighted-average interest rates on 3-month Certificates of Deposit (CDs) were around 1.68% from 2024 to 2025, the calibration is within a reasonable range. The annual policy interest rate target ($\bar{R}$) is set to 3%, which is greater than the steady state retail deposit interest rate. This figure is supported by the median of the longer-run FOMC Summary of Economic Projections for the Fed Funds Rate from 2019 to 2025, which is between 2.4% to 3%.³² To calibrate the borrowing interest rate of the banker from the household in steady state ($\bar{R}^b$), I assume that the steady state spread of the interest rate on the banker's bond is 0.4% less in

³²The interest on 3-month CD (NDR3MCD) and the longer run Fed Funds Rate (FEDTARMLDR) are retrieved from the FRED database.

annual terms compared to the policy rate.³³

The steady state spreads on loans target the spreads between the lending rates and the policy rate from the data. The spread on retail loans to intermediate good producers is calibrated to match the average annual C&I loan spread of 1.75% calculated by Pandolfo (2025). Next, the spread on retail loans to the shadow bank is set to 1.92% in annual terms, based on the average annual spread on private debt funds provided by Berrospide et al. (2025).³⁴ The steady state spread on loans from shadow banks is calibrated at 6% in annual terms as calculated by Pandolfo (2025).

Finally, the steady state spread on the return on capital for startups is set to 10% in annual terms. I consider return on equity as the proxy for the return on capital, and target a reasonable range using the data from Damodaran (2025). From 2016 to 2025, the U.S. average market return on equity adjusted for R&D while excluding the financial sector was between 6.8% to 15.3% in annual terms.³⁵ I select a relatively conservative figure of 10% to be consistent with the calibration of parameters in the startup sector in Gasparini et al. (2024). The share of monitoring cost ($\mu_n = 0.125$) targets the range between 10% and 15% and the exit rate of startups ($\chi_n = 0.0245$) targets the range between 1% and 5%.³⁶

The remaining structural parameters are set as follows. First, the steady state share of deposits in total funding is calculated to be 0.696 based on the data from the Z.1 Federal Reserve's Financial

³³This is a rather strong assumption since the model does not allow the spread to be positive. Historically, the yield of the banker's bond is usually higher than the policy rate. From 1954:Q3 to 2026:Q1, the average spread of the yield on the Moody's Aaa corporate bond over the federal funds rate is around 2% in annual terms. However, a negative spread is not uncommon in the recent period of high policy rates. From 2023:Q2 to 2024:Q3, the spreads were negative and the average spread was around -0.29% in annual terms. The data are retrieved from the FRED database.

³⁴Following Berrospide et al. (2025), the average interest rate on bank loans to private debt funds in 2024:Q3 was 6.6%. The spread is calculated using the federal funds rate, which was 4.68% in the same period.

³⁵R&D expenditures are treated as capital instead of expenses.

³⁶One caveat is the calibration of steady-state debt to capital ratio for startups. The ratio is calibrated to be less than 5% (0.0455), which indicates excellent financial stability. Although the model does not allow the ratio to exceed 50%, the calibrated value is still low compared with the real economy. While other parameters of the economy lie within a reasonable range, a realistic calibration of the debt-to-capital ratio for startups remains challenging.

Accounts of the United States. Second, the steady state utilization rate ($\bar{u}^f$) is set to 0.4 parsimoniously from Greenwald et al. (2020).³⁷ Third, the share of rented capital from startups in total capital (ζ) is set to 0.20. Given the depreciation rate of internal and external capital ($\delta = 0.025$ and $\delta^n = 0.0375$), the calibration targets the total investment-to-GDP ratio with a reasonable range between 15% to 20%.³⁸ Fourth, the curvature parameters of the remuneration cost functions are set to be 2 following the literature.

Two calibration choices warrant caution. The first concerns the risk weight process, equation (16). The main problem is the parameterization of the response parameter for loans to shadow banks (χ_{sb}), which is set to two-thirds of the response parameter for loans to intermediate good producers (χ_f). It is inevitable to make a strong assumption on the parameter given the lack of the data and the difficulty in identifying the parameter. The details of the calibration of the risk weight process are provided in Appendix C.1.

The second concerns the committed credit process, equation (34). I calibrate the value of $\theta_\pi = 0.5$ and obtain the estimates of ρ_ℓ from the model estimation. There are two challenges in the calibration of those parameters. First, it is difficult to obtain reliable aggregate time-series data to estimate the process. Firm-level data that track the committed credit and EBITDA of firms from an identical sample over a long period are hard to obtain. Second, the identification of both parameters is difficult. Since the coefficient on the EBITDA difference in the process is compounded by the parameters, they are weakly identified, which destabilizes the estimation. To circumvent the challenges, I pick the value of $\theta_\pi = 0.5$, where ρ_ℓ is well-identified and the estimation is stable.

³⁷Greenwald et al. (2020) report that the lowest 80% of the firms in their data utilize between 40% and 50% of the committed credit line they have. Their data covers 207,505 distinct taxpayers from 2012:Q3 to 2020:Q4, although not all firms are tracked for the entire period.

³⁸The depreciation rate of internal capital follows Gerali et al. (2010) and the depreciation rate of external capital is set to be 15% in annual terms, which is the conventional assumption in the R&D literature (Li and Hall (2020)).

Appendix G. Decomposition of the Capital-Asset Ratio under the Household Financial Distress Shock

This appendix expands the discussion of the capital-asset (CA) ratio response in Section 5.1. The wholesale bank's CA ratio in equation (16) can be written, in log deviations from the steady state, as

$$\Delta \log CA_t = \Delta \log (X_t^b + \rho_e e_t^b) - \Delta \log RWA_t, \quad RWA_t \equiv \rho_{re} re_t + \omega_t^f \tilde{\ell}_t^{f,b} + \omega_t^{sb} b_t^{sb}.$$

Three forces operate simultaneously after the shock. (i) The *equity-inflow* channel raises the numerator: $X_t^b + \rho_e e_t^b$ rises by 2.86% at impact and by 2.77% at horizon 5. (ii) The *deleveraging* channel shrinks total assets by 0.14% at impact and 0.33% at horizon 5. (iii) The *risk-density* channel changes the average risk weight RWA_t/A_t . Through the first five quarters, reserves contract less than loans and even edge up by 0.02% on impact while loans fall, so the share of low-risk reserves in assets rises and the density falls by 0.18% at impact and 0.34% at horizon 5. The endogenous decline in ω_t^f and ω_t^{sb} from the output expansion contributes an additional 0.07–0.16% to the density compression. All three forces therefore push the actual ratio in the same direction at short horizons, and the 0.37 pp widening of the CA-ratio gap at horizon 5 understates rather than overstates the equity-side improvement. The picture flips later: by horizon 20, the asset purchase rule pulls reserves down by 0.31% while loans return toward the steady state, raising density by 0.11% and partially offsetting the now-decaying equity inflow.

The discrepancy in retail rate responses reported in the main text reflects the estimated risk-weight processes. With $(1 - \rho_f)\chi_f = -0.43$ and $(1 - \rho_{sb})\chi_{sb} = -0.29$, the risk weight on firm loans is more sensitive to output than that on shadow-bank loans. Quantitatively, ω^f falls by 0.17% at impact and reaches a trough of -0.50% at horizon 4, whereas ω^{sb} falls less (-0.12% at impact, trough -0.33% at horizon 4). The retail rate on firm loans therefore rises by 0.053 pp at horizon 4, compared with 0.055 pp for shadow-bank loans. The share of loans to shadow banks relative to

firm loans troughs at -0.10% at horizon 4 and reaches $+0.11\%$ at horizon 20.

The reserve dynamics underlying the density channel deserve a brief comment. The estimated asset purchase rule has $q_\pi = 0$ and a moderate response to committed loans ($q_\ell = 1.158$). On impact, committed nominal credit $\ell_t^{f,b}$ rises only marginally, but it is deflated by the expected inflation increment Π_{t+1} , which dominates the small endogenous response of EBITDA in the committed-credit process. As a result, real reserves rise slightly for the first four quarters. From horizon 5 onward, the EBITDA-driven increase in committed credit becomes large enough to dominate, and the counter-cyclical rule pulls reserves below steady state.

Appendix H. Impulse Response Functions for Financial Distress Shocks

The following figures present the impulse response functions (IRFs) for a negative shock to the household financial distress cost. The variables are selected to supplement the IRFs in the main text and to provide a more comprehensive picture of the transmission mechanism of financial distress shocks.

There are remarks on the presentation of the IRFs. First, the variables are in real terms and minuscule responses are neglected³⁹. Second, the names of the variables are listed at the top of each panel, and the following abbreviations are used in the figures:

- SB.: Shadow bank.
- ST.: Startups.
- inv.: Investment.
- HH.: Households.
- F.: Intermediate good producers.
- invest.: Investment.
- Target C-A ratio: Target capital-asset ratio of wholesale bank.
- Target B-I: Target leverage ratio of banker.

³⁹The whole responses of a variable to a shock are set to zero when the absolute value of its maximum response is less than 10^{-10} .

- HH. F. distress cost: Household financial distress cost.
- R. weight on L-to-F: Risk weight on loan to intermediate good producers.
- R. weight on L-to-SB: Risk weight on loan to shadow bank.
- MC of production: Marginal cost of production.

Finally, posterior mode values of the parameters and the estimated standard deviations of shocks are used for generating the IRFs. A one standard deviation shock is applied to the economy and the standard deviation is set to be the same as the estimated value in Table 7.

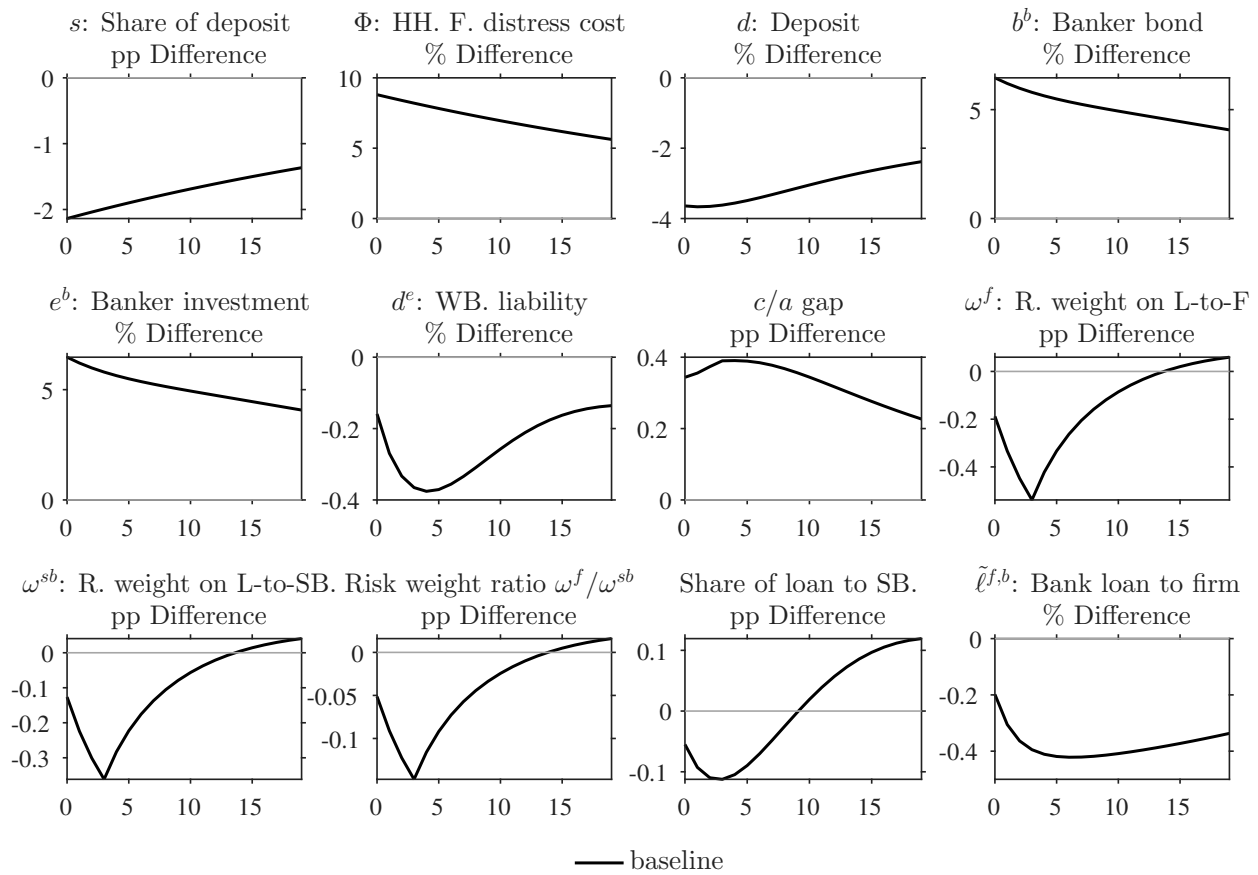


Figure 23: Financial distress shock 1

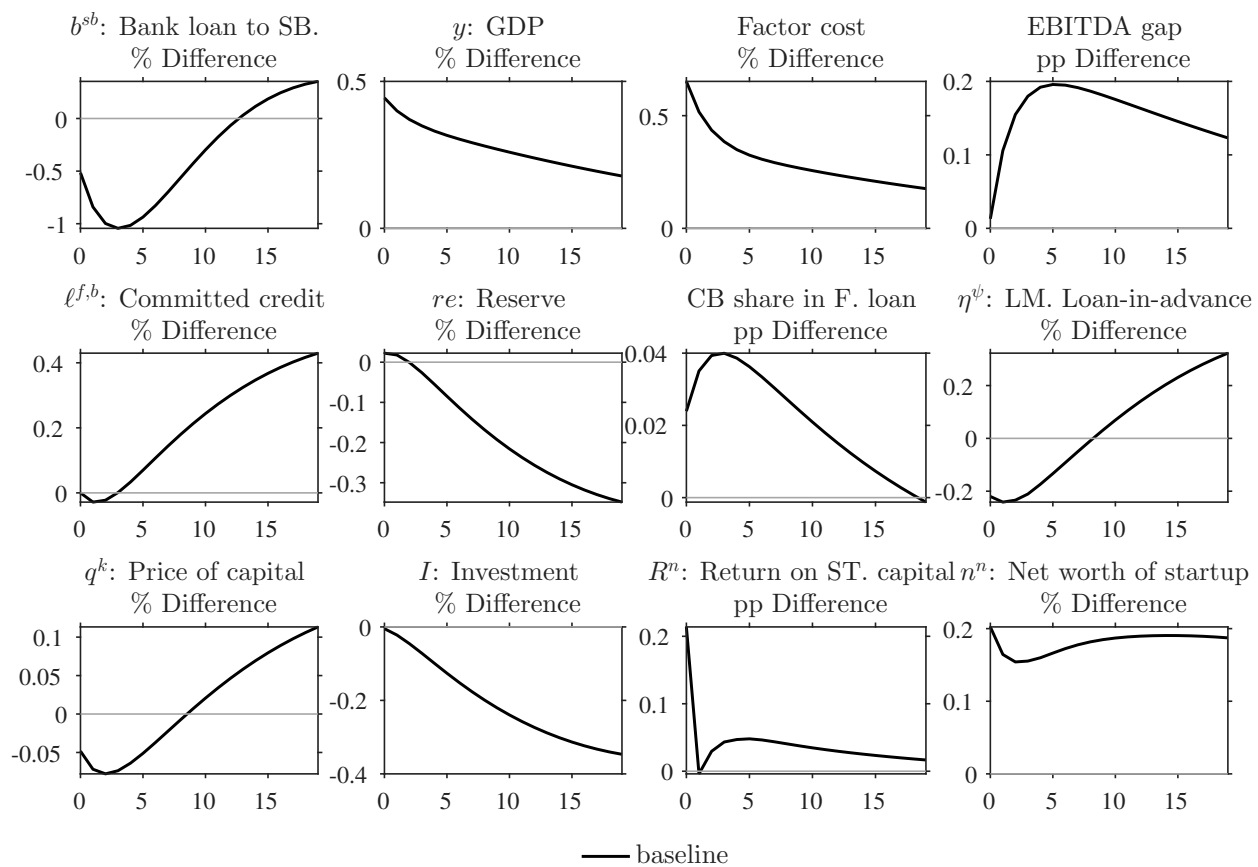


Figure 24: Financial distress shock 2

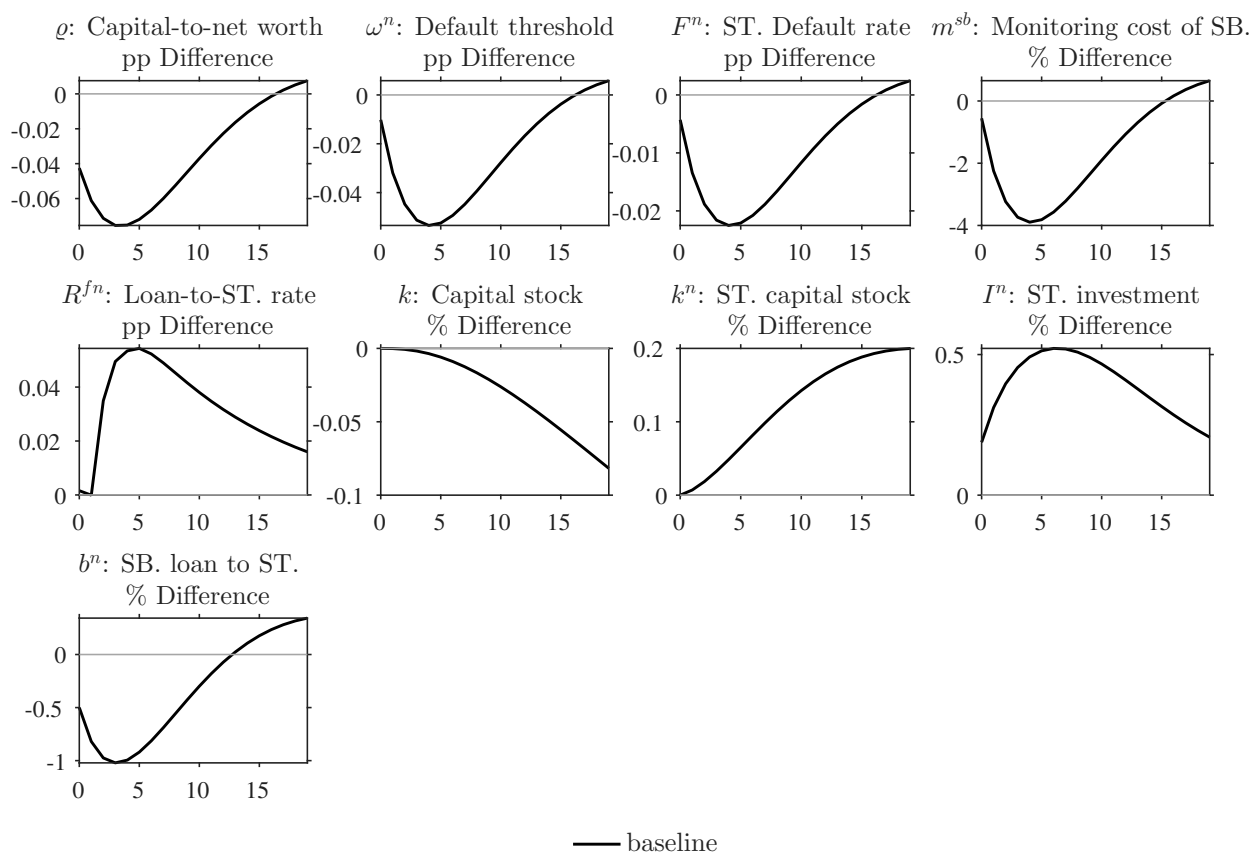


Figure 25: Financial distress shock 3

Appendix I. Impulse Response Functions

The following are the impulse response functions (IRFs) to the rest of the shocks in the model. They encompass the responses of selected macroeconomic variables to technology shocks, monetary policy shocks, asset purchase policy shocks, startups' return shocks, debt-investment shocks, and two macroprudential policy shocks for banker leverage ratio and wholesale bank capital-asset ratio.

I.1 IRFs for Technology Shocks

The following figures present the impulse response functions (IRFs) for a positive shock to the technology.

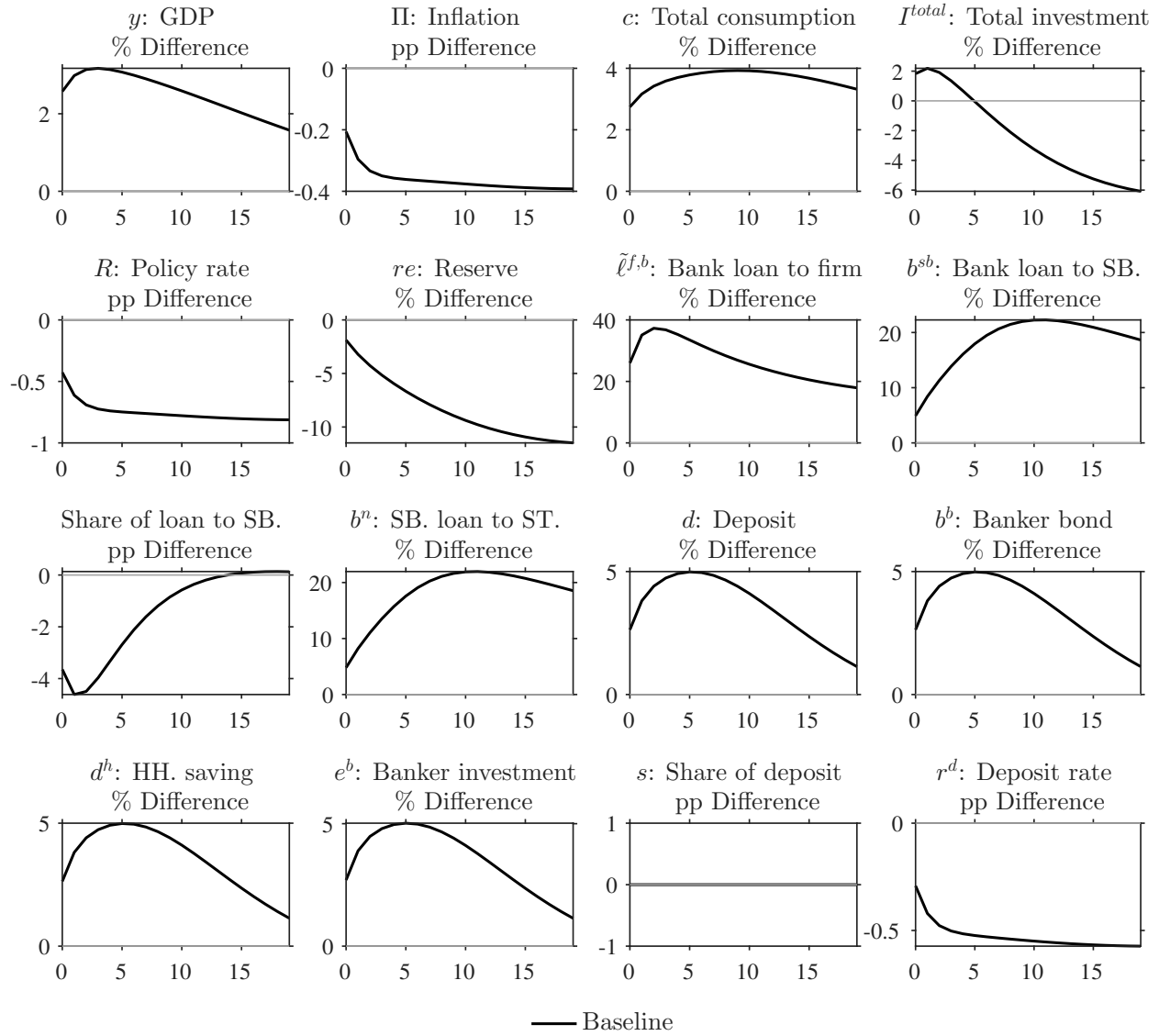


Figure 26: Technology shock 1

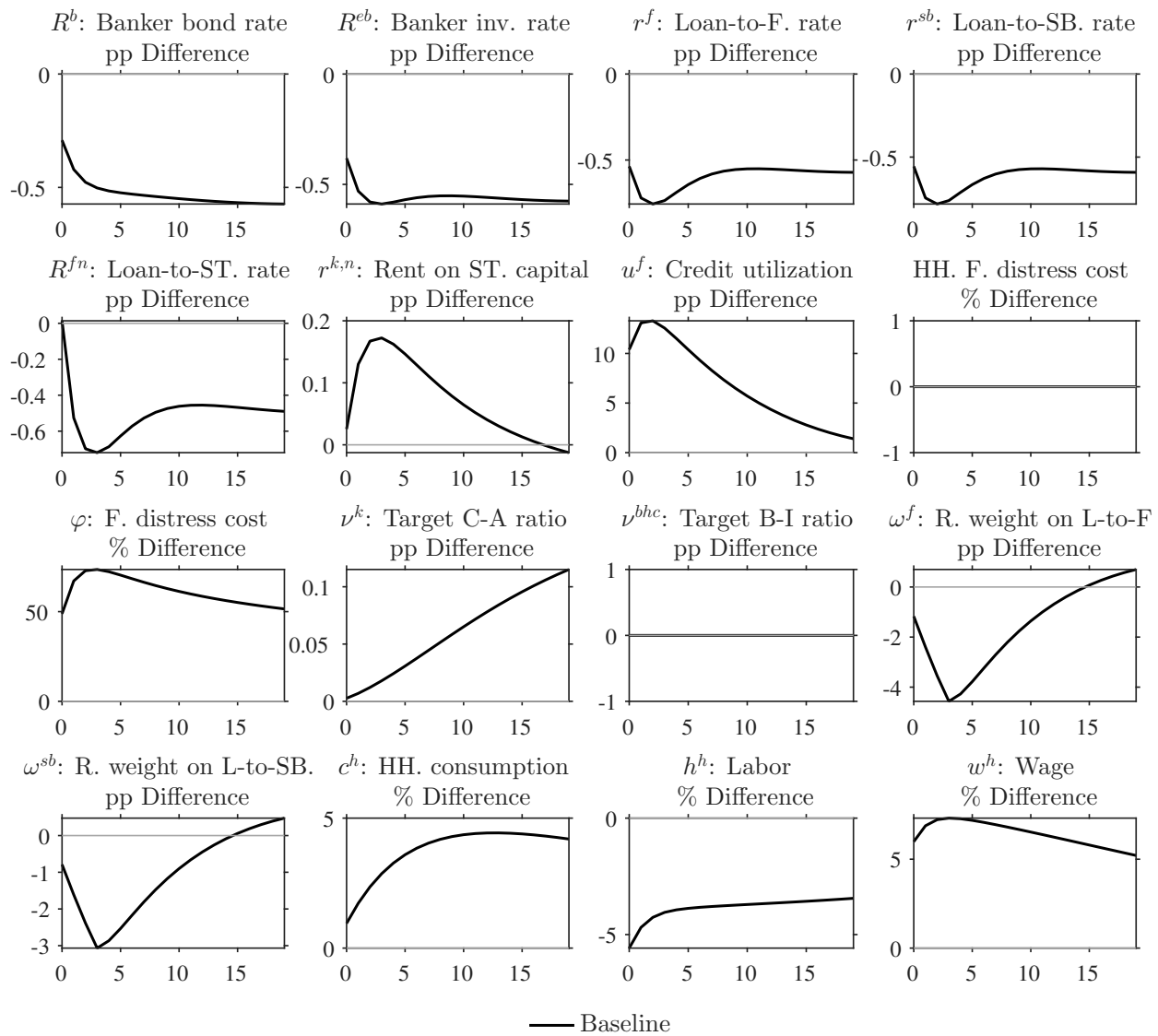


Figure 27: Technology shock 2

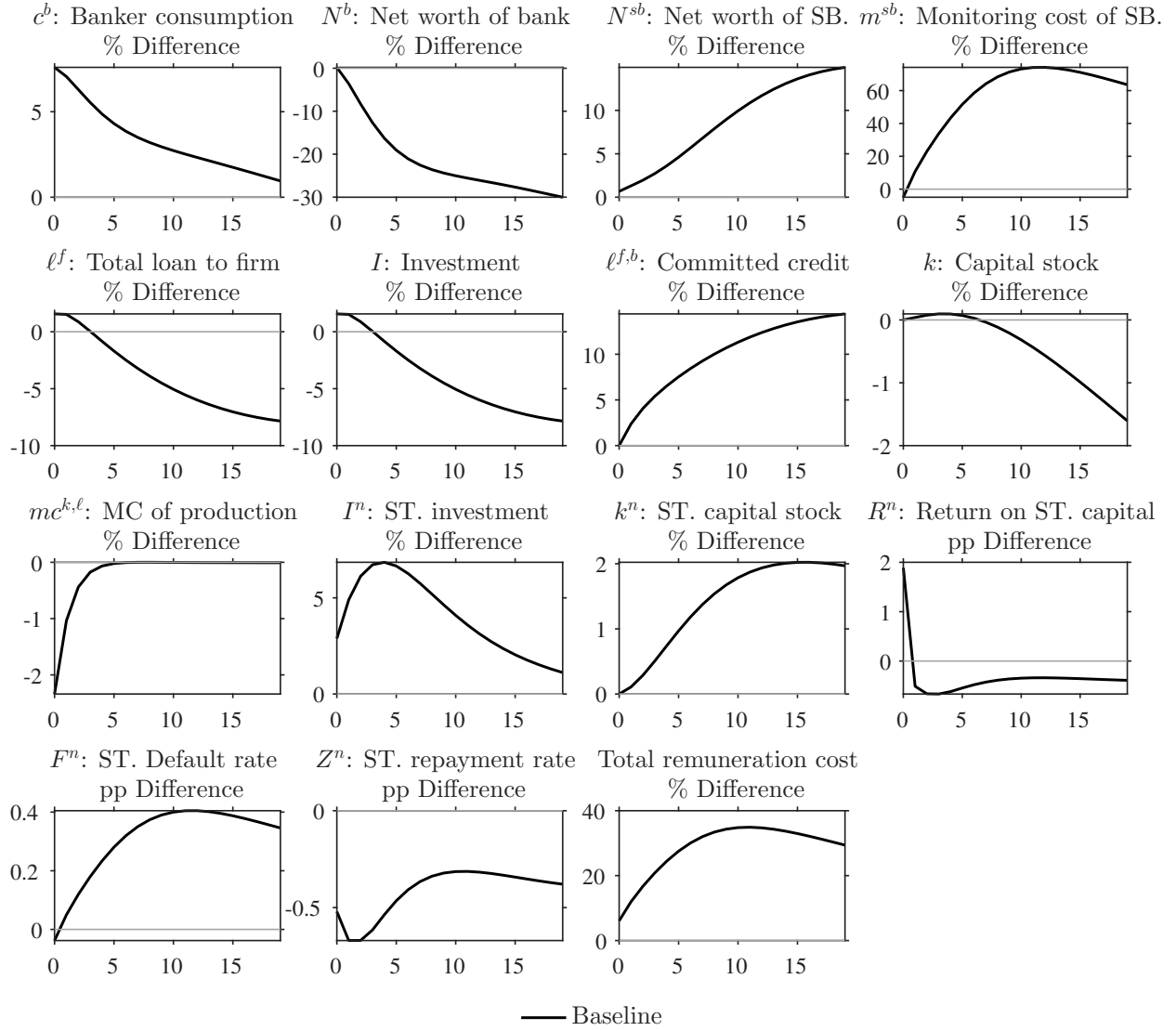


Figure 28: Technology shock 3

I.2 IRFs for Monetary Policy Shocks

The following figures present the impulse response functions (IRFs) for a positive shock to the policy rate.

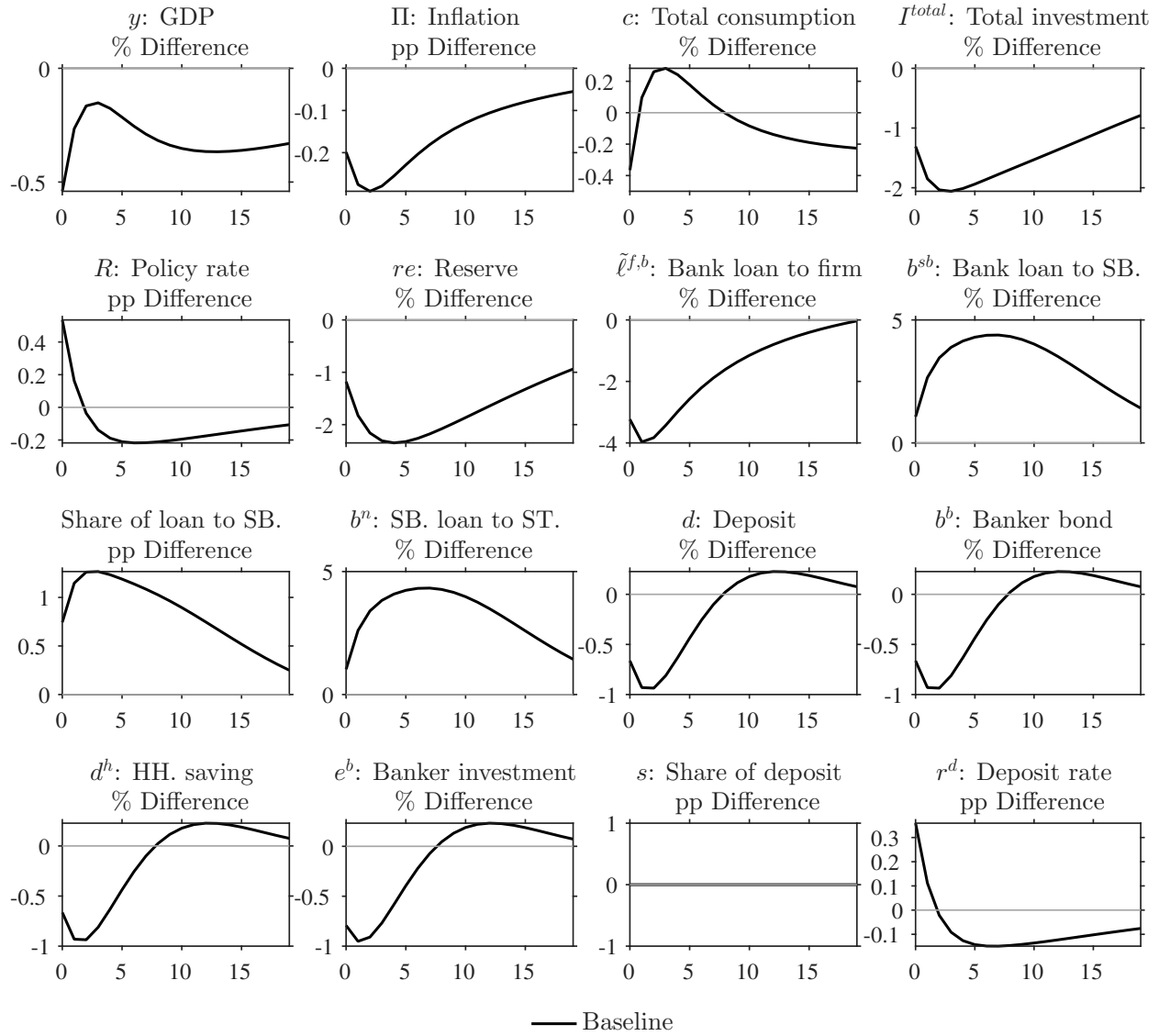


Figure 29: Monetary policy shock 1

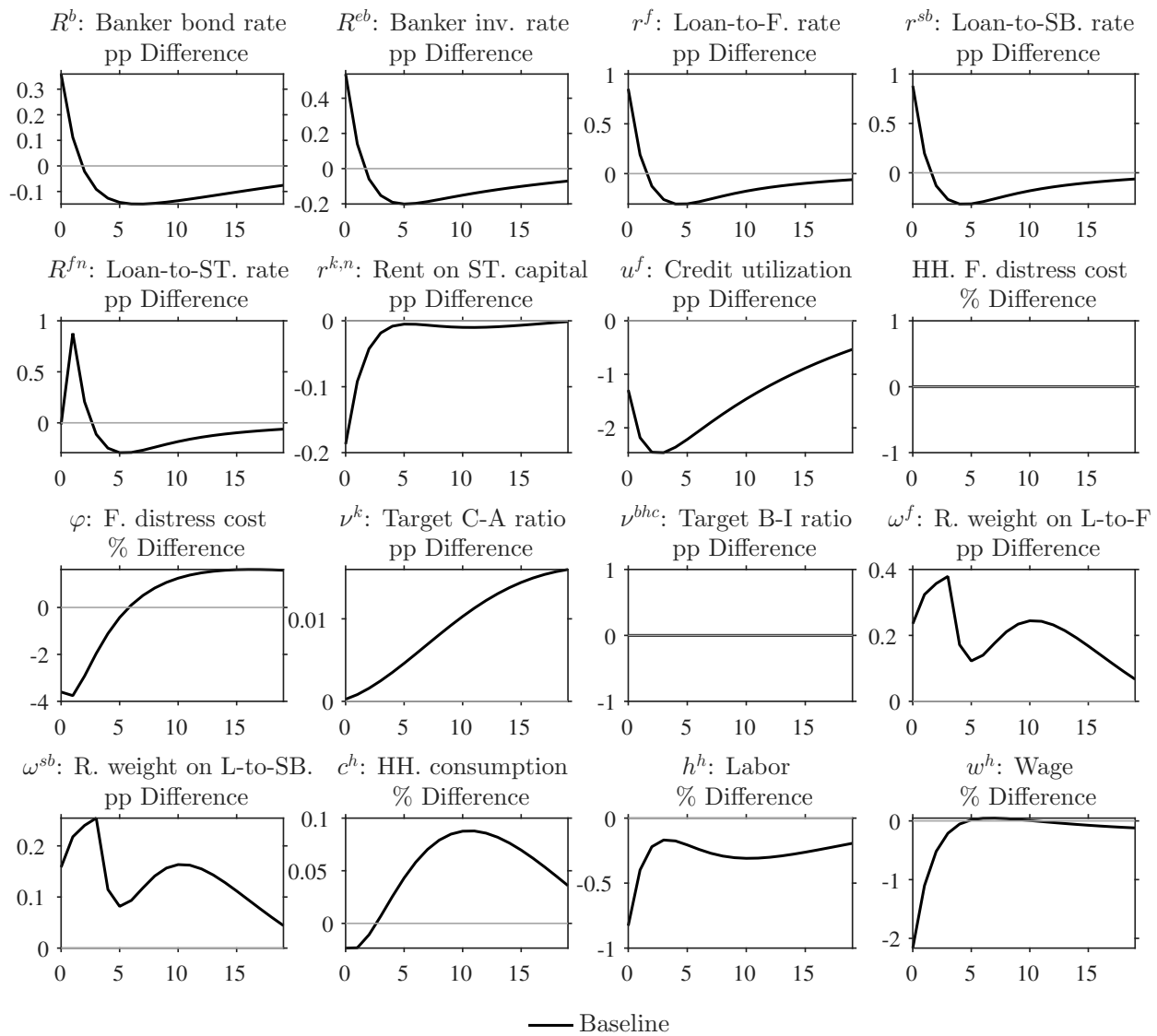


Figure 30: Monetary policy shock 2

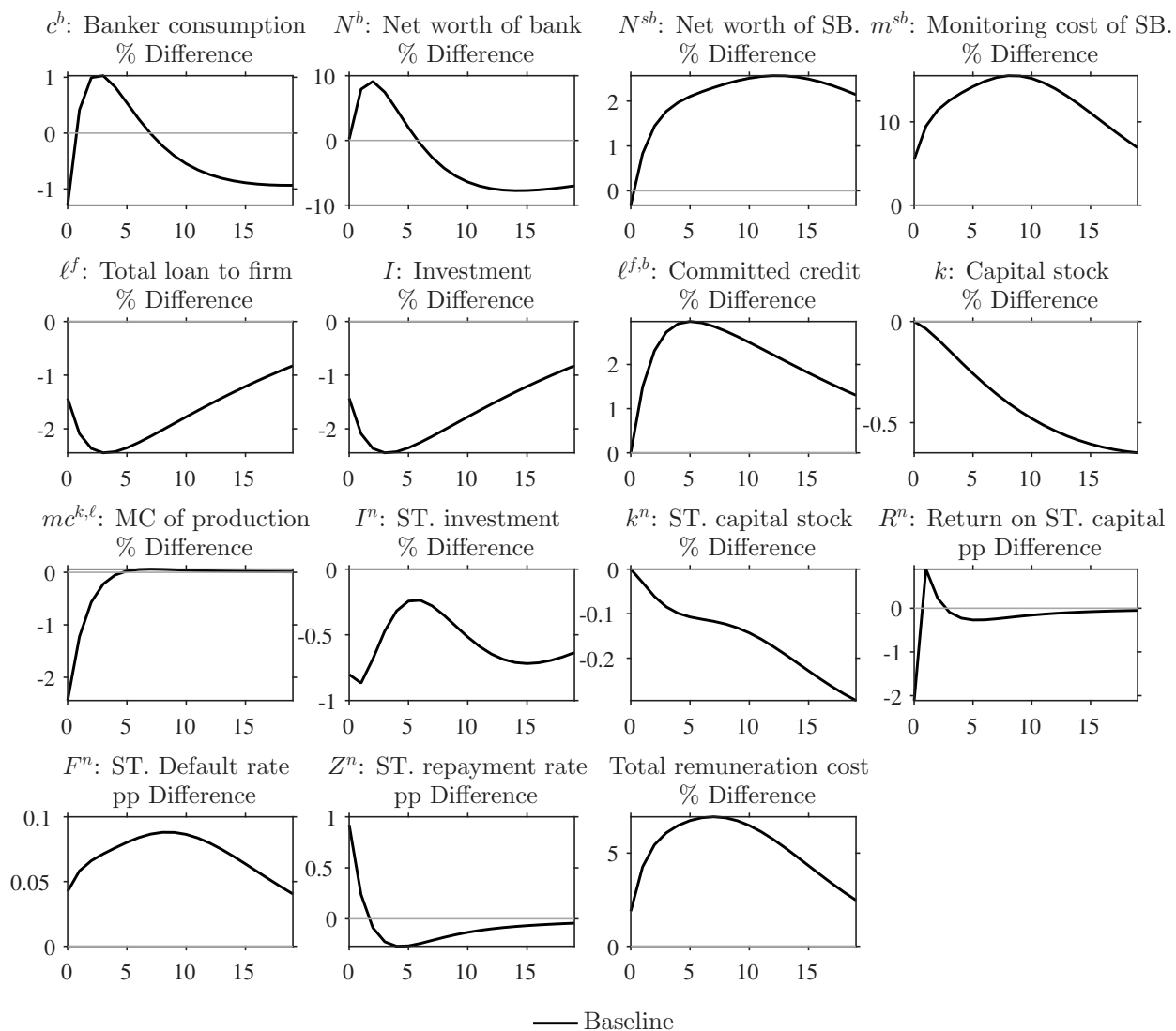


Figure 31: Monetary policy shock 3

I.3 IRFs for Asset purchase policy Shocks

The following figures present the impulse response functions (IRFs) for a positive shock to the asset purchase policy.

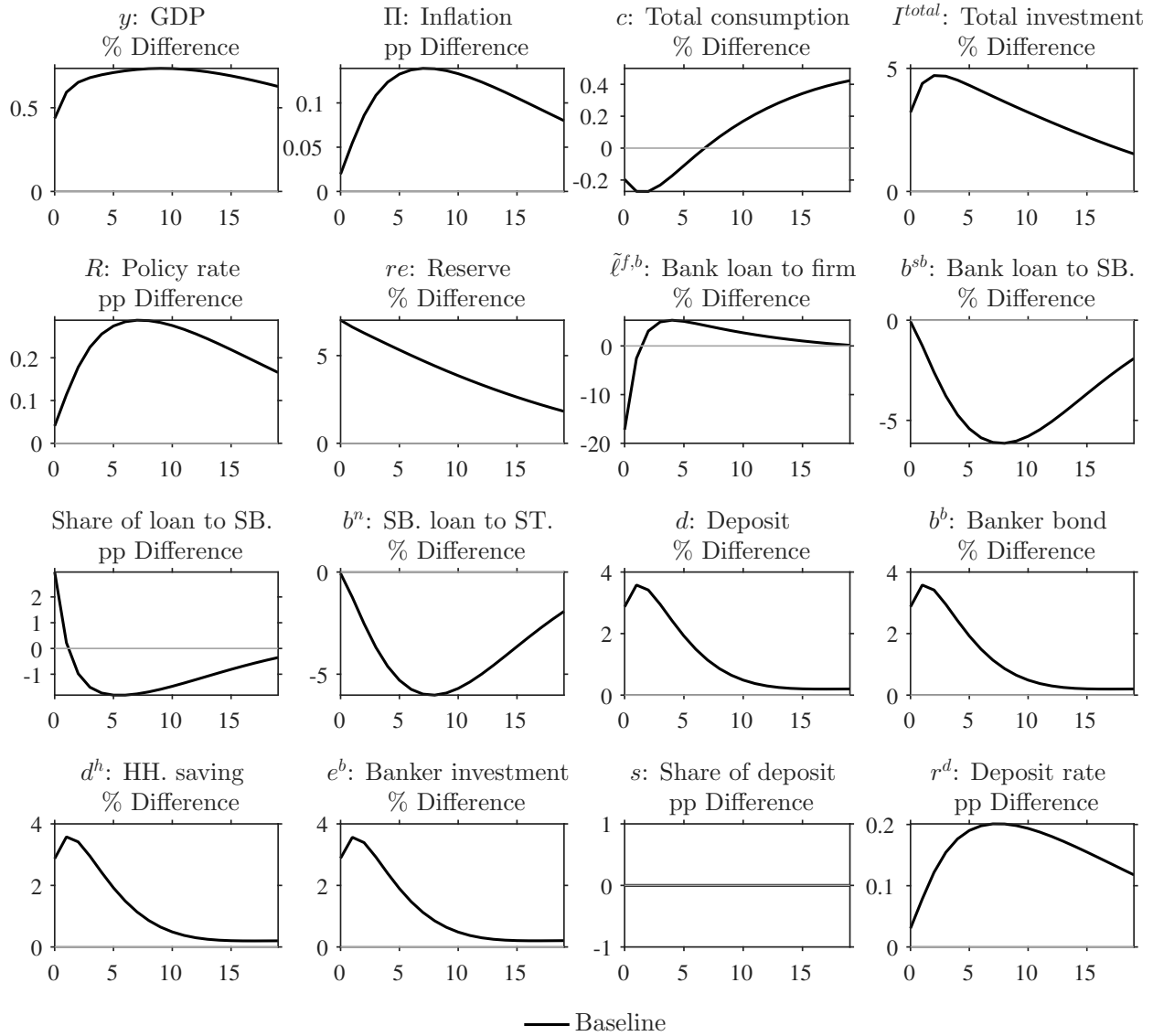


Figure 32: Asset purchase policy shock 1

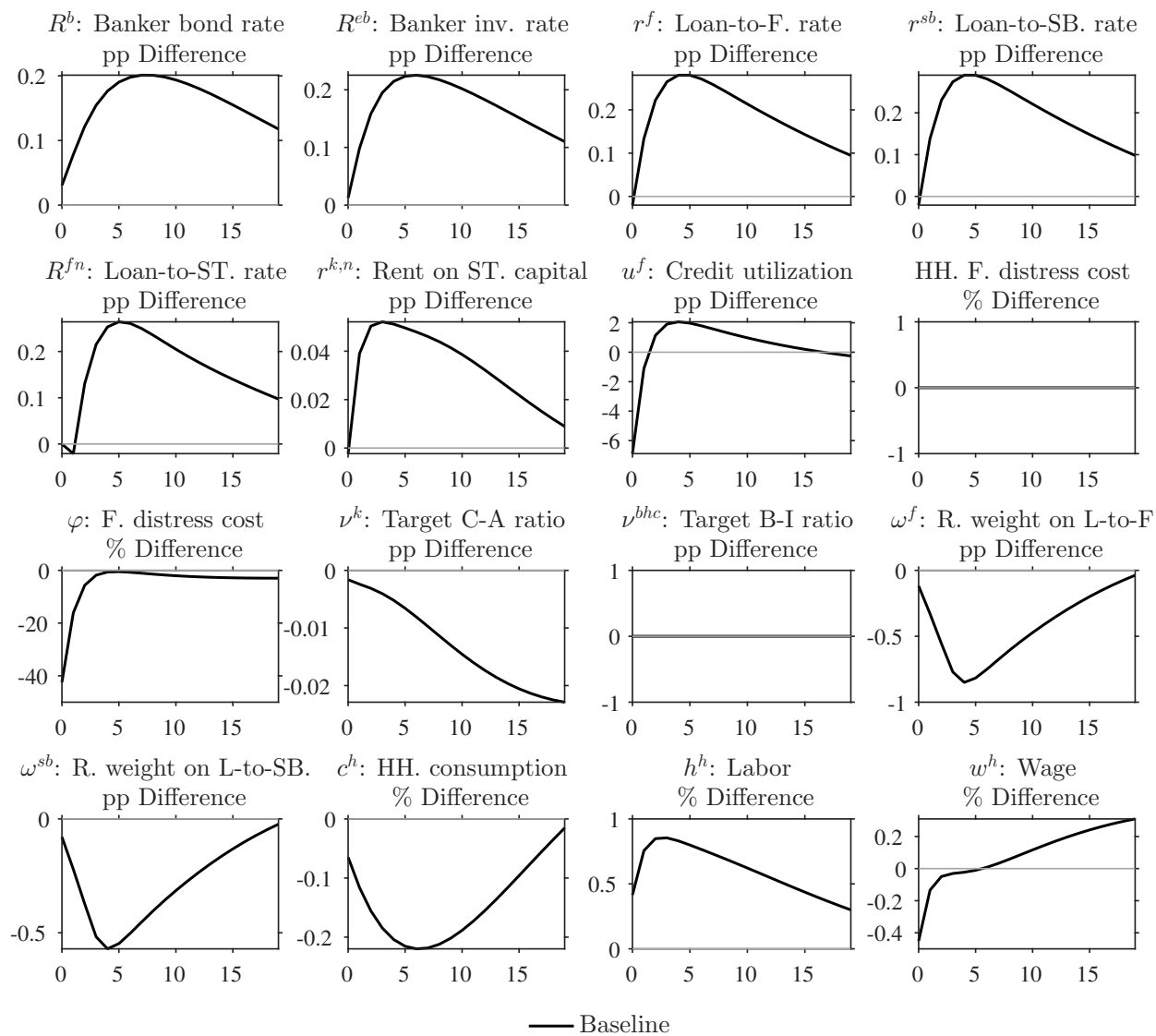


Figure 33: Asset purchase policy shock 2

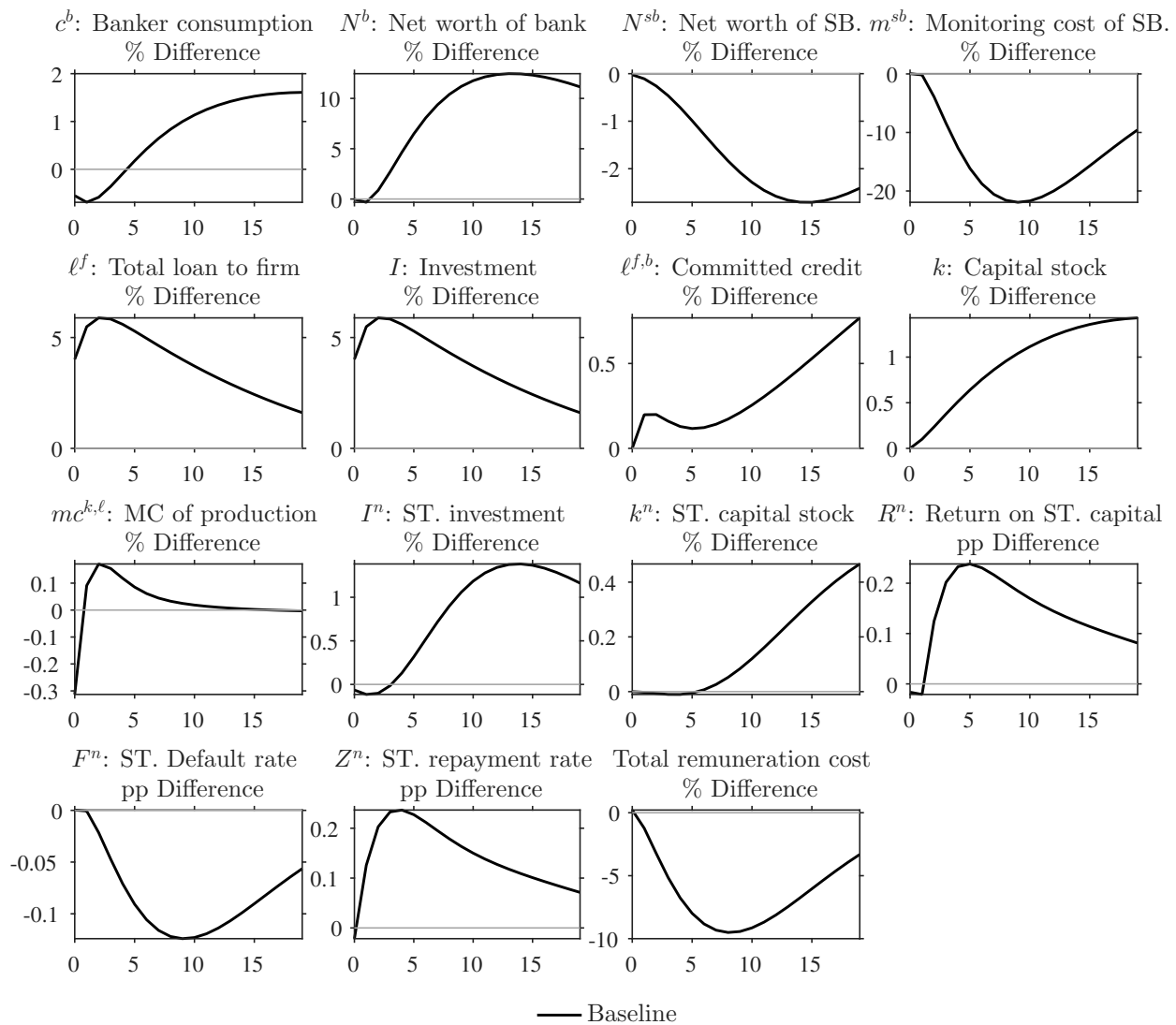


Figure 34: Asset purchase policy shock 3

I.4 IRFs for Return Shocks

The following figures present the impulse response functions (IRFs) for a positive shock to the startups' return.

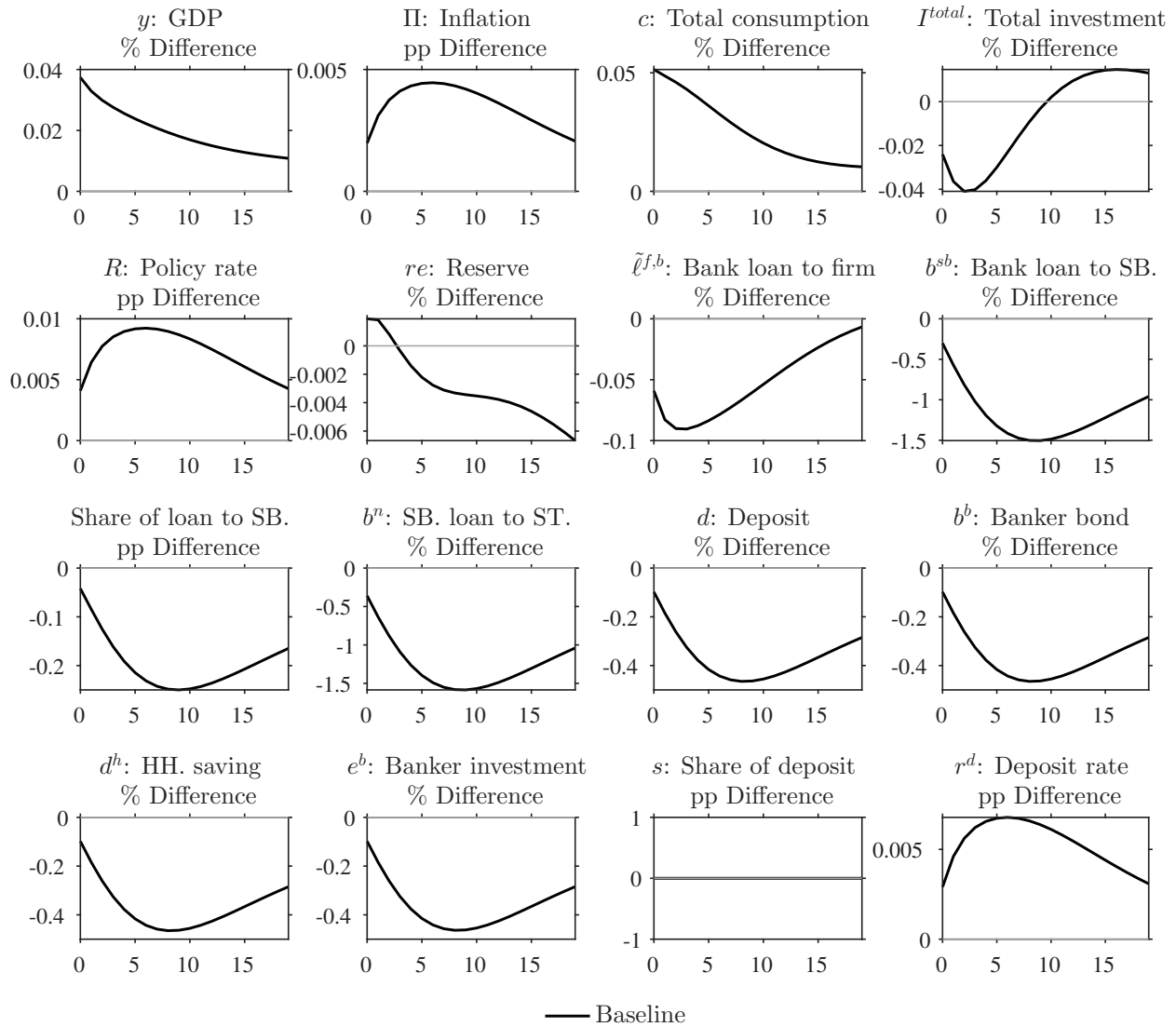


Figure 35: Return shock 1

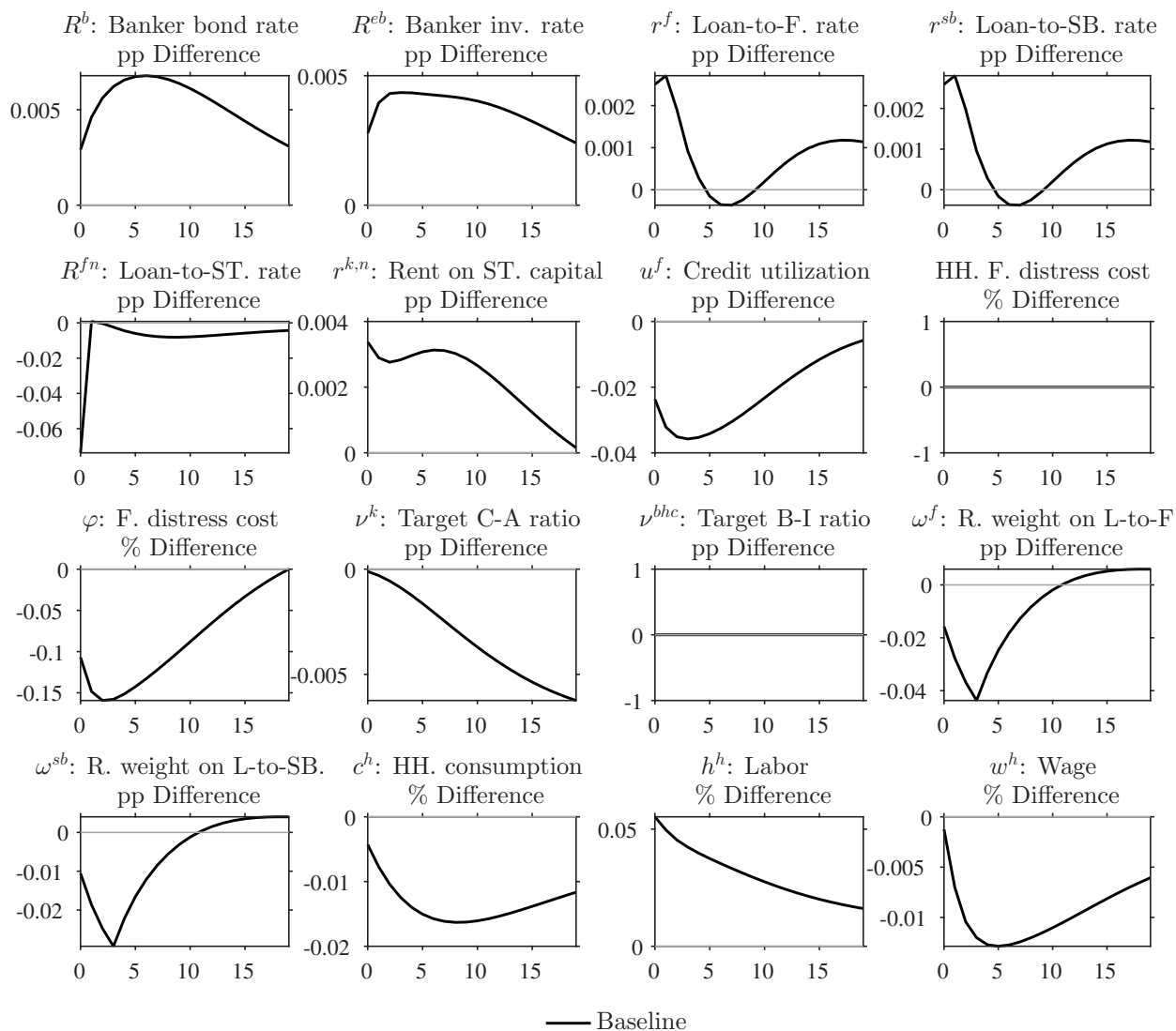


Figure 36: Return shock 2

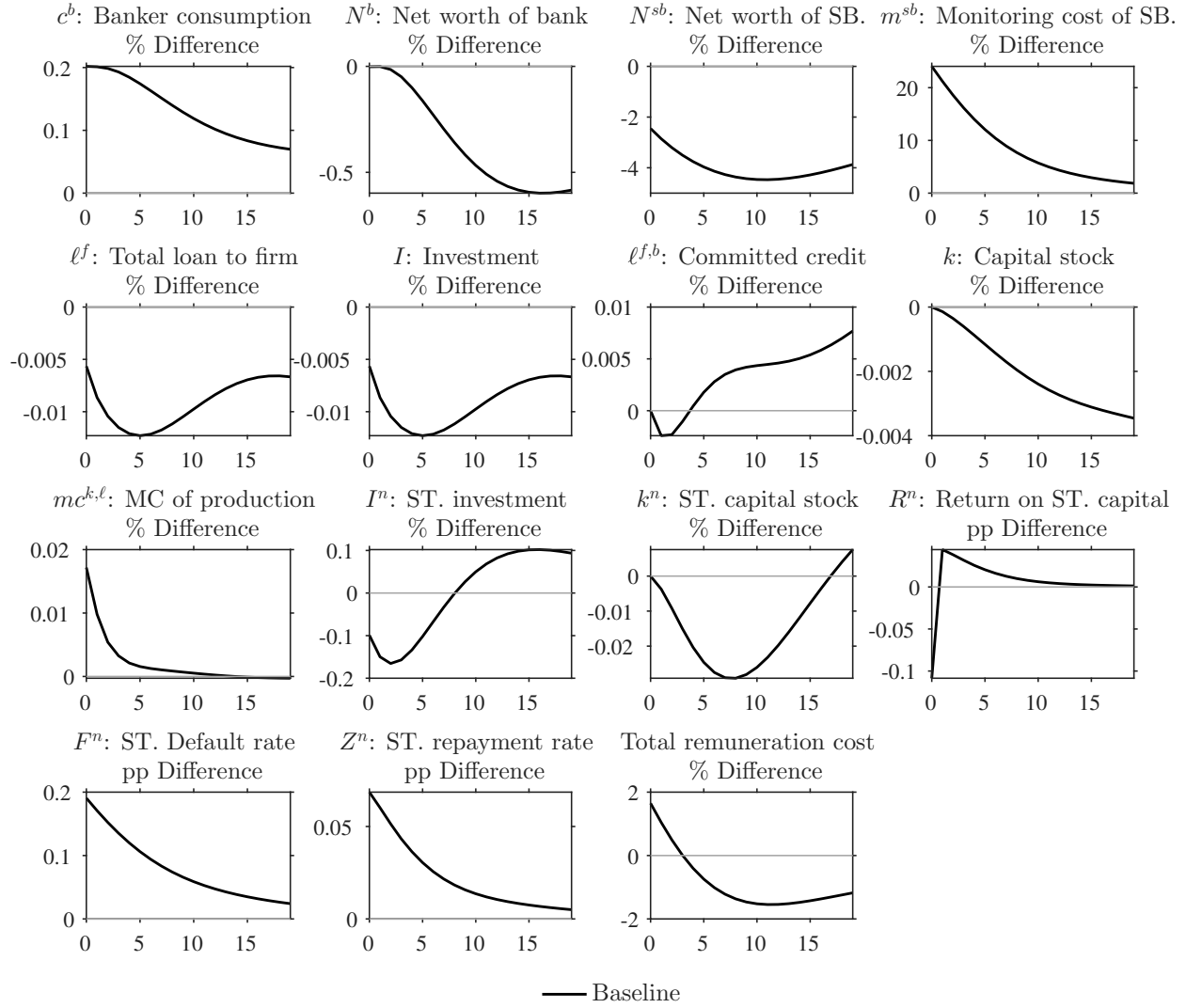


Figure 37: Return shock 3

I.5 IRFs for Debt-Investment Shocks

The following figures present the impulse response functions (IRFs) for a positive shock to the intermediate good producers' debt-investment multiplier.

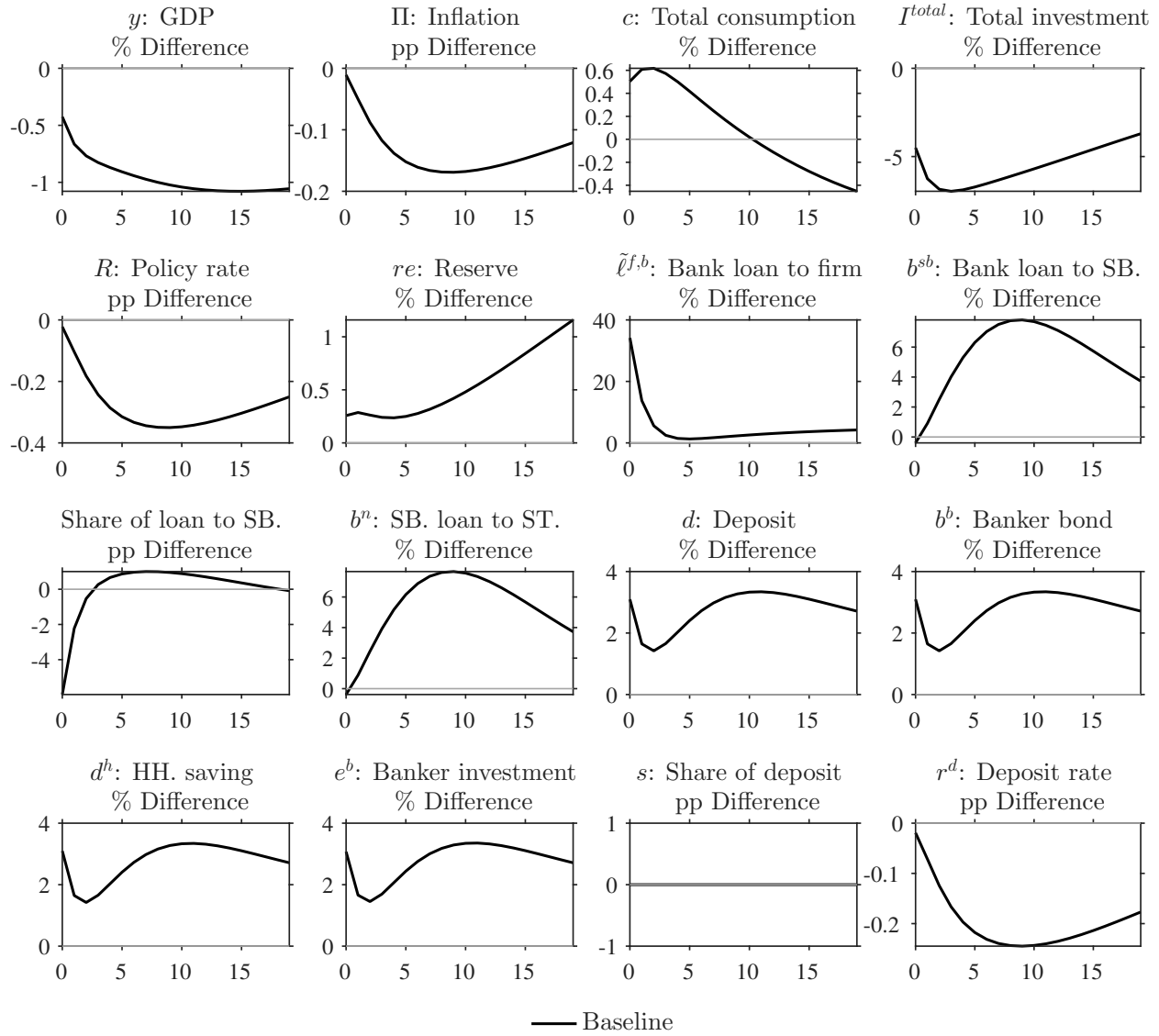


Figure 38: Debt-investment shock 1

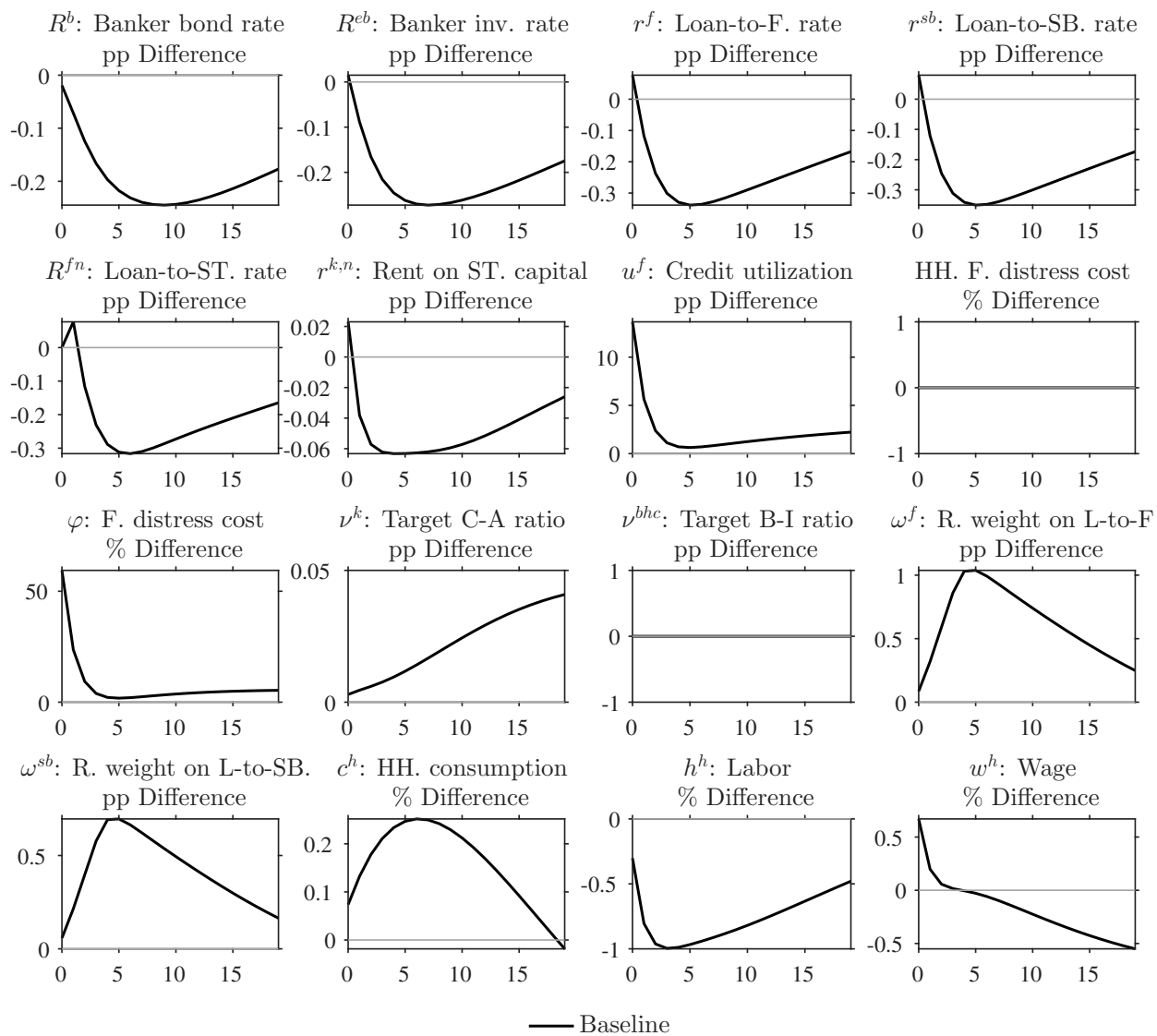


Figure 39: Debt-investment shock 2

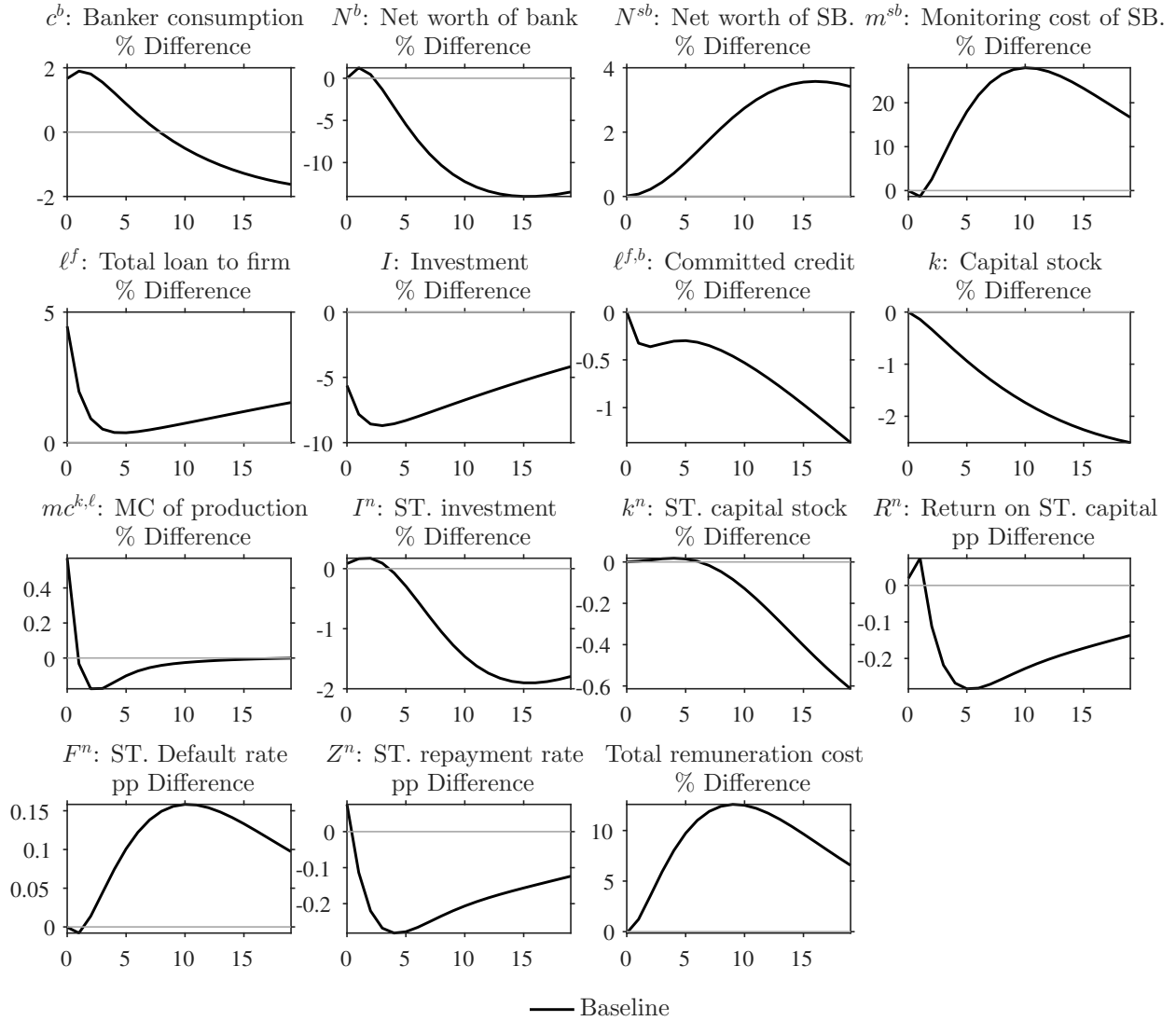


Figure 40: Debt-investment shock 3

I.6 IRFs for Macprudential Policy Shocks 1: Capital-Asset Ratio of Wholesale Bank

The following figures present the impulse response functions (IRFs) for a positive shock to the capital-asset ratio of the wholesale bank.

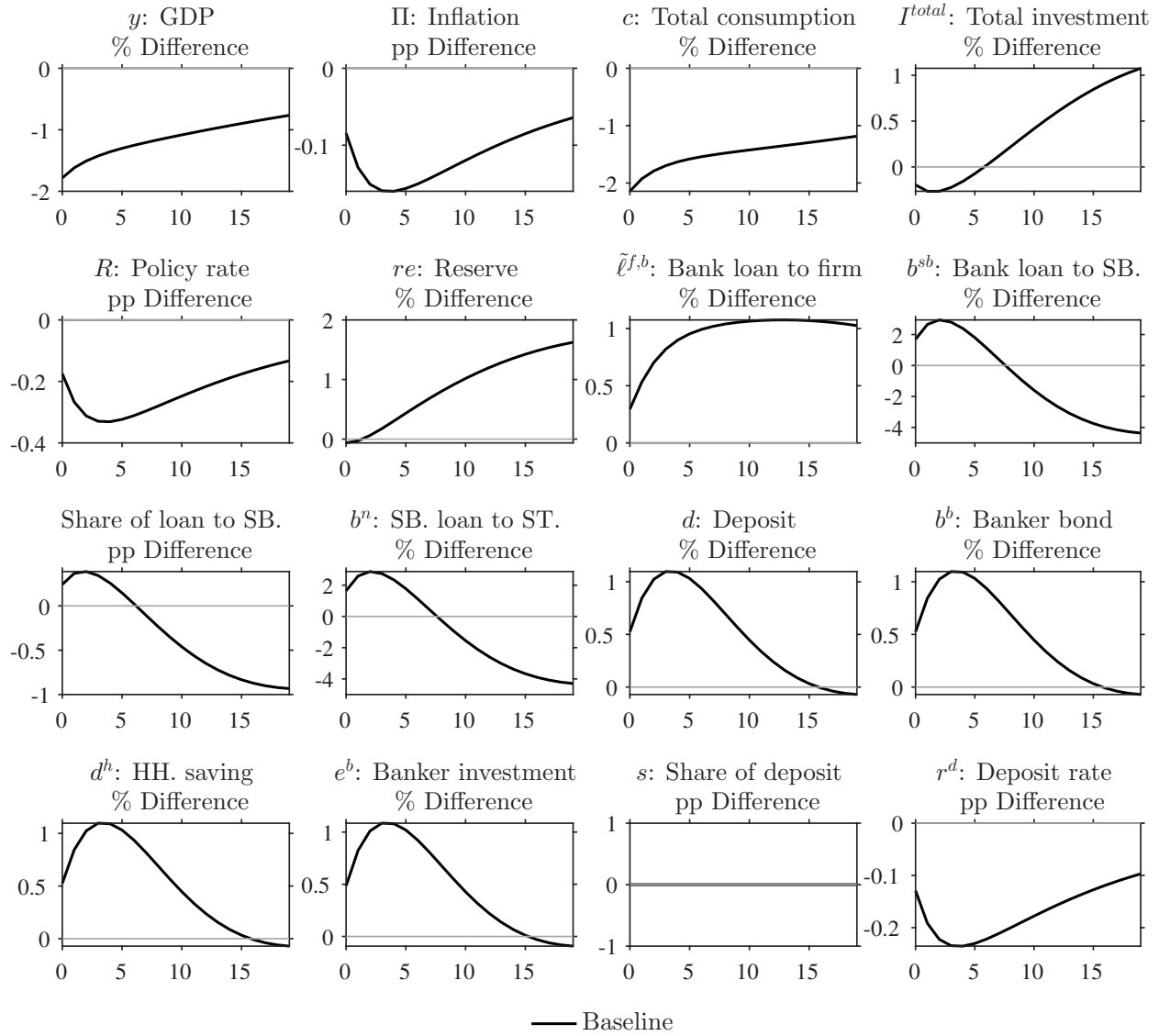


Figure 41: Macprudential policy shock 1

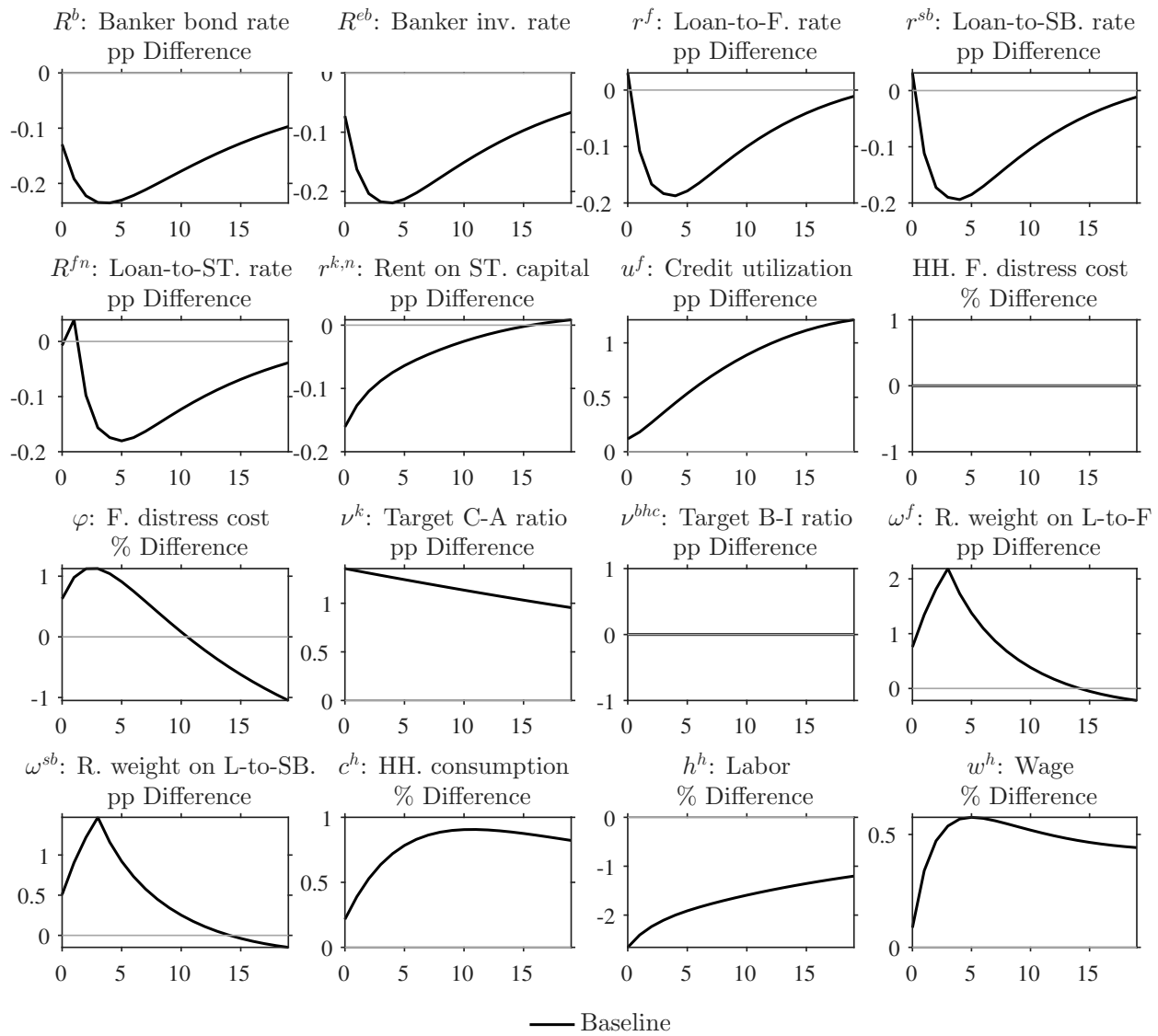


Figure 42: Macprudential policy shock 2

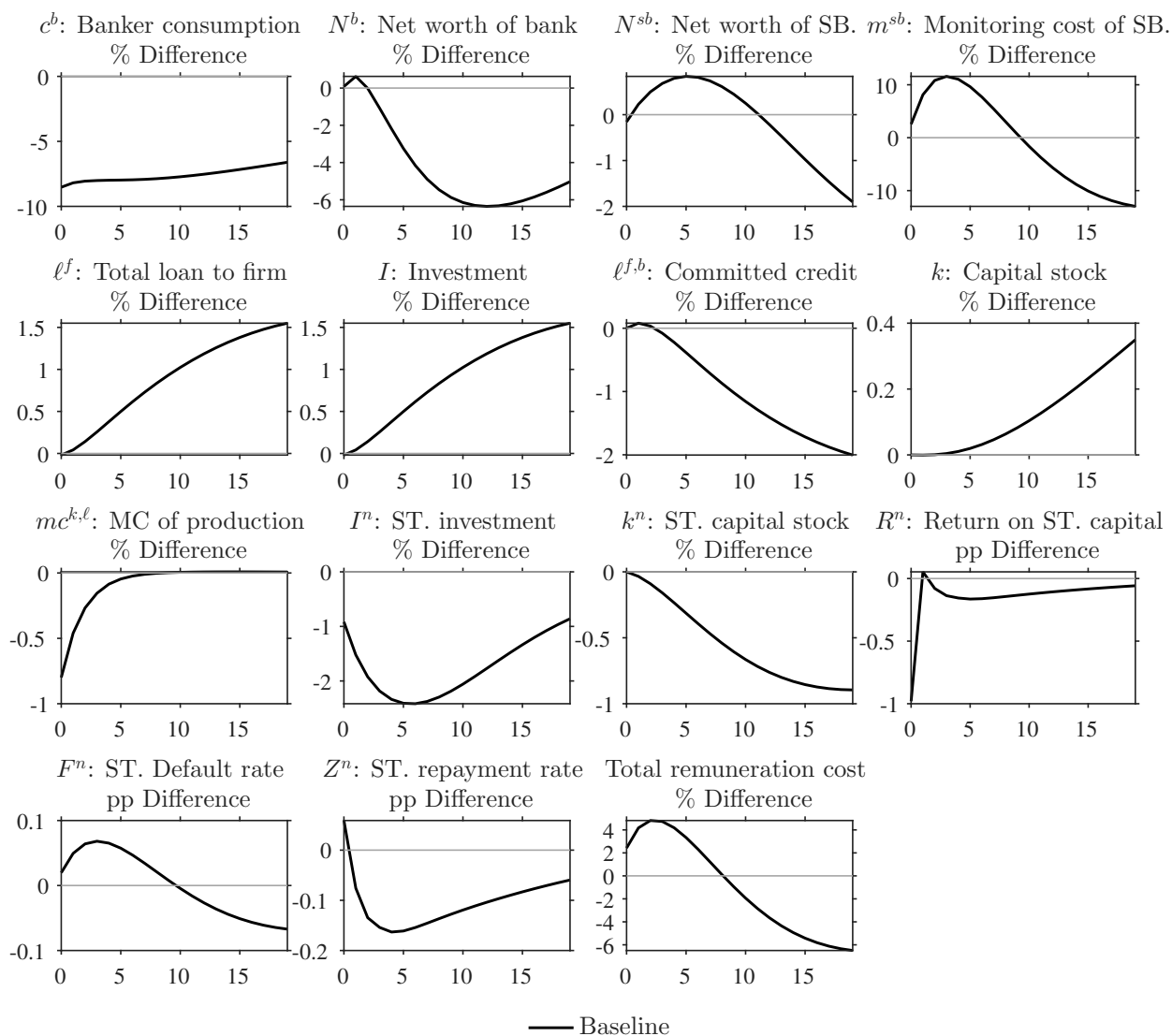


Figure 43: Macroprudential policy shock 3

I.7 IRFs for Macprudential Policy Shocks 2: Leverage Ratio of Banker

The following figures present the impulse response functions (IRFs) for a positive shock to the leverage ratio of the banker.

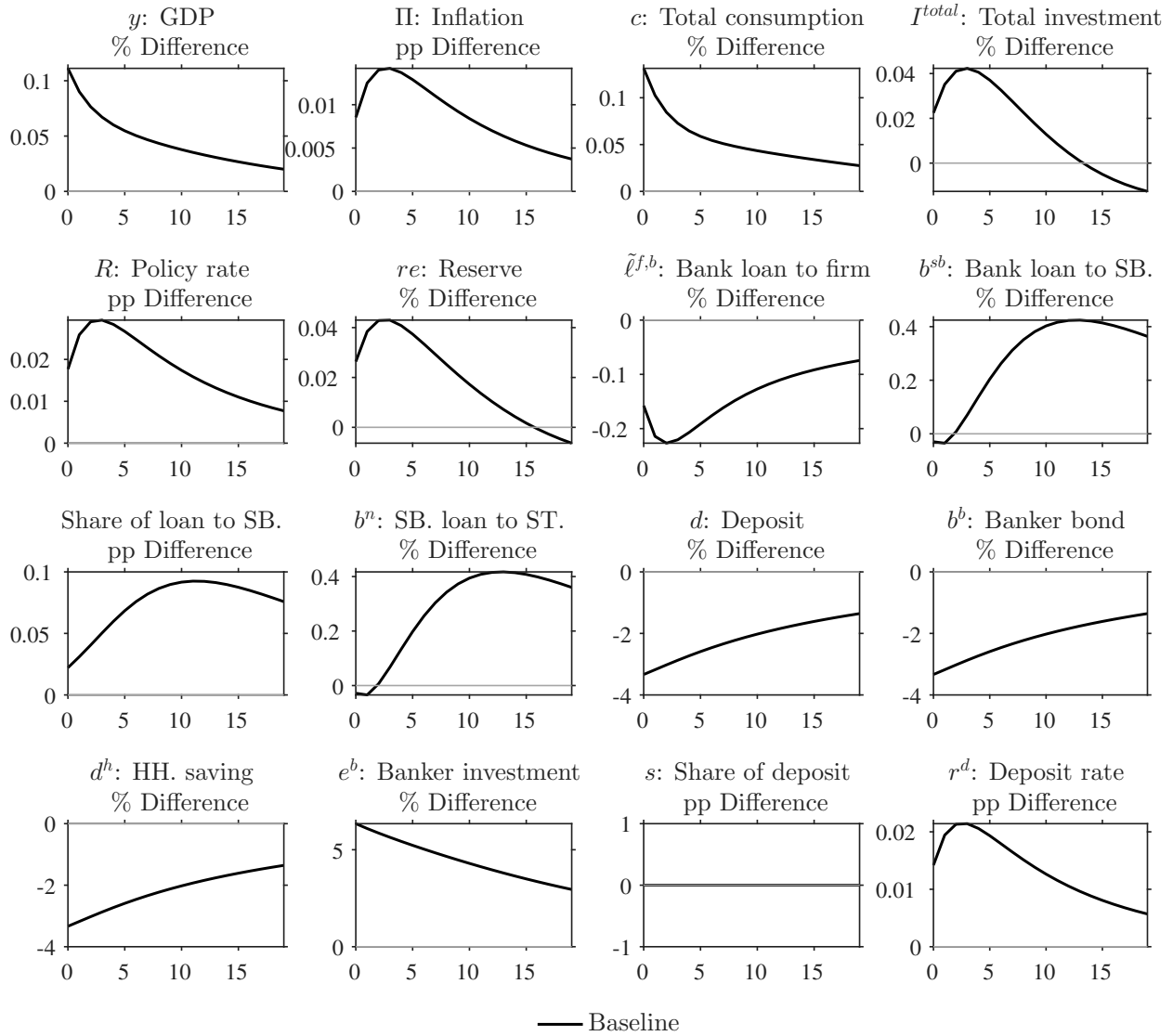


Figure 44: Macprudential policy shock 4

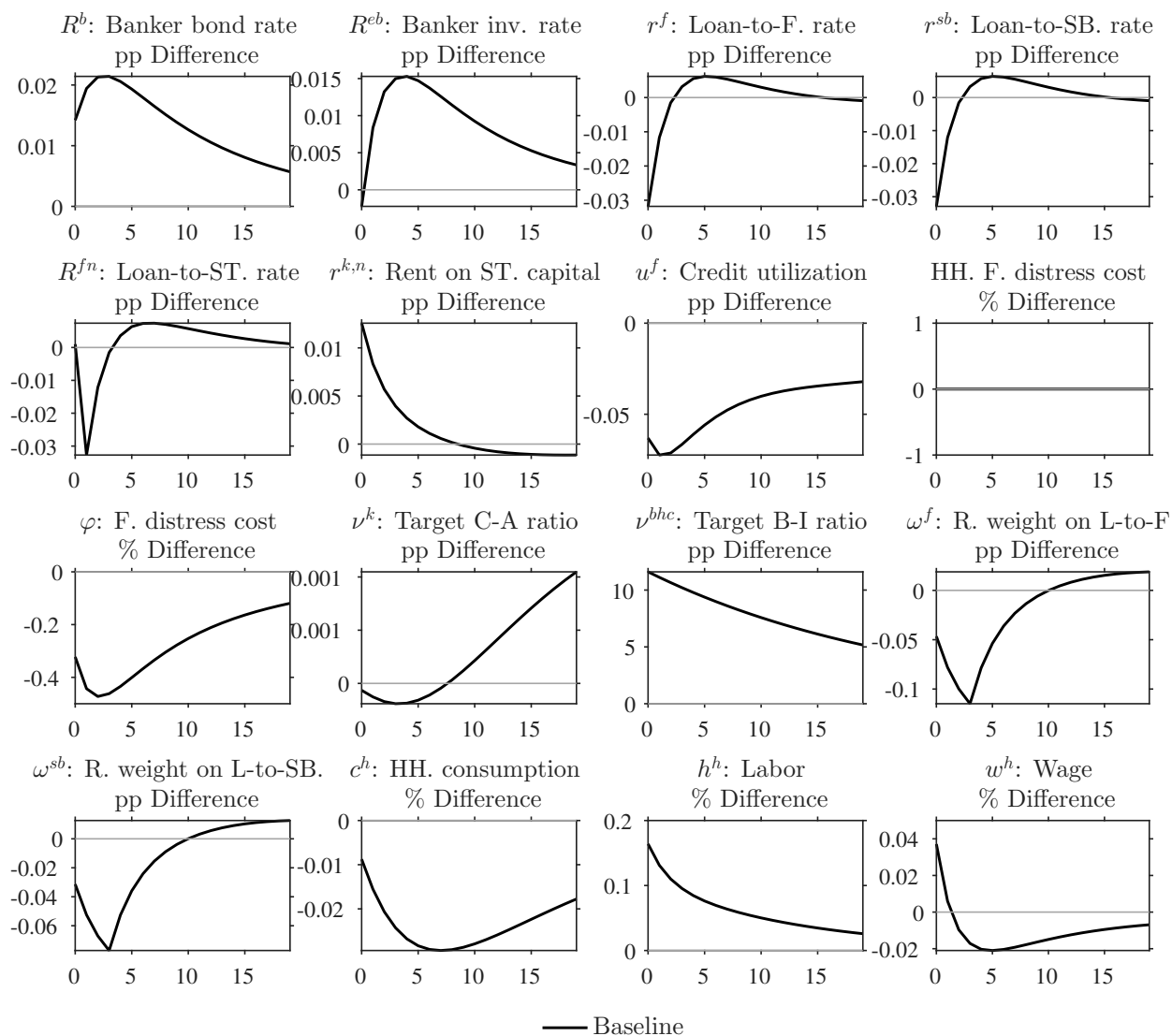


Figure 45: Macprudential policy shock 5

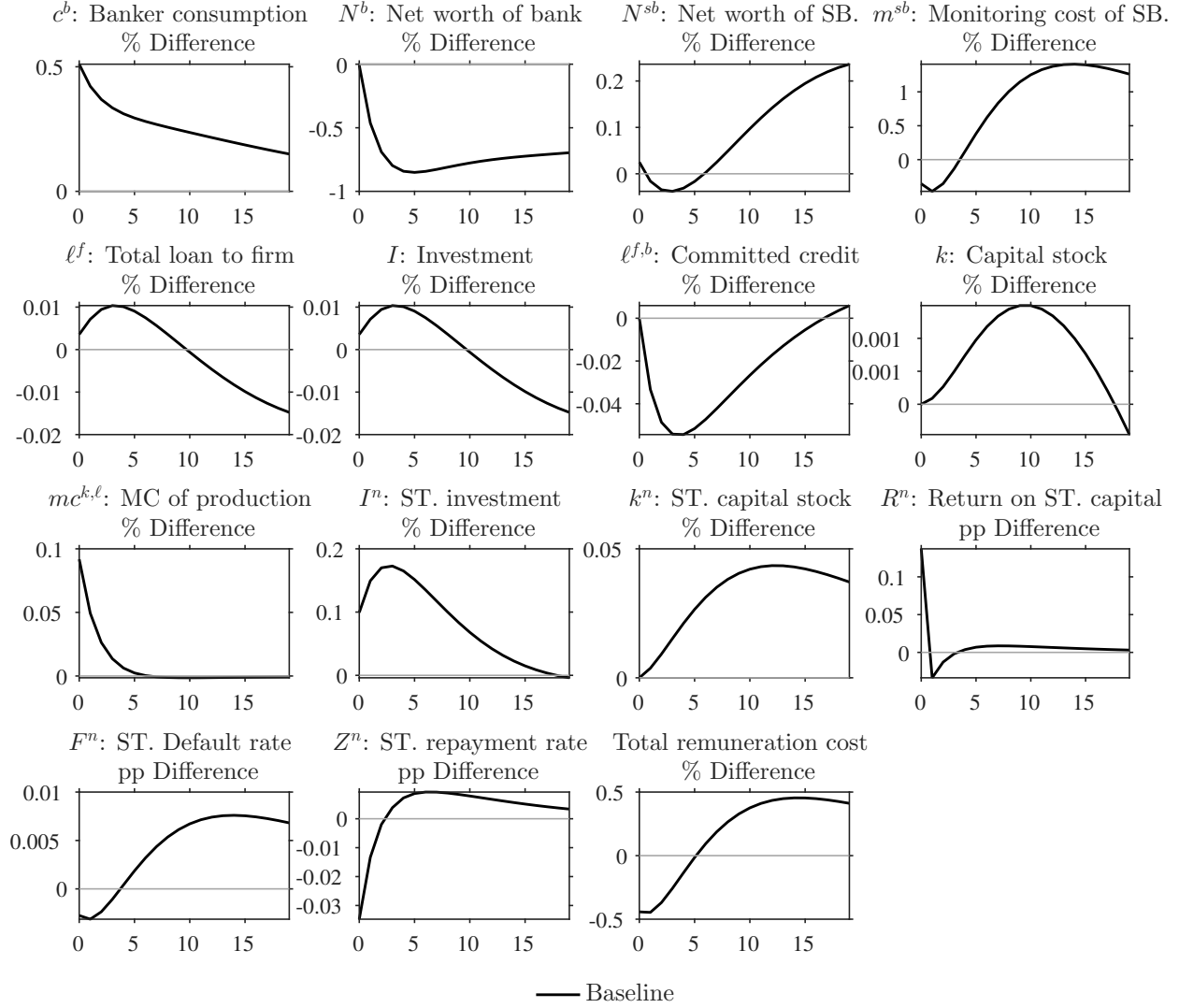


Figure 46: Macprudential policy shock 6